



**School of Computing, Engineering and Mathematical Science  
Department of Engineering**

# **Design and Evaluation of a Geographic Information System-Assisted Hybrid Communication and Segmented Operational Technology Network Architecture for a Lagoon-Based Wastewater Treatment Plant**

A thesis submitted in partial fulfilment of the requirements of the degree of  
Master of Engineering (Telecommunication and Networking)

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**Date of submission: 7 July 2026**

## Certificate of Authorship/Originality

I certify that the work presented in this thesis of which I am the sole author is, to the best of my knowledge and belief, original, except as acknowledged in the text, and that the material has not been submitted, either in whole or in part, for a degree at this or any other university.

In addition, I declare that all information sources and literature used are indicated in the thesis. I certify that I have complied with the rules, requirements, procedures and policies of the University.

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## Abstract

Large lagoon-based wastewater treatment plants contain widely distributed sensors, pipelines, remote terminal units, closed-circuit television (CCTV) systems and central supervisory control infrastructure. Their communication requirements differ substantially in data rate, latency, range, energy use and resilience, which makes a single-technology network difficult to justify. This study designed and evaluated a geographic information system (GIS)-assisted hybrid communication and segmented operational technology architecture using a Western Treatment Plant-style case study represented by the hypothetical name Greenstock Water Corporation. Quantum Geographic Information System (QGIS) was used to model the site, prepare 423 infrastructure nodes and generate 395 communication links. The links were assigned to low-power wide-area network (LPWAN), Wi-Fi HaLow, Wi-Fi HaLow backhaul, fifth generation (5G) mobile communication or fibre, and fibre backhaul according to operational role. A 120-second network simulator 3 (ns-3) model measured packet delivery ratio, delay, throughput and packet loss under baseline, obstruction, traffic-load and combined sensitivity scenarios. The baseline achieved an overall authorised packet delivery ratio of 99.48%. Fibre produced 100% delivery with 1.103 ms average delay, while LPWAN produced 99.00% delivery with 200.401 ms delay, reflecting its low-rate telemetry role. Under doubled obstruction and tripled traffic, overall delivery remained 98.98%, although loss increased. A separate flat-versus-segmented security model showed that nine unauthorised flows could reach the supervisory control and data acquisition (SCADA) zone in the flat scenario, whereas all nine were blocked by the segmented policy without reducing normal traffic performance. A Packet Tracer implementation then validated Virtual Local Area Network (VLAN) separation, core redundancy, gateway routing, an operational technology (OT) demilitarized zone (DMZ) and restricted firewall paths. The findings indicate that a GIS-assisted hybrid design can match communication technologies to operational needs while security zoning limits lateral movement. The results remain simulation-based and require field measurements, protocol-specific testing and production-grade security validation before deployment.

## Acknowledgements

First, I would like to thank my academic advisor, Dr. Alex Stumpf, during this journey for his constant help, advice and feedback for this thesis project. I am very appreciative of the direction he provided to develop the research focus, improve methodology, and keep the project moving.

Next, I would like to thank, with great sincerity, Leonie Williams, at the time with the Western Treatment Plant as the Environment and Compliance Officer at Melbourne Water, for her field site guidance and clarifying the setting. It was extremely helpful to me as it framed the field conditions, aspects of operation, and the environment surrounding the research.

I would also like to thank Henry Cooper from Melbourne Water and the Western Treatment Plant for providing me the necessary geographic information system (GIS) layers to this project. The layers helped me build a better understanding of the WTP and helped construct the spatial model to the methodology.

Thank you to La Trobe University and all those who helped me with this capstone research project.

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## Abbreviations

| Abbreviation | Meaning                                        |
|--------------|------------------------------------------------|
| 3GPP         | 3 <sup>rd</sup> Generation Partnership Project |
| 5G           | 5 <sup>th</sup> Generation                     |
| ACL          | Access Control List                            |
| ASA          | Adaptive Security Appliance                    |
| CCTV         | Closed-Circuit Television                      |
| CRS          | Coordinate Reference System                    |
| GIS          | Geographic Information System                  |
| HSRP         | Hot Standby Router Protocol                    |
| IACS         | Industrial Automation and Control System       |
| IEC          | International Electrotechnical Commission      |
| IoT          | Internet of Things                             |
| LPWAN        | Low-Power Wide-Area Network                    |
| NIST         | National Institute of Standards and Technology |
| OT           | Operational Technology                         |
| PDR          | Packet Delivery Ratio                          |
| QGIS         | Quantum Geographic Information System          |
| QoS          | Quality of Service                             |
| RTU          | Remote Terminal Unit                           |
| SCADA        | Supervisory Control and Data Acquisition       |
| VLAN         | Virtual Local Area Network                     |
| WTP          | Western Treatment Plant                        |

# 1. Introduction

## 1.1 Background and research context

Wastewater treatment is an essential public service that protects human health, receiving environments and the long-term sustainability of urban water systems. Modern plants do more than move wastewater through a fixed sequence of physical and biological processes. They operate as distributed cyber-physical systems in which sensors, controllers, communication links, supervisory platforms and human operators continuously interact. Measurements such as flow, pressure, level, dissolved oxygen, temperature, pH, turbidity, gas concentration and equipment state are needed to support process control, compliance monitoring, maintenance and incident response. The practical value of these measurements depends on whether they can be delivered reliably from the field to remote terminal units (RTUs), gateways and supervisory control and data acquisition (SCADA) systems.

The communication problem becomes more difficult in a lagoon-based treatment environment. Lagoon plants may extend across many square kilometres and include open water bodies, wetlands, embankments, long pipelines, pump stations, electrical assets and central control facilities. Sensors installed around lagoons or pipeline corridors can be far from mains power and fixed communications. Some devices send only small periodic measurements, while CCTV and other high-bandwidth systems may require sustained data rates. Alarm and control messages can be more time-sensitive than routine environmental observations. A communication system designed around one technology can therefore perform well for one class of device but poorly for another.

The Western Treatment Plant in Werribee, Victoria, provides a useful spatial and operational reference for this problem. Melbourne Water describes the site as a large treatment facility that combines sewage treatment with an extensive lagoon, wetland and agricultural environment (Melbourne Water, 2023). The scale and variety of the site create a realistic context for examining long-range low-power sensing, moderate-rate process telemetry, high-bandwidth monitoring and fibre-connected control infrastructure. This thesis did not use live operational systems or confidential plant information. Instead, the site was represented as the hypothetical Greenstock Water Corporation using public spatial data, satellite interpretation and research-generated assumptions.

Digitalisation has increased the importance of plant-wide data availability. Internet of Things (IoT) platforms, edge computing and digital twins can improve monitoring and decision support

only when field data are sufficiently reliable, timely and contextualised (Gubbi et al., 2013; Shi et al., 2016; Tao et al., 2019; Fuller et al., 2020). A digital twin of a wastewater process cannot compensate for persistent communication gaps, missing observations or insecure pathways. For this reason, communication architecture should be treated as part of the operational engineering problem rather than as an independent information technology layer.

## **1.2 Communication heterogeneity in wastewater operations**

The devices used in wastewater operations do not have uniform communication requirements. A lagoon-level sensor may transmit a short measurement every several minutes and remain battery powered for long periods. A process instrument associated with a pipeline, pump or treatment stage may send data more frequently and require faster event reporting. An RTU aggregates data and exchanges status or control information with higher-level systems. A CCTV stream has a much larger bandwidth requirement than a pH or temperature measurement. The central server and SCADA environment require stable, high-capacity links and controlled access.

LPWAN technologies are attractive for low-rate field sensing because they prioritise range and energy efficiency rather than throughput (Raza et al., 2017; Centenaro et al., 2016; Mekki et al., 2019). Wi-Fi HaLow, based on IEEE 802.11ah, operates below 1 GHz and provides a middle position between very low-rate LPWAN and conventional higher-rate wireless systems. It can support longer range and improved penetration compared with 2.4 GHz Wi-Fi while retaining IP-oriented networking and higher data capacity than many LPWAN solutions (IEEE, 2016; Austin & Ancillotti, 2018; Lee et al., 2021). Private 5G and fibre offer greater capacity and lower delay for high-bandwidth or critical backhaul applications, but they have different cost, deployment and coverage constraints (Andrews et al., 2014; 3GPP, 2022a).

These trade-offs suggest that the architecture should be hybrid. Hybridisation is not simply the use of several technologies at one site. It requires a defensible method for assigning each technology to a device class, linking field zones through aggregation points, supporting reliable backhaul and maintaining manageable security boundaries. The same physical environment also influences different wireless links in different ways. Distance, vegetation, built areas, water bodies and local obstruction can affect received power and packet error. A spatially abstract model may therefore overlook important relationships between assets and communication paths.

A treatment-plant communication path normally contains several layers. Field instruments measure physical variables. RTUs or programmable controllers collect those measurements, apply local logic and communicate with higher-level systems. Gateways bridge wireless and wired networks, aggregate traffic and provide protocol or security functions. The SCADA layer presents process state, alarms and trends to operators. Historical systems, engineering workstations and analytics platforms use the collected data for reporting and optimisation (Stouffer et al., 2023).

A weakness at one layer affects the entire chain. A sensor may be accurate but operationally ineffective if its packets are frequently lost. A gateway may have adequate radio coverage but become a bottleneck if aggregated traffic exceeds its backhaul. A SCADA server may be well protected but still exposed if field, vendor and management networks are not separated. The communication architecture must therefore be evaluated as an end-to-end system (Boyes et al., 2018; Gungor et al., 2011; Stouffer et al., 2023).

The hierarchy also influences where performance should be measured. A short sensor-to-RTU link can perform well while the RTU-to-gateway or gateway-to-server path fails. For this reason, the GIS link model did not treat every device as a direct client of the central server. It represented access, aggregation and backhaul roles. The Packet Tracer model used the same logic by separating sensor switches, a control switch, a gateway aggregation switch, the WTP gateway, the ASA firewall and central servers (Boyes et al., 2018; Gungor et al., 2011).

### **1.3 Operational technology security context**

Wastewater communication systems are also operational technology systems. Their security objective is not limited to data confidentiality. Availability, integrity, safe process operation and controlled recovery are central concerns. A flat network can allow a compromised device or vendor endpoint to reach systems beyond its operational purpose. Security guidance for industrial control systems therefore emphasises network segmentation, zones, conduits, least privilege, controlled remote access and monitoring (Stouffer et al., 2023; IEC, 2020; IEC, 2013).

The relationship between communication performance and security is important. Excessive isolation can make operations difficult, while unrestricted connectivity creates lateral movement and attack paths. A practical design must preserve authorised traffic, such as RTU-to-SCADA communication, while blocking unnecessary direct access from sensors or vendor

devices. This study therefore treated security segmentation as a second design dimension rather than adding it after the performance model was complete.

### **1.3.1 Western Treatment Plant-style spatial environment**

The Western Treatment Plant-style environment was selected because it exposes communication issues that are less visible in a compact building. Large lagoons create long line-of-sight paths but also place sensors far from power and fixed networks. Wetlands and vegetation introduce variable attenuation. Pipeline corridors create elongated deployment patterns rather than regular grids. Central control facilities contain power, fibre and staff access, while remote field assets may be visited only periodically.

These conditions make technology allocation important. A conventional office network design cannot simply be extended across the site. The architecture needs low-power coverage for remote measurements, moderate-capacity links for operational telemetry, high-capacity services for video and a stable backbone for central systems. It must also tolerate the fact that some assets are easier to maintain than others. A battery-powered sensor may need years of operation, whereas a gateway installed at a pump station can use mains power and a larger antenna.

The case study was not intended to represent Melbourne Water's current network. It was a research abstraction that used the site's scale and visible infrastructure to test a general method. This distinction protected confidentiality and prevented the thesis from implying that observed public features reveal internal OT design.

## **1.4 Research problem**

The research problem was that a large lagoon-based wastewater treatment plant required communication across geographically dispersed and operationally different devices, yet the available technologies had conflicting characteristics. Long-range low-power links could not provide the same bandwidth and delay as fibre or 5G. High-capacity infrastructure could not be economically extended to every remote sensor. Spatial obstruction and link distance created further uncertainty. At the same time, a communication design that connected every device through a common network would not provide adequate OT security separation (Raza et al., 2017; Gungor et al., 2011; Stouffer et al., 2023).

Previous research has provided strong evidence about individual technologies, IoT architectures, edge computing and industrial cybersecurity (Al-Fuqaha et al., 2015; Shi et al.,

2016; Boyes et al., 2018; Stouffer et al., 2023). However, many studies use idealised topologies, examine one communication family, or treat cybersecurity separately from physical network planning. The problem addressed in this thesis was therefore not to identify a universally superior technology. It was to determine how several technologies could be combined in a spatially grounded architecture, how that architecture behaved under changing conditions, and whether a segmented implementation could preserve authorised operation while reducing unauthorised reach (Gungor et al., 2011; Boyes et al., 2018; Stouffer et al., 2023).

## **1.5 Research gap**

The literature revealed four connected gaps. First, technology comparisons commonly report general range, energy, throughput and latency characteristics without mapping those characteristics to the operational roles of a lagoon-based treatment plant. Second, simulation studies frequently use random, grid or small test topologies that do not reflect the geographic relationships between lagoons, pipelines, RTUs, gateways and a central control area. Third, sensitivity to obstruction and traffic growth is not always evaluated alongside the baseline architecture. Fourth, communication and cybersecurity evaluations are often separated, even though the physical topology determines the available attack paths and the location of security controls.

This thesis responded by integrating QGIS-based spatial modelling, role-based technology assignment, ns-3 performance simulation, sensitivity testing, policy-level security experiments and a Packet Tracer implementation. The novelty was therefore integrative rather than a claim of inventing a new protocol. The study connected methods that are normally evaluated separately and applied them to one coherent case-study architecture.

### **1.5.1 Why integrated evaluation was required**

The four research gaps were connected. A technology-selection study without geography can recommend a range that is unsuitable for the actual distances. A GIS model without traffic analysis can show where devices are located but not whether links can carry their data. A performance model without security can optimise paths that create excessive reach. A security model without operational traffic can block functions needed by operators.

Integrated evaluation created trade-offs that would otherwise remain hidden. For example, a gateway reduced the number of long backhaul links but concentrated traffic and security responsibility. A high-bandwidth CCTV zone improved surveillance but could also generate

the most damaging unauthorised traffic in a flat network. Fibre provided reliable long-distance backhaul but created dependence on physical routes. These interactions justified an architecture-level research question rather than a narrow protocol benchmark.

## **1.6 Research Question**

The research question was:

How can a GIS-assisted hybrid communication architecture combining LPWAN, Wi-Fi HaLow, high-bandwidth communication, gateway aggregation and segmented OT networking improve communication suitability, reliability and operational resilience in a large lagoon-based wastewater treatment plant?

## **1.7 Research Aim**

The aim was to design and evaluate a spatially grounded hybrid communication and segmented OT architecture for a Western Treatment Plant-style environment.

## **1.8 Research Objectives**

The objectives were to:

1. Construct a GIS representation of distributed sensors, RTUs, gateways, communication links and the central server;
2. Assign communication technologies according to location, traffic type, range, latency and power constraints;
3. Implement a reproducible ns-3 model and measure packet delivery ratio, delay, throughput and packet loss;
4. Evaluate robustness under increased obstruction, increased traffic and a combined stress scenario;
5. Compare a flat network with a segmented zone-and-conduit model under unauthorised traffic;
6. Implement the logical design in Packet Tracer using VLANs, redundant core switching, gateway routing, an ASA firewall and an OT DMZ; and
7. Identify limitations and provide recommendations for field validation and future deployment.



## 1.9 Scope and boundaries

The study included GIS-assisted node placement, link generation, technology assignment, performance simulation, sensitivity analysis, security zoning and Packet Tracer validation. The model included lagoon and level sensors, process and pipeline sensors, RTUs, CCTV or other high-bandwidth nodes, gateways and a central server. It evaluated LPWAN, Wi-Fi HaLow, Wi-Fi HaLow backhaul, 5G or fibre, and fibre backhaul as functional technology classes.

The work did not include physical radio measurements, live access to SCADA equipment, production firewall testing, full protocol-stack emulation for every wireless technology, detailed energy-harvesting design, commercial procurement analysis or machine-learning model training. Edge processing was represented as gateway aggregation and prioritisation at an architectural level, not as a trained anomaly-detection model. The results therefore support comparative design decisions rather than certify a deployment.

## 1.10 Contribution and significance

The study contributed a repeatable workflow that links spatial infrastructure planning to network simulation and OT security design. The QGIS model preserved the positions and relationships of 423 nodes and 395 links. The ns-3 experiments produced technology-level and scenario-level performance evidence. The security model quantified the difference between flat and segmented reach. Packet Tracer then demonstrated that the planned zones could be translated into VLANs, routing, redundancy and firewall rules.

For practitioners, the work offers a way to screen technologies and identify where field measurements are most valuable before capital is committed. For research, it shows how GIS context, sensitivity analysis and security policy can be incorporated into one communication-architecture evaluation. For the case-study environment, it provides a reasoned basis for using different communication methods for low-rate lagoon sensing, process telemetry, high-bandwidth monitoring and central backhaul.

### 1.10.1 Relationship to engineering practice

The workflow reflects an early engineering design process. Public and inferred information was converted into a spatial asset model. Requirements were translated into technology roles. Simulation tested whether the design behaved consistently with those requirements. Sensitivity analysis examined uncertainty. Security zoning converted risk principles into permitted and

prohibited paths. A smaller implementation then tested whether the logical design could be configured.

This sequence is useful because infrastructure decisions are expensive to reverse. A GIS-assisted simulation cannot remove the need for field trials, but it can reduce the number of unsuitable options carried into procurement. It can also expose missing return routes, single points of failure and overly broad security paths before hardware is installed.

### **1.11 Key terms and interpretation**

Several terms are used consistently throughout the thesis. “Hybrid communication architecture” refers to a planned combination of communication technologies assigned to different operational roles. It does not mean that every node can use every technology. “GIS-assisted” means that node and link positions, distances and obstruction context were derived from geographic layers rather than generated as a random abstract topology. “Operational resilience” refers to the ability to maintain useful monitoring and authorised communication when traffic, obstruction or security conditions change. It does not imply uninterrupted service under every fault.

“Segmentation” refers to the separation of devices into logical security zones and the restriction of traffic between them. VLANs provide logical separation, but the security effect depends on routing and firewall policy. “Flat network” describes a scenario in which candidate field devices can reach the SCADA server without zone-based policy restrictions. “Edge processing” refers to functions located at or near gateways. In this project, edge behaviour was limited to architectural aggregation and prioritisation concepts.

The distinction between architecture-level and protocol-level results is also important. Architecture-level results show how configured technology classes and paths behave in the model. Protocol-level results would require detailed implementation of LoRaWAN, IEEE 802.11ah, 5G scheduling and industrial application protocols. The present thesis primarily reports architecture-level evidence.

### **1.12 Research assumptions at the problem-framing stage**

The study began with several framing assumptions. The case-study area was treated as predominantly open and relatively flat, although wetlands, vegetation, buildings and embankments created local obstruction. Sensor traffic was assumed to be mostly periodic, with higher-rate or urgent traffic assigned to different classes. Gateways were assumed to be placed

at powered and maintainable locations. External interference was not modelled as a dominant baseline factor. These assumptions reduced complexity and made the project feasible, but they also defined the boundary of the conclusions.

The assumptions were not hidden after results were generated. They were represented in GIS attributes, scenario parameters and methodological limitations. Sensitivity testing then challenged two of the most influential assumptions: obstruction severity and traffic volume. This approach supported a more critical interpretation than presenting one baseline as a precise forecast.

### **1.13 Thesis structure**

Chapter 1 introduced the research context, problem, gap, question, objectives, scope and contribution. Chapter 2 reviews wastewater monitoring, LPWAN, Wi-Fi HaLow, 5G and fibre, edge computing, GIS-assisted planning, ns-3 simulation and OT security. Chapter 3 explains the quantitative design-based methodology, data preparation, simulation configuration, sensitivity scenarios, security modelling and Packet Tracer validation. Chapter 4 presents and critically discusses the GIS architecture, baseline performance, sensitivity results, security results and practical implementation. Chapter 5 answers the research question, summarises the contribution, identifies limitations and provides recommendations for deployment and future research.

*Table 1 Research objectives and evidence.*

| Objective | Purpose                                                                   | Evidence                                                           |
|-----------|---------------------------------------------------------------------------|--------------------------------------------------------------------|
| 1         | Construct a GIS representation of sensors, RTUs, gateways and the server. | QGIS node and link datasets, maps and counts.                      |
| 2         | Assign technologies according to operational role and site conditions.    | Technology allocation table and link distribution.                 |
| 3         | Measure communication performance.                                        | ns-3 PDR, delay, throughput and packet-loss results.               |
| 4         | Test robustness under changed assumptions.                                | Obstruction, traffic and combined sensitivity scenarios.           |
| 5         | Evaluate flat and segmented security modes.                               | Attack-flow reach, blocked packets and normal traffic performance. |
| 6         | Validate the logical network design.                                      | Packet Tracer VLAN, routing, redundancy and ACL tests.             |

## **2. Literature Review**

### **2.1 Review purpose and organisation**

The literature review examined the technical and methodological foundations required to design a communication architecture for a lagoon-based wastewater treatment plant. The review was organised around the main connected themes: smart wastewater monitoring, heterogeneous IoT communications, long-range low-power technologies, Wi-Fi HaLow, high-capacity backhaul, GIS-assisted modelling, network simulation and OT cybersecurity. The purpose was not to rank technologies in isolation. It was to identify the conditions under which each technology is useful, examine how previous studies have evaluated those conditions, and determine where an integrated approach was missing.

The review prioritised peer-reviewed journal and conference papers, standards, technical reports and official documentation. Seminal sources were retained where they defined IoT, edge computing, digital twins, LPWAN or industrial security concepts. More recent sources were used for technology implementation and current standards. The literature was evaluated by comparing assumptions, deployment context, performance metrics and methodological limitations. This was important because results from a small indoor IoT trial cannot be transferred directly to an open, multi-kilometre wastewater facility without considering topology and propagation.

### **2.2 Smart wastewater monitoring and digitalisation**

Wastewater treatment has traditionally relied on instrumentation and SCADA, but the growth of IoT technologies has expanded the number and location of possible measurements. IoT architectures connect physical devices, communications, data platforms and applications, allowing observations to be collected beyond conventional control-room boundaries (Atzori et al., 2010; Gubbi et al., 2013; Al-Fuqaha et al., 2015). In a wastewater context, distributed sensing can improve visibility of lagoon conditions, pipeline behaviour, pump status and environmental discharge. It may also support preventive maintenance and faster identification of abnormal conditions.

The engineering value of additional sensing depends on data quality and communication reliability. Adding devices without ensuring connectivity can create a larger but incomplete information system. This matters for digital twins, which depend on timely and consistent data to maintain a useful relationship between the physical asset and its digital representation. Tao

et al. (2019) described digital twins as an integration of physical entities, virtual models, services, data and connections. Fuller et al. (2020) similarly identified connectivity, data management and model integration as core enabling technologies. These frameworks show that a digital twin is not only a simulation. It requires an operational data path that can deliver measurements with known delay, reliability and context.

Wastewater processes also create different temporal requirements. Many environmental variables change gradually and can be observed using periodic low-rate sampling, while alarms, equipment trips and control events need faster reporting. Tchobanoglous et al. (2014) showed that wastewater treatment combines biological, chemical and hydraulic processes with different monitoring needs. This diversity supports an architecture that separates routine sensing from critical and high-bandwidth services.

The original project proposal emphasised intelligent edge processing as a way to reduce unnecessary traffic and support local decisions. Edge computing moves selected processing closer to data sources, which can reduce backhaul use and response time (Shi et al., 2016; Satyanarayanan, 2017). Boyes et al. (2018) noted that industrial IoT systems require integration between operational assets, communications and analytics while preserving safety and security constraints. In the present study, edge computing was treated conservatively. Gateways were modelled as aggregation and prioritisation points, but no trained machine-learning algorithm was claimed. This distinction was necessary because literature often presents advanced edge analytics conceptually without validating the complete operational chain in a specific site.

### **2.2.1 Smart water networks and monitoring architecture**

Smart-water research has often focused on distribution networks, leakage, metering and quality monitoring. The communication lessons remain relevant to treatment plants: measurements are geographically distributed, some assets are difficult to power, and central applications require dependable data. Lalle et al. (2021) examined LoRaWAN routing for smart water grids and showed the importance of network architecture in addition to sensor selection. Pagano et al. (2024) proposed a smart-water IoT framework for evaluating energy and data in large scenarios, demonstrating the value of simulation when device populations become too large for simple experiments.

A treatment plant differs from a water-distribution network because it has concentrated process areas, open lagoons, safety-related gas measurements and central operational control. It also generates a wider mixture of periodic, event-driven and high-bandwidth traffic. The present

research therefore used smart-water studies as a foundation but adapted the architecture to treatment-specific zones and aggregation.

### **2.3 Communication requirements and the case for a hybrid architecture**

Industrial communication design requires trade-offs between range, energy use, bandwidth, delay, reliability, cost and deployment complexity. Gungor et al. (2011) showed that no single communication technology satisfies every requirement in distributed infrastructure systems. Although their analysis focused on smart grids, the same principle applies to wastewater assets because both environments combine low-rate sensing, control traffic and high-capacity backhaul. Xu et al. (2014) and Boyes et al. (2018) also argued that industrial IoT systems are heterogeneous and must integrate devices with different capabilities and lifecycles.

A single-technology design creates practical compromises. Fibre provides high capacity and predictable performance, but extending fibre to every sensor can be expensive and disruptive. Cellular technologies provide wide-area connectivity but depend on spectrum, coverage, service or private infrastructure. LPWAN provides long range and low power but is not suitable for continuous video or large data volumes. Conventional Wi-Fi offers high throughput but can have limited range and greater power demand in large outdoor environments. Hybrid design therefore becomes a method of matching technology to operational role.

Hybridisation also affects resilience. If all data rely on one radio layer or one gateway, a local failure can isolate many devices. Multiple communication classes and aggregation points can create alternative design options, but they also increase management complexity. Palattella et al. (2016) described the 5G-era IoT as a multi-technology environment in which low-power access, cellular systems and edge services coexist. The benefit is flexibility, but interoperability and architecture become more important than the nominal performance of a single link.

The present research used functional separation. Low-rate lagoon sensors were assigned to LPWAN. Process and pipeline devices were assigned to Wi-Fi HaLow. RTU-to-gateway links used Wi-Fi HaLow backhaul. CCTV and high-bandwidth assets used 5G or fibre. Gateway-to-server links used fibre. This allocation was not intended to prescribe one vendor solution. It translated the comparative properties found in the literature into a site-level architecture that could be tested.

### **2.3.1 Technology selection as a multi-criteria problem**

Technology selection is a multi-criteria engineering problem rather than a simple speed comparison. A useful technology must satisfy the device's reporting interval, message size, range, power supply, availability target, maintenance capability and security role. Cost also appears in several forms: hardware, gateways, spectrum or service, civil works, operations, battery replacement and specialist support.

A decision based only on nominal range can be misleading. LPWAN range may be attractive, but duty-cycle and collision limits matter. A decision based only on throughput can also be misleading because a high-rate radio may consume unnecessary energy and require more infrastructure. Fibre has excellent performance, but trenching and route diversity can dominate cost. Private 5G can provide managed wireless service, but spectrum, core functions and vendor support increase complexity.

The literature tends to compare technologies using tables of general characteristics. These tables are useful for screening but weak for site design because ranges and data rates depend on configuration and environment. The present research converted general properties into operational roles and then tested the complete topology. This is a more defensible use of literature than selecting a technology directly from a generic comparison table.

## **2.4 LPWAN and LoRaWAN for low-rate field sensing**

LPWAN technologies were developed to connect large numbers of low-power devices over longer distances than conventional short-range wireless systems. Raza et al. (2017) described LPWAN as a family of technologies that exchange throughput and latency for coverage, energy efficiency and scalability. Centenaro et al. (2016) highlighted their suitability for smart-city and wide-area IoT applications where devices transmit small amounts of data. Mekki et al. (2019) compared LoRaWAN, Sigfox, NB-IoT and LTE-M, showing that technology selection depends on traffic pattern, mobility, quality of service and infrastructure ownership.

LoRaWAN is particularly relevant to remote sensing because it can operate in unlicensed spectrum and support star-of-stars architectures through gateways. In this thesis, LoRaWAN is considered a low-rate sensing technology that falls under low-power wide-area networking as opposed to a large-scale, plant-wide communication technology. The LoRa Alliance specification defines device classes, adaptive data rate and network behaviour, while Semtech



transceivers provide the physical-layer modulation commonly used by LoRa systems (LoRa Alliance, 2020; Semtech, 2020). The technology is attractive for lagoon sensors because small environmental readings can tolerate low data rates and longer reporting intervals.

However, LoRaWAN has clear limits. Adelantado et al. (2017) demonstrated that duty-cycle restrictions, collisions and downlink limitations constrain scalability and application behaviour. Blenn and Kuipers (2017) observed real-world variation in The Things Network, reinforcing that nominal range does not guarantee consistent delivery. Lalle et al. (2021) reviewed multi-hop approaches for smart water grids and noted that standard LoRaWAN is primarily designed around gateway-based star connectivity rather than general multi-hop routing. The results indicate that if LoRaWAN-based LPWAN links for high-frequency control, video transmission, or bunches of concurrent traffic without an analysis of the limits of capacity, duty-cycle, and traffic-load.

The literature therefore supports LPWAN for periodic lagoon, level and water-quality measurements, but not as a universal plant network. The present study retained this role separation. LPWAN links were expected to have high delivery for modest traffic but greater delay and lower throughput than Wi-Fi HaLow, 5G or fibre. The simulation results were interpreted against that functional expectation rather than judged against high-capacity technologies on identical criteria.

#### **2.4.1 LPWAN reliability, energy and scaling trade-offs**

Energy efficiency is a major reason to use LPWAN, but energy was not measured directly in the final simulation. Banotra et al. (2023) reviewed self-sustainable IoT design and showed that communication duty cycle, processing and energy harvesting interact across layers. This supports long reporting intervals and local filtering for remote sensors. It also shows why packet delivery alone is not enough to select a device: a configuration that retransmits frequently may increase reliability while reducing battery life.

LPWAN scaling is affected by traffic synchronisation and gateway density. A large number of devices can share a channel when messages are short and infrequent, but simultaneous alarm bursts may create collisions. The present deterministic traffic model did not fully represent this behaviour. Its LPWAN result should therefore be treated as a baseline for routine telemetry. Future alarm-burst and shared-channel experiments are needed before control or emergency services are assigned to the same tier.

## 2.5 Wi-Fi HaLow as a middle-range industrial IoT option

Wi-Fi HaLow, which is based on IEEE 802.11ah extends Wi-Fi operation into sub-1 GHz spectrum. The standard was developed for longer-range, lower-power and higher-device-count IoT applications compared with conventional Wi-Fi (IEEE, 2016). Austin and Ancillotti (2018) described 802.11ah as the Wi-Fi approach to IoT, combining familiar WLAN concepts with features intended for sensors and machine-type communication. Lee et al. (2021) demonstrated system-on-chip development and performance evaluation for long-range, low-power use, showing the practical maturity of Wi-Fi HaLow beyond a purely conceptual standard.

Wi-Fi HaLow occupies a useful middle position. It can provide more capacity and native IP-oriented behaviour than many LPWAN systems while maintaining longer range and lower-frequency propagation advantages compared with 2.4 or 5 GHz Wi-Fi. This is relevant to process and pipeline sensors that may generate more frequent telemetry, event messages or larger payloads than lagoon sensors. It is also relevant to RTU-to-gateway aggregation, where several field flows are combined.

The technology still has limitations. Propagation at sub-1 GHz can improve range, but performance remains dependent on channel width, modulation, transmit power, regulatory limits, interference and antenna placement. Commercial availability is growing but remains less universal than conventional Wi-Fi. In simulation, full IEEE 802.11ah behaviour is also more difficult to represent than a simplified link class. Tian et al. (2018) extended the ns-3 802.11ah module and illustrated the effort required to reproduce standard-specific behaviour. This indicates that studies using generic or point-to-point abstractions should be explicit about what is being compared.

The present model represented Wi-Fi HaLow as a technology class with delay, capacity and error assumptions informed by its intended role. It did not claim to reproduce every MAC mechanism of IEEE 802.11ah. This abstraction allowed the study to preserve a 423-node GIS topology and compare system-level performance, but it limited conclusions about contention, restricted access windows and detailed channel behaviour. The literature therefore both justified Wi-Fi HaLow selection and defined an important methodological limitation.

### 2.5.1 Wi-Fi HaLow standard features relevant to the case study

IEEE 802.11ah includes mechanisms intended for large IoT populations, such as hierarchical association identifiers, restricted access windows and narrower channel options (IEEE, 2016). These features can reduce contention and extend operation, but their benefit depends on device

configuration and traffic scheduling. The standard can support many stations, yet deployment density and access timing must be engineered.

For the case study, HaLow was attractive because process sensors were more concentrated than lagoon sensors and were often associated with powered assets or RTUs. This made moderate-rate communication and infrastructure-managed access plausible. The backhaul use was also conceptually different from the access use. RTUs aggregate several devices, so their links can justify larger antennas, mains power and higher capacity.

## **2.6 5G and fibre for high-bandwidth and backbone services**

High-bandwidth and time-sensitive plant services require a different communication tier. Andrews et al. (2014) described the technical direction of 5G in terms of densification, spectrum, multiple antennas and flexible architectures. 3GPP specifications define the New Radio system and channel models used for performance evaluation (3GPP, 2022a; 3GPP, 2022b). In an open treatment-plant environment, private 5G can provide managed wireless capacity for CCTV, mobile assets or locations where fibre is impractical.

5G does not remove the need for fibre. Fibre provides high capacity, low delay, electromagnetic immunity and a strong basis for fixed backbone links. It is especially suitable for connecting gateways, buildings and central servers where trenching or existing conduits make deployment feasible. The limitation is cost and physical installation. Damage to a single fibre route can also create a critical failure unless paths are redundant.

The combined use of wireless access and fibre backhaul is common in industrial and utility architectures because the technologies solve different problems. The present study therefore used a combined “5G or fibre” class for high-bandwidth field links and a dedicated fibre-backhaul class for gateway-to-server communication. This grouping reflected the available data and comparative objective, but it also combined two technologies with different physical behaviour. The results should therefore be read as performance targets for a high-capacity tier rather than as a protocol-level 5G certification.

Wang et al. (2021) compared NB-IoT, LTE Cat-M1, Sigfox and LoRa under mobility, showing that cellular and LPWAN families respond differently to speed, coverage and traffic. Their findings reinforce that cellular systems are useful where managed quality of service and

mobility matter, while low-power systems suit sparse telemetry. The hybrid architecture used these differences rather than forcing all devices into a cellular or unlicensed design.

### **2.6.1 Reliability, redundancy and backhaul design**

Backhaul reliability depends on architecture as well as medium. A fibre link may have very low packet error but still fail if a cable is cut or a switch loses power. A wireless link may have greater variation but provide route diversity where trenching is difficult. Resilient design therefore considers failure domains, redundant paths, power and recovery procedures.

The present ns-3 model focused on link performance and did not simulate physical cable cuts or gateway failure. Packet Tracer added partial redundancy at the core using HSRP, EtherChannel and dual uplinks. The gap between the two models is useful: performance simulation showed how links handled traffic, while the implementation exposed device-level single points of failure. A production architecture would need both communication-quality analysis and reliability engineering.

## **2.7 Edge processing and gateway aggregation**

Edge computing is often proposed as a solution to bandwidth, latency and autonomy challenges in IoT systems. Shi et al. (2016) defined edge computing as computation performed near data sources rather than exclusively in a remote cloud. Satyanarayanan (2017) linked edge computing to reduced latency and context-aware services. Wen et al. (2022) applied edge concepts to remote operation, maintenance and intelligent analysis, showing the relevance of local processing to industrial systems.

For wastewater monitoring, gateways can filter repetitive data, aggregate sensor messages, prioritise alarms and continue limited local operation during backhaul interruption. These functions may improve resilience and reduce central traffic. They also create responsibility: a gateway becomes a security boundary and potential single point of failure. It requires controlled management, logging, update mechanisms and physical protection.

The project proposal initially included data filtering, aggregation and local anomaly detection. The final study implemented the network architecture and treated gateways as aggregation points, but it did not train or validate an anomaly-detection model. This difference is important when comparing the work with edge-computing literature. The thesis demonstrates where edge services could operate and how traffic reaches them. It does not demonstrate the accuracy of an edge analytics algorithm. Future work can build on the communication model by adding

processing delay, data reduction ratios and priority scheduling based on measured edge workloads.

## **2.8 GIS-assisted network planning and propagation context**

Geographic information systems provide a structured way to connect infrastructure location with network design. QGIS supports spatial layers, coordinate systems, geometry processing and export to tabular formats (QGIS Development Team, 2024). For a treatment plant, GIS can represent lagoon boundaries, pipelines, land classes, gateways and node locations. This enables link distance and obstruction context to be derived from the actual layout rather than assumed independently.

Propagation depends on more than Euclidean distance. Vegetation, built areas, terrain and frequency influence path loss. ITU-R P.1411 provides models for short-range outdoor communication, while ITU-R P.833 addresses attenuation in vegetation (ITU-R, 2021a; ITU-R, 2021b). 3GPP TR 38.901 provides channel models for cellular evaluation across 0.5 to 100 GHz (3GPP, 2022b). These standards are not perfect representations of a wastewater site, but they provide recognised foundations for assigning loss and scenario assumptions.

The use of public spatial data also has limitations. Pipeline datasets can be incomplete, coordinates may require cleaning, and visual interpretation of satellite imagery does not reveal all radio obstructions. GIS-derived obstruction classes are therefore better treated as structured assumptions than as substitutes for a radio-frequency survey. The present study used GIS to improve spatial realism and to create sensitivity variables. It did not claim centimetre-level asset accuracy or measured propagation coefficients.

### **2.8.1 Spatial uncertainty and model calibration**

GIS can make a model appear precise because coordinates are displayed on detailed maps. That visual precision should not be confused with measurement accuracy. Public asset data may be generalised, satellite imagery may be dated, and inferred node positions may not match installed equipment. The methodology therefore recorded source and assumption fields in the CSV files. This improves auditability and allows individual links to be revised when better data become available.

Calibration would require field measurements of received signal strength, packet loss and interference at representative frequencies and heights. Those measurements could be used to adjust the GIS obstruction penalties and compare predicted with observed link quality. Without

calibration, the model is strongest for relative spatial reasoning and weakest for exact coverage boundaries.

## **2.9 Network simulation and performance measurement**

Network simulation allows controlled and repeatable evaluation without disrupting critical infrastructure. ns-3 is a discrete-event simulator designed for research and education, with modular protocol models, tracing and configurable topology (Riley & Henderson, 2010; ns-3 Project, 2024a; ns-3 Project, 2024b). (Campanile et al., 2020) reviewed ns-3 research and identified its broad use in communication studies, while also noting that model validity depends on configuration and interpretation.

FlowMonitor provides flow-level statistics such as transmitted packets, received packets, loss, delay and throughput. Carneiro et al. (2009) introduced the framework as a monitoring method for ns-3. These metrics are directly useful for comparing architecture classes, but they do not automatically prove real-world performance. A simulator executes the assumptions encoded by the researcher. Validation therefore requires checking parameter sources, comparing trends with literature, repeating scenarios and explaining simplifications.

The present work used ns-3 because it allowed the 423 GIS-derived nodes and 395 links to be represented in a reproducible experiment. Technology classes were implemented through configurable links rather than complete protocol stacks. This choice prioritised site-scale comparison over detailed MAC-layer behaviour. It was suitable for questions about relative delay, throughput, delivery and obstruction sensitivity, but it limits conclusions about spectrum contention, handover, adaptive data rate and device energy consumption.

Sensitivity analysis is valuable because a single baseline can appear precise while depending heavily on assumed error or traffic. The study therefore doubled obstruction-related error, tripled traffic and combined both changes. This approach did not produce a statistical confidence interval, but it tested whether the architecture's conclusions remained stable when key assumptions were stressed.

### **2.9.1 Methodological critique of simulation studies**

Simulation studies can create false confidence when parameter choices are not transparent. Results may appear quantitative and precise even though delay, error and capacity were selected by the researcher. Reproducibility therefore requires source code, raw outputs, parameter tables and clear statements about what was not modelled.

Another issue is scale. Detailed protocol simulation may be realistic for a small network but computationally difficult for hundreds of heterogeneous links. Simplified links can represent a large architecture but lose protocol detail. There is no universally correct level of abstraction. The choice should follow the research question.

For this thesis, the research question concerned role allocation, topology, sensitivity and segmentation across a large plant. A simplified architecture-level model was therefore justified. The limitation was managed by avoiding claims about exact LoRaWAN capacity, HaLow restricted-access behaviour or 5G scheduling. The literature review was used to validate relative trends, while Packet Tracer tested logical connectivity and policy.

## **2.10 OT cybersecurity, segmentation and controlled remote access**

Industrial systems face security risks that differ from conventional enterprise IT because process availability, safety and long equipment lifecycles are central. NIST SP 800-82 Revision 3 recommends defence in depth, segmentation, controlled remote access, monitoring and risk-based security for OT environments (Stouffer et al., 2023). IEC 62443-3-2 defines security risk assessment and system design, while IEC 62443-3-3 specifies system security requirements and security levels (IEC, 2020; IEC, 2013). A key concept is the division of assets into zones connected by controlled conduits.

Research on SCADA security has repeatedly identified flat networks, weak access control, legacy protocols and poor separation as major concerns. Nicholson et al. (2012) examined SCADA security in the context of cyber conflict. Knowles et al. (2015) reviewed cybersecurity management in industrial control systems and highlighted governance and lifecycle problems. Cherdantseva et al. (2016) reviewed risk-assessment methods for SCADA and found that no single method covers every operational and cyber dimension. Humayed et al. (2017) similarly emphasised the coupled physical and cyber consequences of attacks.

Segmentation does not guarantee security, but it reduces unnecessary reach and creates enforcement points. VLANs separate broadcast domains, while routed access-control lists and firewalls determine permitted traffic. IEEE 802.1Q defines VLAN bridging, and Cisco documentation provides practical mechanisms for HSRP, EtherChannel and access lists in the Packet Tracer implementation (IEEE, 2022; Cisco Systems, 2020a; Cisco Systems, 2020b; Cisco Systems, 2023). An OT DMZ and jump server provide a controlled location for vendor access rather than allowing direct connections to SCADA.

The present security experiment compared a flat scenario with a segmented policy. In the segmented case, unauthorised flows from field zones to SCADA were not installed. This was an idealised representation of policy enforcement, not an exploit simulation. Packet Tracer then provided a device-level demonstration in which authorised pings succeeded and prohibited paths failed. The two methods complemented each other: ns-3 quantified attack reach at scale, while Packet Tracer demonstrated a concrete logical configuration.

### **2.10.1 Defence in depth and availability**

Defence in depth is especially important in wastewater operations because a single control failure can affect environmental compliance and public service. CISA guidance and NIST OT guidance recommend layered controls rather than reliance on one firewall (CISA, 2016; Stouffer et al., 2023). Layers can include physical security, asset inventory, secure configuration, segmentation, authentication, monitoring, backups and incident response.

Availability complicates security decisions. An aggressive rule can interrupt required process communication, while a permissive rule can expose SCADA. The correct policy must therefore be derived from operational flows. The Packet Tracer test matrix used this principle: the RTU and CCTV had defined SCADA paths, management had broader approved access, and vendor access terminated in the DMZ. Sensors were not allowed to bypass aggregation and communicate directly with SCADA.

### **2.10.2 VLANs, firewalls and the limits of segmentation**

VLAN separation is sometimes described as a security control by itself, but VLANs primarily organise Layer 2 traffic. Security depends on the routing boundary and the rules applied there. An incorrectly configured multilayer switch can route between VLANs without restriction, and an overly broad firewall permit can defeat the intended zone model. Segmentation must therefore include address design, route control, ACLs, firewall policy, management restrictions and verification.

Firewalls are also not a complete solution. Allowed protocols can still be exploited, credentials can be compromised, and devices within one zone can attack each other. Defence in depth requires endpoint hardening, secure remote access, monitoring and recovery. The present Packet Tracer model demonstrated only the network-policy layer. Its value was to prove the intended traffic matrix and expose return-path requirements, not to reproduce every security control.



## 2.11 Synthesis and research gap

The literature strongly supports the individual components of the proposed design. LPWAN is appropriate for sparse low-rate telemetry but has capacity and duty-cycle limits. Wi-Fi HaLow provides a useful sub-1 GHz middle tier, but detailed performance depends on implementation and channel conditions. 5G and fibre suit higher-capacity services, although their cost and deployment models differ. Edge gateways can aggregate and prioritise data, but they become critical assets. GIS improves spatial context, while ns-3 supports reproducible experimentation. OT standards support segmentation, zones, conduits and controlled vendor access.

The unresolved issue is how these components perform when combined in one large, geographically grounded lagoon-based architecture. Existing work does not provide a complete answer because technology papers commonly isolate one protocol, GIS studies may not include operational security, and cybersecurity studies often assume a network topology rather than deriving it from infrastructure. The present study addressed this gap by creating a single workflow from GIS to performance simulation, sensitivity analysis, security modelling and Packet Tracer validation.

This synthesis also shaped the research claims. The work does not claim that one technology is universally superior or that simulation proves field performance. It argues that a role-based hybrid architecture is more defensible than a uniform design, and that segmentation can reduce unauthorised reach without materially reducing authorised traffic in the model. These propositions were tested in the methodology described in Chapter 3.

### 2.11.1 Conceptual framework derived from the review

The literature synthesis produced a five-layer conceptual framework.

- The first layer was the physical and spatial environment: lagoons, pipelines, vegetation, distance and asset location.
- The second layer was device function: sensing, control, aggregation, surveillance and central processing.
- The third layer was communication technology, selected according to the first two layers.
- The fourth layer was performance, measured using delivery, delay, throughput and loss.
- The fifth layer was security, which restricted communication according to operational need.

This framework guided the thesis structure. GIS represented the first layer. Node types represented the second. Link classes represented the third. ns-3 metrics represented the fourth. The flat-versus-segmented model and Packet Tracer represented the fifth. The contribution was the explicit connection between the layers.

## **3 Research Methodology**

### **3.1 Chapter overview**

This chapter describes the design and evaluation processes for the communication and operational technology (OT) security architecture at lagoon-based wastewater treatment plants and the related materials and methods used. There are two main sections to this chapter. The first section explains the study materials and includes the case study environment, QGIS spatial data, node and link data, the ns-3 simulation environment, and the Packet Tracer validation tool. The second section describes the methods for the GIS model, the assignment of communication technologies, the baseline and sensitivity simulations, security segmentation modelling, and the validation of the architecture with Packet Tracer.

The overall workflow of the methodology is presented in Figure 1. It illustrates the progression of the study from the literature review and GIS modelling to the simulations, security modelling, validation with Packet Tracer, and the closing interpretations.

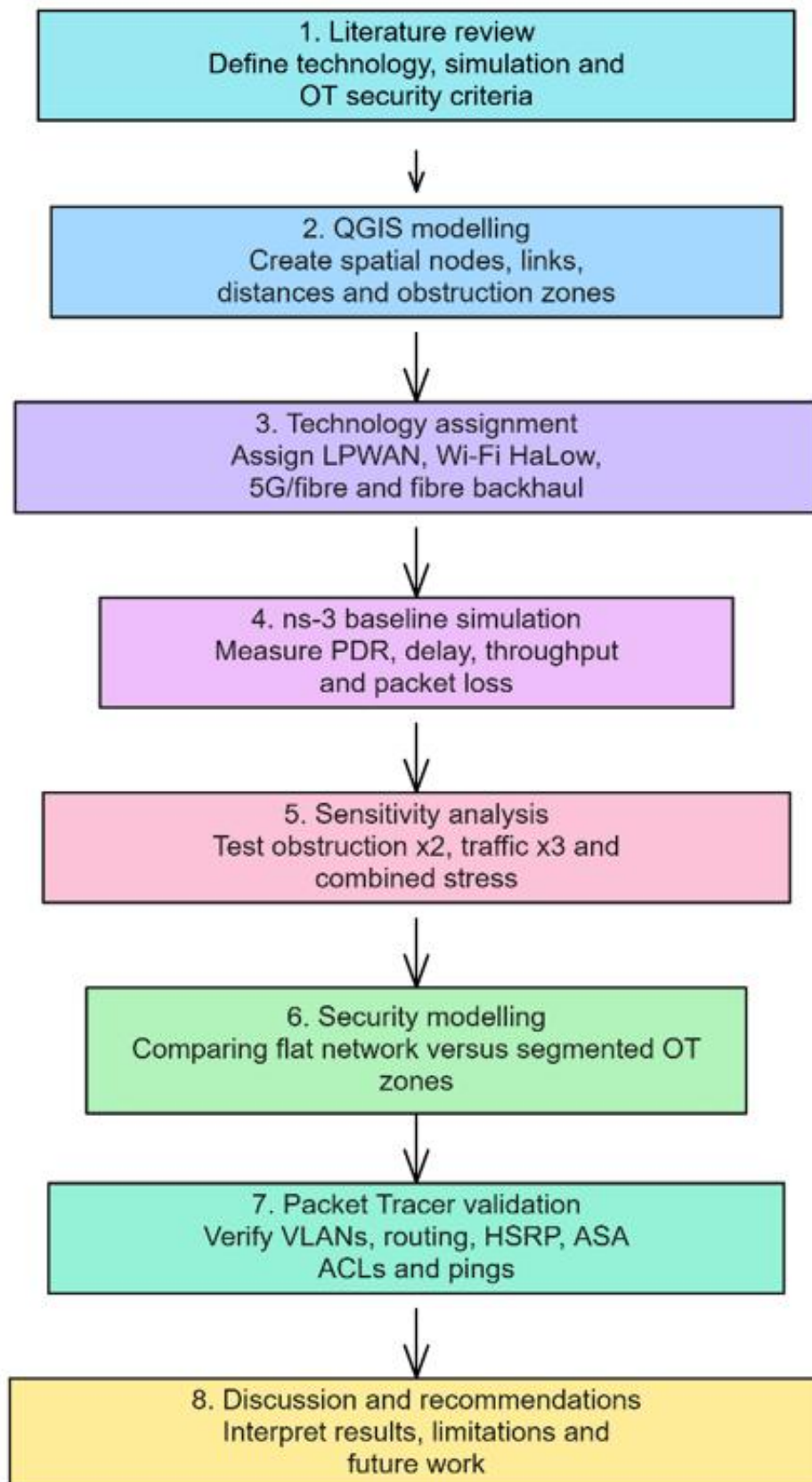


Figure 1: Research workflow used in the study (source: author's work).

## 3.2 Materials

### 3.2.1 Case-study environment

The lagoon-based wastewater treatment plant is the chosen case study for this research. These plants have a geographically dispersed collection of assets, which include lagoons, sample and control locations, process areas, control units, and central operational systems. These dispersed and varied (in function) assets change the communication design in contrast to the more compact and contained designs of indoor industrial networks. Factors such as distance, terrain, and vegetation impact the design and selection of communication technology.

The case study environment illustrates the specific example communication problem associated with wastewater treatment that includes low-rate sensors, process control, CCTV, and other high bandwidth devices, gateways, and central SCADA systems. These components require different communication networks. This example illustrates the need for a communication architecture using multiple technologies.

### 3.2.2 QGIS spatial data

Spatial models of the treatment plant environment were created in QGIS. The QGIS model helped in the placement of communication nodes and the drawing of links in the field between devices, aggregation points, gateways, and central systems. The spatial model was necessary because communication in a large, outdoor treatment plant is influenced by distance, terrain, and obstructions.

QGIS data included mapped nodes and links and grouped technologies. The model helped the communication design to be approached as a spatial problem and not a logical network design problem. Figure 2 shows the QGIS site model used to illustrate the communication nodes and the obstruction design environment.

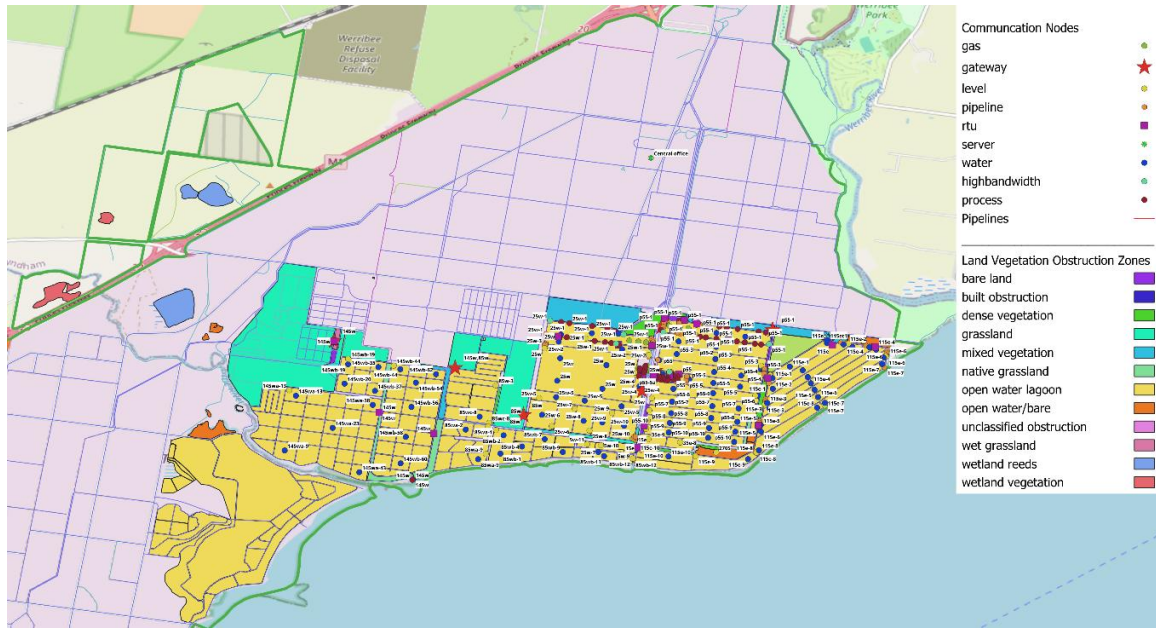


Figure 2: QGIS site model showing the GIS-assisted node and obstruction environment (source: author's work).

### 3.2.3 Node and link datasets

From the QGIS model, datasets for the nodes and links needed for the simulation were generated. The node datasets included the groups of devices considered in the study, such as lagoon assets, process sensors, RTUs, CCTV or other high bandwidth devices, gateways, and central systems. The link datasets included the connections between nodes, the type of technology, the distance, and the role of the link in the communication architecture.

The final model contained 423 nodes and 395 communication links. The communication links were mapped to several communication technologies: LPWAN, Wi-Fi HaLow, Wi-Fi HaLow backhaul, 5G/fibre, and fibre backhaul. These datasets were the primary input to the ns-3 simulation model. The node attribute table is shown in Figure 3, and the communication-link attribute table is shown in Figure 4.

Nodes old — Features Total: 423, Filtered: 423, Selected: 0

|    | fid | type     | name   | height_m | height_ref | height_note                                                                                     |
|----|-----|----------|--------|----------|------------|-------------------------------------------------------------------------------------------------|
| 1  | 1   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 2  | 2   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 3  | 3   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 4  | 4   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 5  | 5   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 6  | 6   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 7  | 7   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 8  | 8   | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 9  | 9   | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 10 | 10  | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 11 | 11  | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 12 | 12  | level    | p55-5a |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 13 | 13  | level    | p55-5a |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 14 | 14  | level    | p55-5  |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 15 | 15  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 16 | 16  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 17 | 17  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 18 | 18  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 19 | 19  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 20 | 20  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 21 | 21  | pipeline | p55-5a |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |

Show All Features

Figure 3: QGIS node attribute table showing 423 node features with node identifiers, device type, name and antenna-height assumption fields.

Figure 3 confirms that the node layer included fields for node identifiers, device type, device name and antenna-height assumption. This was necessary since the same node data were subsequently exported from QGIS and used to build the ns-3 simulation.

auto\_connections\_tech copy — Features Total: 395, Filtered: 395, Selected: 395

|    | from_id | to_id | from_type | to_type | technology  | distance_m        | final_error | fid | freq_mhz | snr_db            | tx_power | noise_fig | bandwidth_hz | noise_floor       | path_loss         | total_loss        |
|----|---------|-------|-----------|---------|-------------|-------------------|-------------|-----|----------|-------------------|----------|-----------|--------------|-------------------|-------------------|-------------------|
| 1  | 1       | 38    | gas       | rtu     | Wi-Fi HaLow | 67.805624761625   | 0.0035      | 1   | 915      | 56.70626388038... | 20       | 8         | 1000000      | -106              | 68.29373631961... | 69.29373631961... |
| 2  | 2       | 38    | gas       | rtu     | Wi-Fi HaLow | 20.17191599736... | 0.0035      | 2   | 915      | 67.23663509981... | 20       | 8         | 1000000      | -106              | 57.76336490018... | 58.76336490018... |
| 3  | 3       | 330   | gas       | rtu     | Wi-Fi HaLow | 65.22516845003... | 0.0035      | 3   | 915      | 57.04327383108... | 20       | 8         | 1000000      | -106              | 67.956726068917   | 68.956726068917   |
| 4  | 4       | 38    | gas       | rtu     | Wi-Fi HaLow | 80.62509271068... | 0.0035      | 4   | 915      | 55.20217357790... | 20       | 8         | 1000000      | -106              | 69.79782642209... | 70.79782642209... |
| 5  | 5       | 330   | gas       | rtu     | Wi-Fi HaLow | 80.566023537478   | 0.0035      | 5   | 915      | 55.20853934511... | 20       | 8         | 1000000      | -106              | 69.7914605488...  | 70.7914605488...  |
| 6  | 6       | 346   | gas       | rtu     | Wi-Fi HaLow | 75.28671209871... | 0.0035      | 6   | 915      | 55.7972114954388  | 20       | 8         | 1000000      | -106              | 69.2027885045612  | 70.2027885045612  |
| 7  | 7       | 330   | gas       | rtu     | Wi-Fi HaLow | 105.6944845225... | 0.0035      | 7   | 915      | 52.85053161829... | 20       | 8         | 1000000      | -106              | 72.14946838170... | 73.14946838170... |
| 8  | 8       | 330   | water     | rtu     | LPWAN       | 103.2310335333... | 0.0105      | 8   | 915      | 58.08627247786... | 14       | 6         | 125000       | -117.030898869... | 71.9446273920588  | 72.9446273920588  |
| 9  | 10      | 330   | water     | rtu     | LPWAN       | 117.1333312266... | 0.0105      | 9   | 915      | 56.9886809601...  | 14       | 6         | 125000       | -117.030898869... | 73.0420317739025  | 74.0420317739025  |
| 10 | 12      | 330   | level     | rtu     | LPWAN       | 61.76407539028... | 0.0105      | 10  | 915      | 62.54775910048... | 14       | 6         | 125000       | -117.030898869... | 67.4831407694362  | 68.4831407694362  |
| 11 | 13      | 330   | level     | rtu     | LPWAN       | 18.33409990213... | 0.0105      | 11  | 915      | 73.09728611894... | 14       | 6         | 125000       | -117.030898869... | 56.93361375097... | 57.93361375097... |
| 12 | 14      | 400   | level     | rtu     | LPWAN       | 222.401353732283  | 0.02        | 12  | 915      | 42.41972946018... | 14       | 6         | 125000       | -117.030898869... | 78.6111704097336  | 88.6111704097336  |
| 13 | 15      | 38    | water     | rtu     | LPWAN       | 169.2723348425... | 0.0105      | 13  | 915      | 53.79075829563... | 14       | 6         | 125000       | -117.030898869... | 76.24014157428... | 77.24014157428... |
| 14 | 16      | 38    | water     | rtu     | LPWAN       | 502.8698159120... | 0.0105      | 14  | 915      | 44.33336661943... | 14       | 6         | 125000       | -117.030898869... | 85.69753325048... | 86.69753325048... |
| 15 | 17      | 400   | water     | rtu     | LPWAN       | 414.904760665563  | 0.02        | 15  | 915      | 37.00350962852... | 14       | 6         | 125000       | -117.030898869... | 84.02739024138... | 94.02739024138... |
| 16 | 18      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 172.1562386921... | 0.0035      | 16  | 915      | 48.6131228081222  | 20       | 8         | 1000000      | -106              | 76.3868771918778  | 77.3868771918778  |
| 17 | 19      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 139.1797880066... | 0.0035      | 17  | 915      | 50.46005408141... | 20       | 8         | 1000000      | -106              | 74.53994591858... | 75.53994591858... |
| 18 | 20      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 115.5264242410... | 0.0035      | 18  | 915      | 52.07795149219... | 20       | 8         | 1000000      | -106              | 72.92204850780... | 73.92204850780... |
| 19 | 21      | 330   | pipeline  | rtu     | Wi-Fi HaLow | 46.10629708675... | 0.0035      | 19  | 915      | 60.05637323202... | 20       | 8         | 1000000      | -106              | 64.94362676797... | 65.94362676797... |
| 20 | 22      | 330   | pipeline  | rtu     | Wi-Fi HaLow | 43.56721347635... | 0.0035      | 20  | 915      | 60.54838244464... | 20       | 8         | 1000000      | -106              | 64.4516175553512  | 65.4516175553512  |

Show All Features

Figure 4: QGIS link attribute table showing 395 generated communication links with endpoint IDs, device types, technology classes, distance and link-performance fields.

Figure 4 confirms that the communication-link layer included endpoint IDs, device types, technology classes, distance and link-performance fields. These attributes were necessary since they linked the spatial QGIS model to the inputs for the simulation used for baseline, sensitivity and security analysis.

### **3.2.4 ns-3 simulation tool**

The network simulator 3 (ns-3) environment was used to assess the performance of the developed communication models. It was beneficial to use ns-3 because it can model the delivery of packets, packet loss, delays and throughput, all under controlled assumptions. Simulation was the most suitable approach for this research since deploying an actual treatment plant was beyond the limits of this thesis.

The ns-3 model used QGIS node and link structures to analyse the proposed hybrid communication design. The simulation did not attempt to model the full range of physical radio phenomenon of a real-world implementation. Rather, the simulation provided a means of comparing how different classes of technology responded to baseline and stress testing under controlled conditions.

### **3.2.5 Packet Tracer validation tool**

Cisco Packet Tracer was used to validate the logical design of OT network segmentation. Packet Tracer was able to test and show the accessibility of reachability, filtering and routing at the network layer, firewalls, control lists, and Layer 2 and 3 segmentations in a network environment.

The Packet Tracer model consisted of a number of sensor and control switches, a gateway aggregation switch, a WTP gateway, an ASA firewall, a SCADA server, and a DMZ jump server. The focus of the Packet Tracer validation was not to model all the communication protocols of the wastewater plant. The goal of the validation was to test if the segmented structure would permit communication to SCADA and DMZ systems and block communication for all other traffic.

## **3.3 Methods**

### **3.3.1 Research workflow**

The research followed a methodical, multi-step process. The first step was to review the literature to establish the needs and gaps for communication, simulation, and OT security. The second step was building the spatial node and link model in QGIS. The third step was to assign communication technologies to the links depending on the device role, distance, and expected traffic. The fourth step was to use ns-3 for simulations to establish baseline performance. The fifth step was to change assumptions of obstructions and traffic to test various scenarios. The



sixth step was to compare flat and segmented network designs. The last step was to use Packet Tracer to establish and test VLAN zoning, routing, and access control lists/firewall services.

Figure 1 shows the research workflow. As shown, the research progressed from a literature review, and GIS modelling, to simulations, security modelling, and validation via Packet Tracer, culminating in the final interpretation.

### 3.3.2 QGIS modelling process

The first step in the QGIS modeling process was to locate the primary communication nodes given the case study environment. Each node's role was defined based on its purpose in the treatment plant. Lagoon sensors and process sensors were modeled as low-rate measuring devices. RTUs and control devices were modeled as local control and aggregation devices. CCTV and other high-demand bandwidth devices were used as surrogates for high data devices. Gateways were modeled as aggregation and backhaul devices, while a central server was used to represent SCADA and operational systems.

Based on the designed communication hierarchy, links were established among the devices. The model did not treat all field devices as direct clients of the centralized server. It characterized access, aggregation and backhaul layers. Figure 5 illustrates an example for lagoon monitoring of the recommended node placement, while Figure 6 illustrates the fields of link-quality dedicated to obstruction-based modelling.

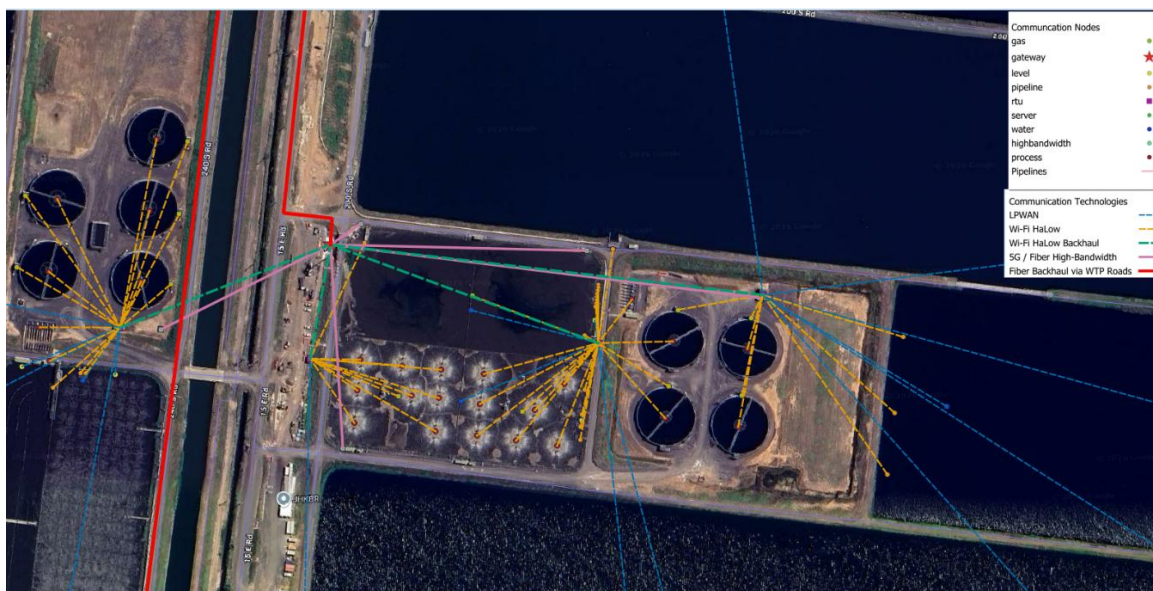


Figure 5: Recommended node placement example for lagoon monitoring (source: author's work).



auto\_connections\_tech copy — Features Total: 395, Filtered: 395, Selected: 395

| id | zwer | noise_fg | bandwidth_hz | noise_floor       | path_loss         | total_loss        | rx_power          | obs_cross | obs_density | veg_loss_db | veg_error | obs_ref            | obs_note            | from_name | to_name | endpoint_check |
|----|------|----------|--------------|-------------------|-------------------|-------------------|-------------------|-----------|-------------|-------------|-----------|--------------------|---------------------|-----------|---------|----------------|
| 1  | 20   | 8        | 1000000      | -106              | 68.29373631961... | 69.29373631961... | -49.2937363196... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55       | p55     | OK             |
| 2  | 20   | 8        | 1000000      | -106              | 57.76336490018... | 58.76336490018... | -38.7633649001... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55       | p55     | OK             |
| 3  | 20   | 8        | 1000000      | -106              | 67.956726068917   | 68.956726068917   | -48.9567260689... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55       | p55-5a  | OK             |
| 4  | 20   | 8        | 1000000      | -106              | 69.79782642209... | 70.79782642209... | -50.7978264220... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55       | p55     | OK             |
| 5  | 20   | 8        | 1000000      | -106              | 69.79146065408... | 70.79146065408... | -50.7914606540... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 6  | 20   | 8        | 1000000      | -106              | 69.2027885045612  | 70.2027885045612  | -50.2027885045... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 7  | 20   | 8        | 1000000      | -106              | 72.14946838170... | 73.14946838170... | -53.1494683817... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 8  | 14   | 6        | 125000       | -117.030899869... | 71.9446273920588  | 72.9446273920588  | -58.9446273920... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 9  | 14   | 6        | 125000       | -117.030899869... | 73.0420317739025  | 74.0420317739025  | -60.0420317739... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 10 | 14   | 6        | 125000       | -117.030899869... | 67.4831407694362  | 68.4831407694362  | -54.4831407694... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 11 | 14   | 6        | 125000       | -117.030899869... | 56.93361375097... | 57.93361375097... | -43.9336137509... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 12 | 14   | 6        | 125000       | -117.030899869... | 78.6111704097336  | 88.6111704097336  | -74.6111704097... | yes       | high        | 10          | 0.01      | ITU-R P.833; SA... | Dense trees or s... | p55-5     | 115e    | OK             |
| 13 | 14   | 6        | 125000       | -117.030899869... | 76.24014157428... | 77.24014157428... | -63.2401415742... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5     | p55     | OK             |
| 14 | 14   | 6        | 125000       | -117.030899869... | 85.69753325048... | 86.69753325048... | -72.6975332504... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5     | p55     | OK             |
| 15 | 14   | 6        | 125000       | -117.030899869... | 84.02739024138... | 94.02739024138... | -80.0273902413... | yes       | high        | 10          | 0.01      | ITU-R P.833; SA... | Dense trees or s... | p55-5     | 115e    | OK             |
| 16 | 20   | 8        | 1000000      | -106              | 76.3868771918778  | 77.3868771918778  | -57.3868771918... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5     | p55     | OK             |
| 17 | 20   | 8        | 1000000      | -106              | 74.53994591858... | 75.53994591858... | -55.5399459185... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5     | p55     | OK             |
| 18 | 20   | 8        | 1000000      | -106              | 72.92204850780... | 73.92204850780... | -53.9220485078... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5     | p55     | OK             |
| 19 | 20   | 8        | 1000000      | -106              | 64.94362676797... | 65.94362676797... | -45.9436267679... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |
| 20 | 20   | 8        | 1000000      | -106              | 64.4516175553512  | 65.4516175553512  | -45.4516175553... | yes       | low         | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... | p55-5a    | p55-5a  | OK             |

Show All Features

Figure 6: QGIS link attribute table showing obstruction crossing, obstruction density, vegetation loss, vegetation error and endpoint-check fields used for link-quality modelling.

### 3.3.3 Technology assignment

Communication technologies were categorized based on the role of each segment. Low-power wide-area network (LPWAN) technologies were employed for long-range, low-rate sensing links. Wi-Fi HaLow was used for medium-range, higher-capacity links. Wi-Fi HaLow backhaul was used for some aggregation links. 5G, along with fibre and fibre backhaul, were used for the higher-capacity links between gateways and the central systems.

This assignment of technologies was based on the primary design principle of the study, which was that no single communication technology was adequate for the entire treatment plant. Instead, each technology was employed in areas that were best suited for its primary characteristics. The communication link map in Figure 7 was used to verify the spatial distribution and the hybrid design of the technologies.

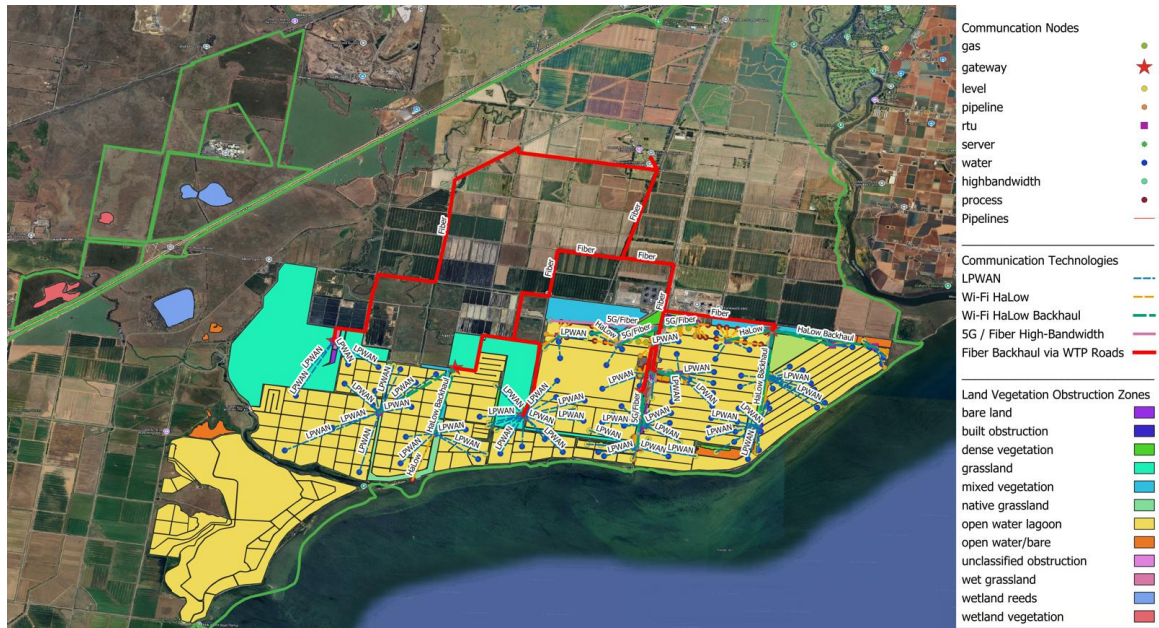


Figure 7: Technology-coloured QGIS communication-link map used during methodology to show the hybrid technology layout and link distribution.

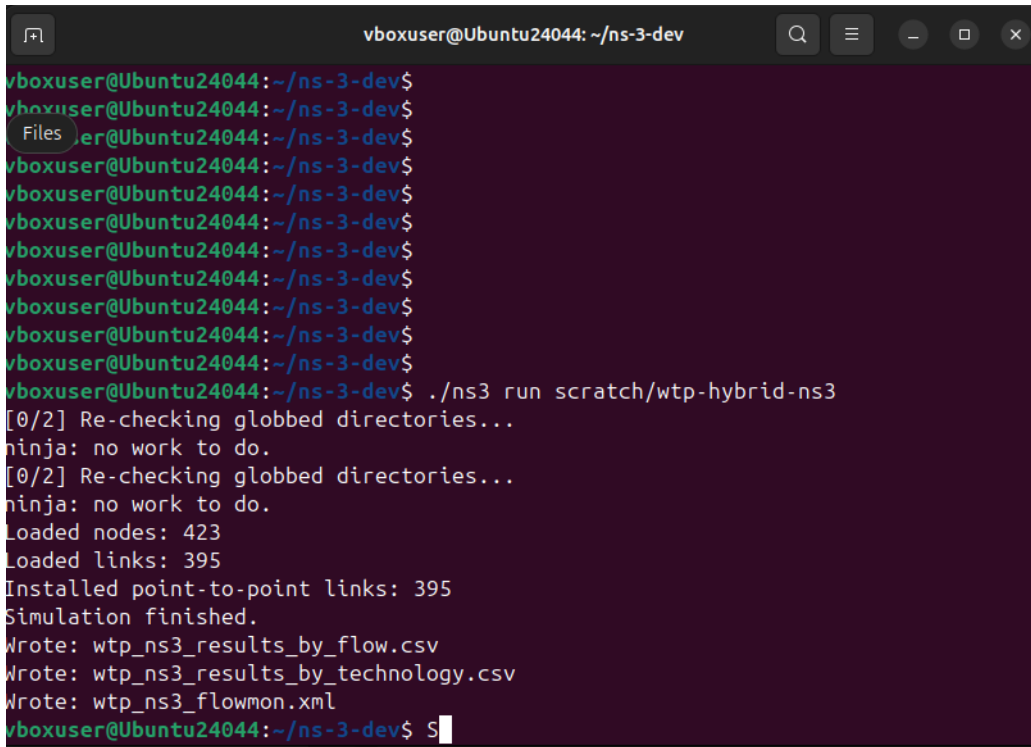
### 3.3.4 Baseline ns-3 simulation

The purpose of the baseline ns-3 simulation was to analyze the performance of the proposed hybrid communication architecture within a defined set of parameters. Four key performance indicators were the focus of the simulation:

- Packet Delivery Ratio (PDR): the proportion of packets sent and later received.
- Average Delay: the average time taken for packets to traverse the link from the source to the destination.
- Throughput: the measure of data received in a defined time period.
- Lost Packets: the measure of packets sent but not received.

These indicators define the key communication requirements for a wastewater monitoring and control system.

Packet loss and PDR indicate communication reliability. Delay indicates communication responsiveness. Throughput indicates whether the links are sufficient. The output files shown in Figure 8 were produced by ns-3 for the baseline, sensitivity and security analyses.

A terminal window titled 'vboxuser@Ubuntu24044: ~/ns-3-dev' with standard window controls. The terminal shows a series of shell commands and their outputs. The commands are mostly repetitions of 'vboxuser@Ubuntu24044: ~/ns-3-dev\$'. The final command is './ns3 run scratch/wtp-hybrid-ns3', which produces several lines of output: '[0/2] Re-checking globbed directories...', 'ninja: no work to do.', '[0/2] Re-checking globbed directories...', 'ninja: no work to do.', 'Loaded nodes: 423', 'Loaded links: 395', 'Installed point-to-point links: 395', 'Simulation finished.', 'Wrote: wtp\_ns3\_results\_by\_flow.csv', 'Wrote: wtp\_ns3\_results\_by\_technology.csv', 'Wrote: wtp\_ns3\_flowmon.xml', and finally 'vboxuser@Ubuntu24044: ~/ns-3-dev\$ S'.

```
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
Files vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$
vboxuser@Ubuntu24044: ~/ns-3-dev$ ./ns3 run scratch/wtp-hybrid-ns3
[0/2] Re-checking globbed directories...
ninja: no work to do.
[0/2] Re-checking globbed directories...
ninja: no work to do.
Loaded nodes: 423
Loaded links: 395
Installed point-to-point links: 395
Simulation finished.
Wrote: wtp_ns3_results_by_flow.csv
Wrote: wtp_ns3_results_by_technology.csv
Wrote: wtp_ns3_flowmon.xml
vboxuser@Ubuntu24044: ~/ns-3-dev$ S
```

Figure 8: ns-3 output-folder evidence showing the generated raw CSV and FlowMonitor XML result files used for baseline, sensitivity and security analysis.

### 3.3.5 Sensitivity scenarios

Testing the architecture aimed to understand how the sensitivities affect the behavior of the system. The sensitivity scenarios consisted of:

- Baseline: normal obstruction and normal traffic.
- Obstruction x2: loss due to obstruction was increased to illustrate worse-case physical conditions.
- Traffic x3: the demand of traffic was increased to illustrate greater demand for communication.
- Combined x2 and x3: obstruction loss and traffic demand were both increased.

The need to incorporate these scenarios is justified, since outdoor wastewater treatment plants are known to vary in link quality and traffic. The purpose of the sensitivity analysis was to test if the architecture would be unstable under more aggressive conditions. The obstruction and vegetation zone layers are shown in Figure 9 and were used to support the obstruction-related sensitivity assumptions.

wdp\_project\_layers — land\_vegetation\_obstruction\_zones\_editable — Features Total: 250, Filtered: 250, Selected: 250

|    | id    | globalid         | veg_type          | density    | obstruction | loss_db | error_add | ref_code           | method_note         | obs_cross | obs_type            | obs_density | obs_level  | veg_loss_db | veg_error | obs_ref            | obs_note            |
|----|-------|------------------|-------------------|------------|-------------|---------|-----------|--------------------|---------------------|-----------|---------------------|-------------|------------|-------------|-----------|--------------------|---------------------|
| 1  | 71369 | 3C48C0F9-88C...  | open_water_or...  | low        | low         | 1       | 0.0005    | Open water/bar...  | Borrow pit or ex... | yes       | Borrow pit          | low         | low        | 1           | 0.0005    | Open water/bar...  | Borrow pit or ex... |
| 2  | 69764 | 3D4875F7-8B5...  | open_water_or...  | low        | low         | 1       | 0.0005    | Open water/bar...  | Borrow pit or ex... | yes       | Borrow pit          | low         | low        | 1           | 0.0005    | Open water/bar...  | Borrow pit or ex... |
| 3  | 40778 | 1C057E1D3-ECF... | open_water_or...  | low        | low         | 1       | 0.0005    | Open water/bar...  | Borrow pit or ex... | yes       | NULL                | high        | high       | 10          | 0.01      | ITU-R P.833; SA... | Dense trees or s... |
| 4  | 56672 | 92EF83F8-977E... | open_water_or...  | low        | low         | 1       | 0.0005    | Open water/bar...  | Borrow pit or ex... | yes       | Borrow pit          | low         | low        | 1           | 0.0005    | Open water/bar...  | Borrow pit or ex... |
| 5  | 56949 | DD55A681-8F7...  | open_water_or...  | low        | low         | 1       | 0.0005    | Open water/bar...  | Borrow pit or ex... | yes       | Borrow pit          | low         | low        | 1           | 0.0005    | Open water/bar...  | Borrow pit or ex... |
| 6  | 39718 | 420F8AF9-7E7...  | wetland_reeds     | medium     | medium      | 6       | 0.005     | ITU-R P.833; we... | Temporary fres...   | yes       | Temporary fres...   | medium      | medium     | 6           | 0.005     | ITU-R P.833; we... | Temporary fres...   |
| 7  | 73314 | 8C37B515-4E7...  | wetland_reeds     | medium     | medium      | 6       | 0.005     | ITU-R P.833; we... | Temporary fres...   | yes       | Temporary fres...   | medium      | medium     | 6           | 0.005     | ITU-R P.833; we... | Temporary fres...   |
| 8  | 59375 | 5A1431A7-6B3...  | wetland_reeds     | medium     | medium      | 6       | 0.005     | ITU-R P.833; we... | Temporary fres...   | yes       | Temporary fres...   | medium      | medium     | 6           | 0.005     | ITU-R P.833; we... | Temporary fres...   |
| 9  | 37139 | 36C3F44F-FA4...  | wet_grassland     | low_medium | low_medium  | 4       | 0.003     | ITU-R P.833; we... | Paddock wetla...    | yes       | Paddock wetland     | low_medium  | low_medium | 4           | 0.003     | ITU-R P.833; we... | Paddock wetla...    |
| 10 | 27051 | 11FA13B4-C5B...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Experimental p...   | yes       | Experimental p...   | low         | low        | 1           | 0.0005    | Open water/lag...  | Experimental p...   |
| 11 | 31076 | 08EB8642-B41...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Experimental p...   | yes       | Experimental p...   | low         | low        | 1           | 0.0005    | Open water/lag...  | Experimental p...   |
| 12 | 88187 | F6192688-6E4F... | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 13 | 99832 | 22926011-83A...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 14 | 00951 | 27B3EE72-5BD...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 15 | 10212 | E69FF850-1859... | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 16 | 41505 | 08F5C3E0-35E...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 17 | 88784 | 5A782E7F-E32...  | open_water_or...  | low        | low         | 1       | 0.0005    | Open water GIS...  | Farm pond or d...   | yes       | Farm pond           | low         | low        | 1           | 0.0005    | Open water GIS...  | Farm pond or d...   |
| 18 | 34923 | 25B6878E-460...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 19 | 11934 | 6E674521-C3E...  | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |
| 20 | 64933 | 5D93A9D6-AB...   | open_water_lag... | low        | low         | 1       | 0.0005    | Open water/lag...  | Waste stabilisat... | yes       | Waste stabilisat... | low         | low        | 1           | 0.0005    | Open water/lag...  | Waste stabilisat... |

Show All Features

Figure 9: QGIS obstruction and vegetation zone layer used to assign additional loss and error assumptions to communication links.

### 3.3.6 Security modelling

The stage of security modelling included the comparison of a flat network design to a segmented OT network design. In the flat design, more devices were in direct or extended reachability to central systems. The segmented design included zone traffic with access-control rules.

This comparison primarily focused on the effects of segmentation on managing unauthorised reachability while still enabling authorised operational communication.

The two sets of variables were a result of the two interconnected experiments. The first one involved the Communication-Performance simulation and included the class of technology, obstruction multiplier, and traffic multiplier. The second one was the Security-Segmentation test, and here the main comparison was a flat network versus a segmented network. By separating the variables, the mixing of Communication-Performance and Security-Policy assumptions was avoided.

### 3.3.7 Packet Tracer validation

Packet Tracer validated the logical implementation of segmented OT architecture for the specified use case. The model included VLANs for lagoon and process sensors, RTUs, CCTV, gateways/backhaul, SCADA, DMZ, and Management and Vendor Access. Layer 3 switching and routing allowed communication between the authorised zones, while the ASA firewall was used to control access to SCADA and DMZ zones.

The following terms were used during the Packet Tracer validation:

*Table 2: Packet Tracer validation terms and addressing notation.*

| Term | Meaning                                         |
|------|-------------------------------------------------|
| VLAN | Virtual local area network                      |
| HSRP | Hot Standby Router Protocol                     |
| ASA  | Adaptive Security Appliance                     |
| ACL  | Access control list                             |
| DMZ  | Demilitarised zone                              |
| /24  | CIDR subnet prefix representing 255.255.255.0   |
| /30  | CIDR subnet prefix representing 255.255.255.252 |

The /24 and /30 designations are CIDR subnet prefixes used for IP addressing and are not specific to Packet Tracer.

The validation aimed to determine whether authorised traffic could reach the designated endpoints and whether unauthorised traffic would be denied.

Traffic types such as RTU and CCTV access to SCADA, management traffic access to SCADA and DMZ, and vendor traffic restricted to the DMZ. Lagoon and process sensors, as well as vendor devices, were denied access to the SCADA server. The segmented design was validated using the Packet Tracer topology in the following figure.

This figure 10 illustrates the use of Packet Tracer to validate SCADA access control, zoning, routing, and gateway-to-firewall lines. An equivalent screenshot was inserted in the methodology as an example showing the logical security-zoning design, which was converted into a Packet Tracer topology to validate the allowed and denied traffic, which is explained in the results section.

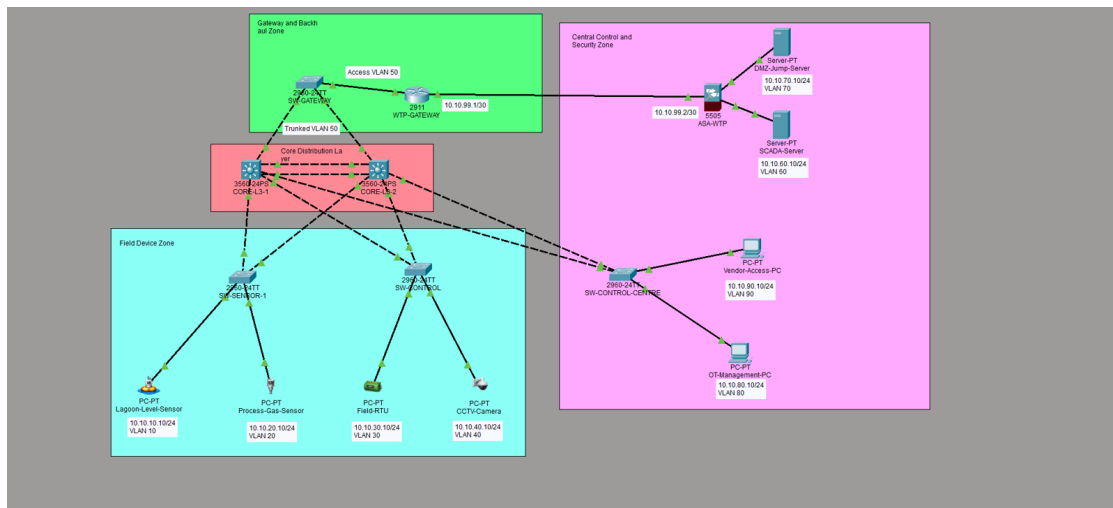


Figure 10: Packet Tracer logical topology used to verify VLAN zoning, routing, gateway-to-firewall transit and controlled SCADA access paths.

### 3.3.7.1 Configuration sequence used in Packet Tracer

The Packet Tracer network was configured in a staged sequence. The first step involved creating the device names and VLANs. This was followed by access port and trunk configuration. Core SVIs, HSRP, Rapid-PVST+, and EtherChannel were configured next. Static routing was implemented between the Core and WTP Gateway. The direct /30 Gateway-to-ASA was brought up next. Following this, ASA VLAN Interfaces, server IPs, and return routes were configured, prior to implementing any restrictive access control lists.

An open baseline access control list was implemented to allow end-to-end troubleshooting. Once all approved paths were validated, the open policy was removed, and the restricted OT access control list was implemented. Testing was done after each major configuration to allow for the faults to be limited to a small number and a working configuration file to be recovered.

### 3.3.8 Validation, reliability and data integrity

Validation was performed at several levels. CSV node and link counts were checked before simulation. Flow-level results were aggregated independently by technology and scenario. Baseline trends were compared with theoretical expectations: fibre should have the lowest delay and highest reliability, LPWAN should have the lowest throughput and greatest delay, and Wi-Fi HaLow should occupy an intermediate role. Sensitivity scenarios checked whether the overall conclusion remained stable when error and traffic assumptions changed.

Packet Tracer validation was staged. Layer 2 connectivity, VLAN membership, trunks, HSRP and core routing were tested before the ASA security rules were restricted. The ASA-to-server paths were tested directly, followed by gateway-to-server and endpoint-to-server paths. A



baseline Packet Tracer file was saved before the final ACL was applied. Allowed and blocked pings were then repeated, and access-list hit counters were checked.

Several troubleshooting findings improved the reliability of the final design. The server default gateways were required for remote return traffic. The gateway needed explicit routes to VLANs 60 and 70, while the ASA needed routes back to internal VLANs. The final topology removed unnecessary direct paths and maintained a single controlled gateway-to-firewall transit link. These corrections were documented because they demonstrate that the final result was validated rather than assumed from configuration alone.

### **3.3.8.1 Packet Tracer addressing and traffic matrix**

The internal addressing scheme used one /24 network for each operational zone: 10.10.10.0/24 for lagoon sensors, 10.10.20.0/24 for process sensors, 10.10.30.0/24 for RTUs, 10.10.40.0/24 for CCTV, 10.10.50.0/24 for gateway aggregation, 10.10.60.0/24 for SCADA, 10.10.70.0/24 for the DMZ, 10.10.80.0/24 for management and 10.10.90.0/24 for vendor access. The core HSRP virtual addresses used .1 in the field and management VLANs.

The direct gateway-to-ASA transit used 10.10.99.0/30. This avoided adding an unnecessary transit switch in the final simplified design. The gateway held routes to SCADA and DMZ through the ASA. The ASA held return routes to all internal VLANs through the gateway. This symmetric routing was necessary for the final tests.

### **3.3.9 Methodological limitations**

The methodology had five main limitations. First, wireless technologies were represented as functional link classes rather than complete protocol stacks, so contention, adaptive rate control, retransmission behaviour and spectrum sharing were simplified. Second, GIS obstruction values were assumptions derived from land classes rather than site measurements. Third, node and gateway positions were plausible but not optimised mathematically. Fourth, fixed propagation delays and adequate queue capacity reduced the visibility of delay variation under stress. Fifth, Packet Tracer validated segmentation logic but did not reproduce all production OT protocols or industrial firewall behaviour.

These limitations affect absolute predictions but do not remove the comparative value of the study. The same topology, metric definitions and scenario logic were applied consistently. The results are therefore most reliable as evidence about relative technology roles, sensitivity and segmentation, not as a final bill of materials or guaranteed service-level agreement.

Table 3: Node groups and communication technology allocation.

| Node group                        | Count | Technology role              | Communication requirement                        |
|-----------------------------------|-------|------------------------------|--------------------------------------------------|
| Lagoon and level sensors          | 162   | LPWAN                        | Low-rate, periodic, remote and power-constrained |
| Process, pipeline and gas sensors | 218   | Wi-Fi HaLow                  | Moderate-rate telemetry and event reporting      |
| RTUs                              | 21    | Wi-Fi HaLow backhaul         | Aggregated field traffic                         |
| CCTV/high-bandwidth devices       | 14    | 5G or fibre                  | High-rate and lower-delay traffic                |
| Gateways                          | 7     | Fibre backhaul               | Stable central aggregation                       |
| SCADA server                      | 1     | Central fibre-connected zone | Monitoring, storage and supervisory functions    |



## 4. Results and Discussion

### 4.1 Chapter overview

This chapter presents the outcomes of the GIS model, baseline ns-3 simulation, sensitivity experiments, security comparison and Packet Tracer validation. The results are interpreted against the research question rather than treated as isolated performance statistics. The discussion focuses on what each communication class contributed, where the architecture remained sensitive, how segmentation changed unauthorised reach, and which conclusions can be transferred to a real design.

### 4.2 GIS-derived architecture

The completed GIS model contained 423 nodes and 395 links. The topology was substantially larger and more spatially varied than the small grid or random layouts often used in comparative IoT simulation. It represented lagoon and water-quality monitoring, pipeline and process sensing, RTU aggregation, high-bandwidth devices, gateway backhaul and a central server.

The node distribution was dominated by process and lagoon sensors. The process sensor zone contained 218 nodes, and the lagoon sensor zone contained 162. Together they represented almost 90% of the nodes. This was operationally reasonable because a large treatment site contains many more measurement points than aggregation or server devices. The field-control zone contained 21 RTUs, the high-bandwidth zone 14 nodes, the gateway zone 7 nodes and the server zone one node.

The link distribution reflected the technology-allocation method. Wi-Fi HaLow was assigned to 201 links and LPWAN to 152 links. These two access technologies therefore served 353 of 395 links. 21 Wi-Fi HaLow-backhaul links connected RTUs to gateways. 14 high-bandwidth links used the 5G-or-fibre class, while 7 fibre links connected gateways to the server. This distribution demonstrates why a single-technology comparison would not fully answer the design problem. Most links required low-power or moderate-rate wireless access, while a much smaller number carried aggregated or high-bandwidth traffic.

The spatial model also revealed that link distance was technology dependent. Baseline grouped results showed an average distance of approximately 423.5 m for LPWAN, 137.1 m for Wi-Fi HaLow, 667.7 m for Wi-Fi HaLow backhaul, 425.7 m for 5G or fibre and 5,532.2 m for fibre backhaul. The long fibre average reflected the centralised backbone rather than a claim that

every fibre link followed a straight physical route. It nevertheless illustrated that the backhaul tier served a different geographic role from the dense access tier.

The GIS process improved traceability. Every simulated link could be associated with source and destination IDs, distance, obstruction and technology. This did not make the model equivalent to an as-built survey, but it reduced the arbitrary placement that would occur in a random topology. The model was therefore more useful for architectural reasoning even though its propagation parameters remained assumptions.

#### **4.2.1 Architecture-level flow paths**

The topology implemented a hierarchy rather than direct sensor-to-server links. Lagoon and process sensors sent traffic toward RTUs or controlled aggregation. RTUs used the Wi-Fi HaLow-backhaul tier to reach gateways. Gateways used fibre or high-capacity links to the central server. CCTV followed a separate high-bandwidth path. This hierarchy reduced the number of central connections and created natural security boundaries.

The architecture also separated traffic by consequence. A lost routine environmental update has a different operational impact from a lost alarm or video stream. The technology roles reflected that distinction. The result is not strict quality-of-service enforcement, but it is a foundation for later prioritisation.

### **4.3 Baseline performance results and comparison with literature**

The baseline simulation for the proposed hybrid communications architecture assigned distinct technologies to distinct operational roles while still maintaining a high overall packet delivery ratio. The overall packet delivery ratio was 99.482% with a throughput of 1,829.957 kbps and 217 packets lost. This indicates that the modelled architecture was able to sustain the anticipated communications demand for the baseline scenario.

The requirements for a hybrid design are evident in the results at the technology level. Fibre was able to achieve a 100.00% packet delivery ratio with an average delay of 1.103 ms and a throughput of 589.120 kbps. These results are in line with the role of fibre as a high capacity and low variability backhaul option for central and/or critical infrastructure links. Strong results were also recorded in the 5G/Fibre category with a 99.18% packet delivery ratio, an average delay of 2.423 ms, and a throughput of 584.311 kbps. These results corroborate the requirement for high-capacity infrastructure for gateway to control center or centralized operational traffic as opposed to high-capacity infrastructure for low-rate remote sensors.

Wi-Fi HaLow's packet delivery ratio, average delay and throughput were 99.58%, 12.288 ms and 454.746 kbps, respectively. Given these metrics, Wi-Fi HaLow could be used in scenarios where a wireless solution is needed to bridge the gap between LPWAN and fibre. Based on its documentation, Wi-Fi HaLow can be compared to traditional short-range Wi-Fi in that an 802.11ah network will support a lower power, longer range and more dense (higher capacity) network for IoT, versus low power wide area networks (Austin & Ancillotti, 2018; Institute of Electrical and Electronics Engineers, 2016; Lee et al., 2021).

LPWAN's packet delivery ratio, average delay and throughput are 99.00%, 200.401 ms and 22.150 kbps, respectively. While these metrics are not ideal, they do not preclude the use of LPWAN in the plant. In their current state, the metrics suggest that LPWAN should be used for low-rate periodic sensing, as opposed to high-frequency control, CCTV, or other types of dense traffic. This fits within expected behavior for LPWAN and LoRaWAN, which are designed to operate under the trade-off of long range, low power, high latency, low throughput and capacity constrained traffic (Adelantado et al., 2017; Mekki et al., 2019; Raza et al., 2017).

Baseline results further validate the main claim of the thesis, that the solution to the communication problem is not the selection of a single dominant design. What these results illustrate is that, within the context of the proposed architecture, each technology plays its own distinctive role. Fibre and 5G/fibre technologies are best suited for high-capacity aggregation and backhaul, Wi-Fi HaLow is optimal for mid-range communications and LPWAN is best for low-rate distributed measurements. This aligns with the Industrial Internet of Things literature, wherein the roles of devices and the type of traffic and communications as well as the required range and reliability dictate the network design (Boyes et al., 2018; Gungor et al., 2011).

The baseline results should, however, be analyzed within the context of the limitations of the model. Because the simulation made controlled assumptions and technology-specific parameters were locked, the results should not be viewed as a reliable prediction of what would happen in the field. Rather, the results indicate a comparative assessment of the potential roles of the respective technologies in the proposed architecture. This is also aligned with the literature on network simulations, where a simulation is useful for controlled comparisons, but is heavily reliant on defined parameters and model calibration (Campanile et al., 2020).

Table 4: Baseline performance by communication technology.

| Technology           | Links | PDR     | Average delay | Throughput   | Lost |
|----------------------|-------|---------|---------------|--------------|------|
| LPWAN                | 152   | 99.00%  | 200.401 ms    | 22.150 kbps  | 32   |
| Wi-Fi HaLow          | 201   | 99.58%  | 12.288 ms     | 454.746 kbps | 85   |
| Wi-Fi HaLow backhaul | 21    | 99.00%  | 5.869 ms      | 179.630 kbps | 41   |
| 5G or fibre          | 14    | 99.18%  | 2.423 ms      | 584.311 kbps | 59   |
| Fibre                | 7     | 100.00% | 1.103 ms      | 589.120 kbps | 0    |

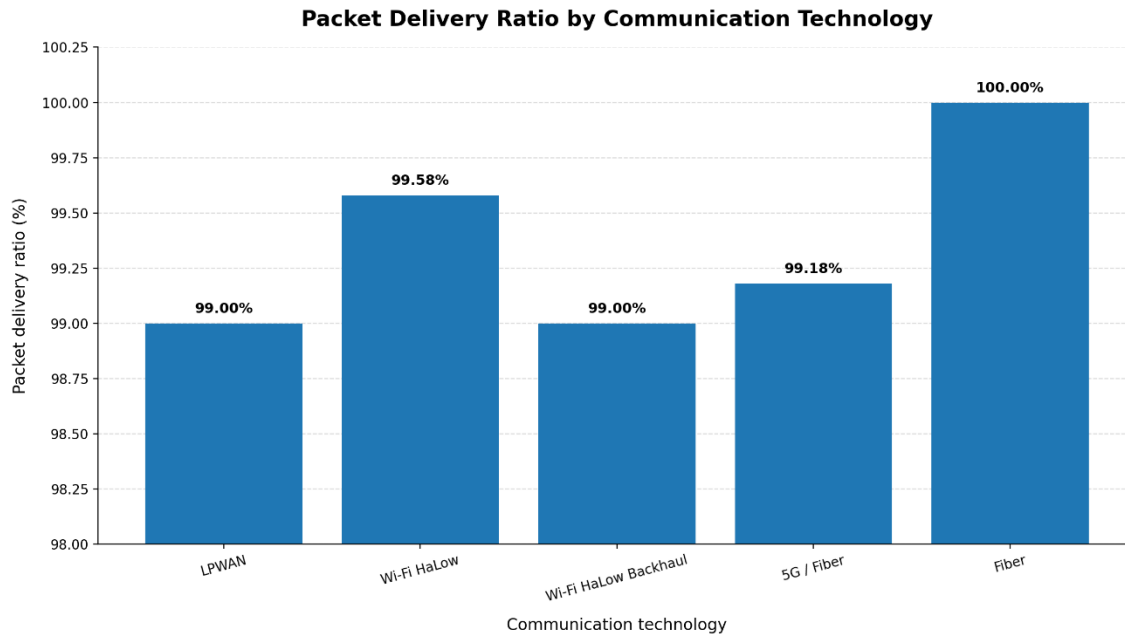


Figure 11: Baseline packet delivery ratio by communication technology (source: author's ns-3 results).

#### 4.4 Sensitivity analysis results

The security results highlighted differences between flat and segmented networks. For the flat network, all nine attack candidate flows were allowed, 90,000 attack packets were generated, and 59,807 attack packets were delivered to the SCADA server. An attack delivery rate of 66.452% was calculated. For the segmented network, all nine attack candidate flows were denied, 90,000 attack packets were dropped, and SCADA received zero attack packets. These results indicate that network segmentation allows the preservation of legitimate traffic while minimizing the unauthorised reach of attack traffic to the core of the operational technology (OT) systems.

In the lagoon-zone attack, only 12 out of 20,000 packets were successfully transmitted in the two lagoon-zone flows. This indicates that in the flat model, low-rate sensing links were not the primary attack-throughput paths. This result does not mean that the lagoon-zone sensors are secure. An attacker can compromise a sensor to manipulate and degrade data quality, generate false data and readings, or serve as an attack vector to the system. Maintaining confidence in the lagoon-zone sensors is a priority and therefore monitoring and segmentation are required.

The results align with the OT Cybersecurity Framework recommending the reduction of lateral movement by the segmentation and requirement of field/vendor/management/control zone communication (International Electrotechnical Commission, 2013, 2020; Stouffer et al, 2023). It also confirms that security is not to be an afterthought once the communication network design is complete.

As part of this thesis, segmentation was examined within the context of architectural design. This is due to the fact that a functional network could still pose risks to SCADA if it allows unrestricted paths to be created.

*Table 5: Overall sensitivity scenario results.*

| Scenario       | TX      | RX      | Lost  | PDR     | Throughput    |
|----------------|---------|---------|-------|---------|---------------|
| Baseline       | 43,407  | 43,182  | 217   | 99.482% | 1829.957 kbps |
| Obstruction x2 | 43,407  | 42,925  | 458   | 98.890% | 1820.817 kbps |
| Traffic x3     | 130,221 | 129,570 | 620   | 99.500% | 5490.901 kbps |
| Combined x2 x3 | 130,221 | 128,891 | 1,266 | 98.979% | 5466.448 kbps |

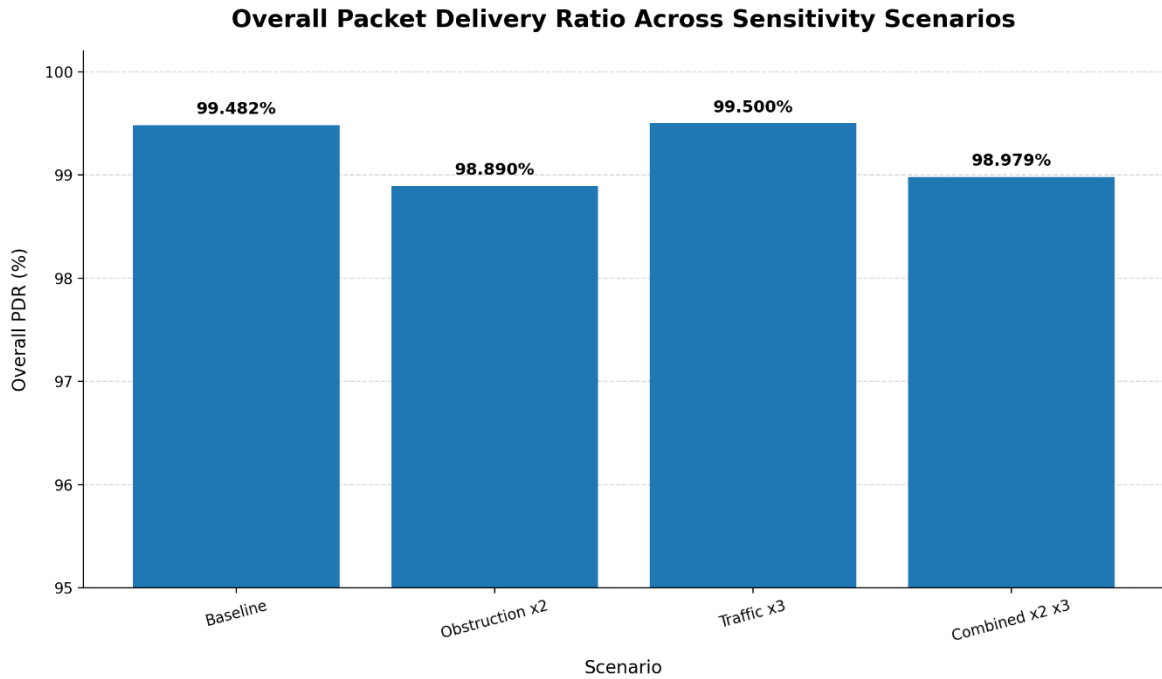


Figure 12: Overall packet delivery ratio across sensitivity scenarios (source: author's ns-3 results).

The obstruction x2 scenario helps us understand what happens to an architecture with more vegetation or obstruction. In this case, the reduction of PDR shows that wireless links need to be validated in the field with sufficient margins. The traffic x3 scenario shows that even with an increased load, the model still has enough capacity, as indicated by an increase in throughput and a minor decrease in PDR. The combined x2 x3 scenario is the most important case, as it shows how the architecture reacts when both physical obstruction and additional monitoring loads are present.

LPWAN and wireless-network literature show a similar pattern. As an example, physical obstruction and increased channel load will impact delivery performance for LPWAN and wireless networks, while the performance of a fibre network will remain stable, even under similar traffic conditions.

#### 4.4.1 Technology-level sensitivity interpretation

The technology-level sensitivity results add detail to the overall PDR. Under obstruction x2, LPWAN lost 70 packets compared with 32 at baseline. Wi-Fi HaLow loss increased from 85 to 199, Wi-Fi HaLow-backhaul loss from 41 to 76, and 5G-or-fibre loss from 59 to 113. Fibre remained lossless. Although Wi-Fi HaLow had more lost packets than LPWAN, it also carried more than six times as many transmissions. Percentage delivery therefore remained higher.

Under traffic x3, throughput scaled to 66.513 kbps for LPWAN, 1,364.022 kbps for Wi-Fi HaLow, 539.753 kbps for Wi-Fi HaLow backhaul, 1,753.253 kbps for 5G-or-fibre and 1,767.360 kbps for fibre. The near-linear scaling confirms that the configured rates were above the offered load. Under combined stress, Wi-Fi HaLow still delivered 99.11%, while LPWAN delivered 97.82%. This suggests that moderate-capacity links offered more error margin for dense process telemetry.

The single lost fibre packet in the combined scenario was rounded to 100.00% PDR in the grouped table. This illustrates the importance of presenting packet counts with percentages. A rounded percentage can conceal a small but non-zero loss.

#### **4.4.2 Implications of obstruction sensitivity for design margins**

The obstruction results indicate that link budgets should include margin rather than use average GIS conditions. LPWAN and wireless backhaul links should be surveyed at the worst practical antenna height and vegetation condition. Gateways should be positioned to reduce repeated obstruction crossings. Critical wireless paths may require directional antennas, higher mounting, redundant gateways or a wired alternative.

The small throughput reduction under obstruction was caused by packet loss rather than lower configured link rate. In a real adaptive radio, modulation and coding may also change, producing a larger throughput penalty. The simulation therefore likely understates some dynamic effects.

#### **4.5 Security segmentation results and attack-flow interpretation**

The security analysis compared a flat network with a segmented policy using the same normal traffic topology. Nine unauthorised candidate flows were defined from field zones toward the SCADA server. In the flat scenario, all nine flows were installed. They attempted 90,000 packets, and 59,807 packets were observed at SCADA. The policy-level attack delivery rate was 66.452% and observed attack throughput was 4,793.673 kbps.

The attack results differed substantially by source zone. Three high-bandwidth-zone flows transmitted 30,000 packets and delivered 29,960, giving 99.867% PDR and approximately 2,401.366 kbps. Three process-zone flows delivered 29,835 of 30,000 packets, giving 99.450% PDR and approximately 2,391.346 kbps. The lagoon-zone attack, which delivered only 12 packets out of 20,000 tries, shows that constrained low-rate sensing links were not the primary attack-throughput path in the flat model. This result does not imply lagoon sensors are safe. A

compromised lagoon sensor can affect the integrity of data, cause the system to report incorrect values, or act as a system breach entry point. The result shows that high-bandwidth and process-zone devices are of greater concern for traffic volume, while low-rate sensors still require a valid authentication, monitoring, and segmentation. This difference reflected the configured technology paths and error rates. It also shows why a flat network is not equally dangerous at every endpoint: high-capacity devices can generate much more effective unauthorised traffic than constrained sensors.

In the segmented scenario, zero attack flows were installed and all nine were recorded as blocked by policy. The 90,000 attempted packets were therefore classified as blocked before application traffic entered the network. No attack packets reached SCADA and attack throughput was zero. Normal traffic performance was not reduced. Normal PDR was 99.48% in the segmented scenario compared with 99.39% under the flat attack scenario, and normal throughput remained approximately 1,829.957 kbps.

The result supports segmentation as a low-overhead control in this model. Authorised flows continued to operate, while prohibited direct paths were removed. This is consistent with NIST and IEC guidance on zones, conduits and least privilege (Stouffer et al., 2023; IEC, 2020; IEC, 2013).

The result must not be overstated. The segmented model blocked flows by not installing them. It did not emulate a firewall under attack, rule bypass, compromised credentials, protocol tunnelling or vulnerabilities in allowed services. The zero-delivery result is therefore an ideal policy outcome. In a real plant, segmentation reduces opportunity but does not eliminate risk. Security still depends on rule maintenance, identity controls, patching, logging, detection, backups and incident response.

The flat-scenario attack PDR was reported as 74.76% for installed attack traffic, while the attempted-packet delivery rate was 66.452%. These values use different denominators. The former used packets actually generated by installed ns-3 attack applications, while the latter compared packets reaching SCADA with the full policy-level attempted count of 90,000. Reporting both makes the result transparent and avoids presenting inconsistent figures as if they measured the same quantity.



Table 6: Security-zone node counts.

| Security zone            | Nodes |
|--------------------------|-------|
| Lagoon sensor zone       | 162   |
| Process sensor zone      | 218   |
| Field control zone       | 21    |
| CCTV/high-bandwidth zone | 14    |
| Gateway/backhaul zone    | 7     |
| SCADA/server zone        | 1     |

Table 7: Flat versus segmented security results

| Mode      | Candidates | Installed | Blocked | Reached SCADA | Attack delivery | Normal PDR |
|-----------|------------|-----------|---------|---------------|-----------------|------------|
| Flat      | 9          | 9         | 0       | 59,807        | 66.452%         | 99.39%     |
| Segmented | 9          | 0         | 9       | 0             | 0.000%          | 99.48%     |

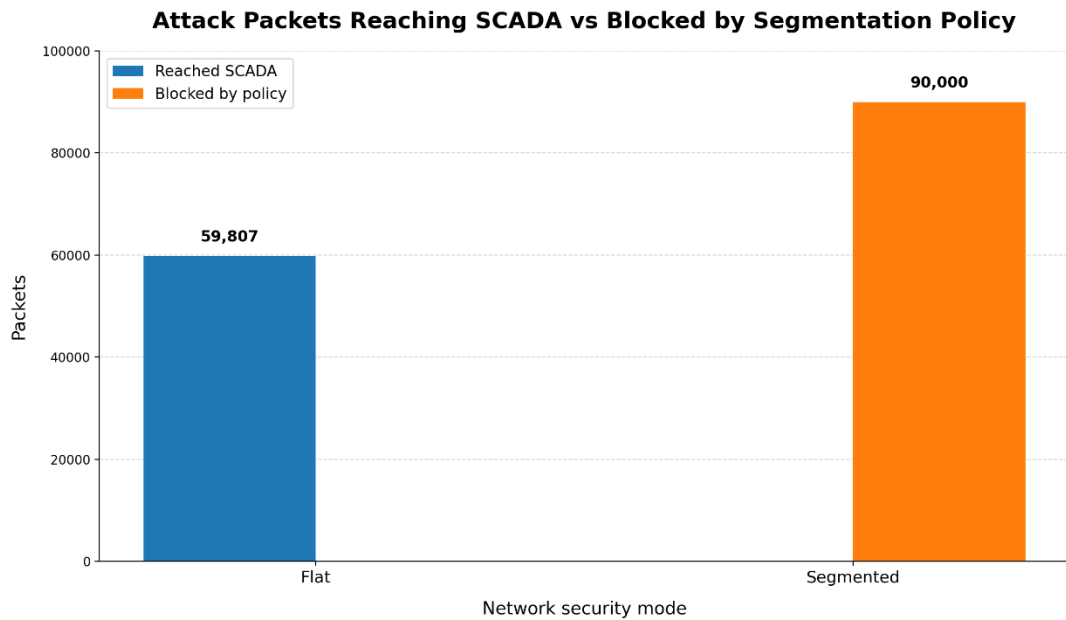


Figure 13: Attack packets reaching SCADA versus packets blocked by segmentation policy (source: author's ns-3 results).

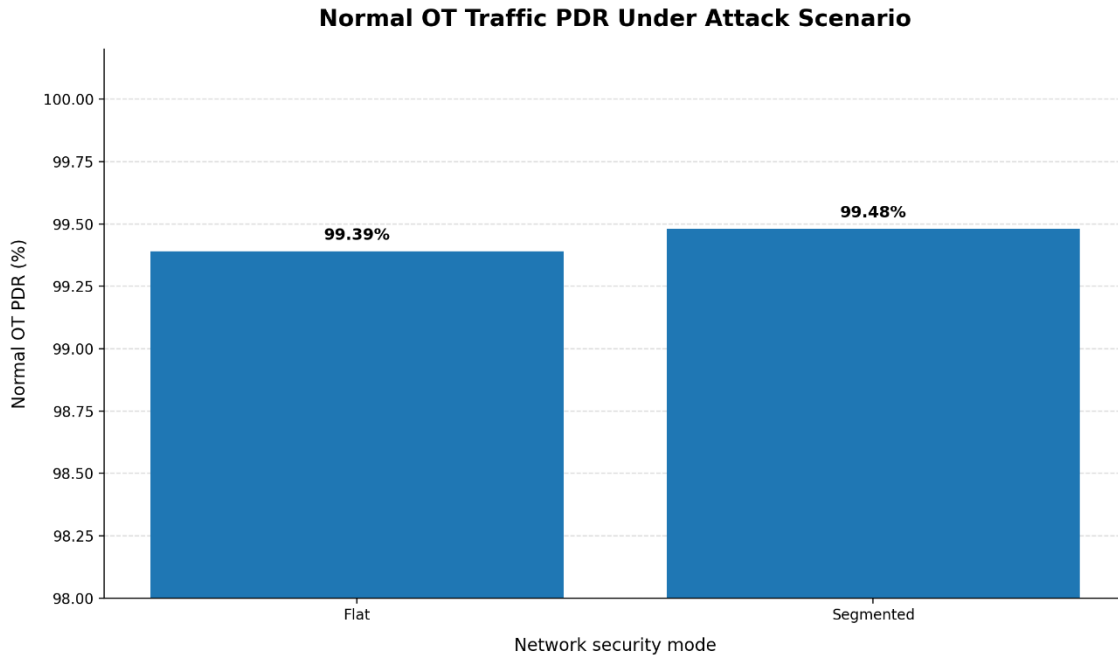


Figure 14: Normal OT traffic PDR under flat and segmented attack scenarios (source: author's ns-3 results).

#### 4.5.1 Security-zone design implications

The node counts also affect security operations. The two sensor zones contained 380 devices, which would create a large management burden if every endpoint were permitted directly into SCADA. Aggregating them through 21 RTUs and seven gateways reduces the number of conduits that require central policy. This is a practical security advantage of the architecture, although gateway compromise then becomes more significant.

The high-bandwidth zone contained only 14 devices but generated the most effective attack traffic in the flat scenario. This demonstrates that risk is not proportional only to device count. A small number of high-capacity cameras or video nodes can create a substantial denial-of-service path. Their isolation and rate monitoring should therefore receive high priority.

#### 4.6 Packet Tracer implementation and validation

The main configuration problems noticed during the validation of the Packet Tracer were as follows:

- The ASA 5505 required VLAN interfaces in order to do Layer 3 addressing.
- The routing of the SCADA and DMZ networks required explicit routes from the WTP gateway.
- The ASA needed return routes to the internal VLANs.

- The SCADA and DMZ servers required correct default gateways.
- For certain ASA tests, Packet Tracer required explicit ICMP return handling.

These problems indicate that the initial focus should be on routing validation, as well as testing firewall rules and return path verification, prior to implementing the solution in a production environment.

The Packet Tracer model translated the architecture into a smaller logical demonstration. Two Cisco 3560 multilayer switches formed the core. EtherChannel connected the cores, dual uplinks connected the access switches, HSRP provided default-gateway redundancy, and Rapid-PVST+ controlled Layer 2 loops. SW-SENSOR-1 connected the lagoon sensor in VLAN 10 and process sensor in VLAN 20. SW-CONTROL connected the RTU in VLAN 30 and CCTV endpoint in VLAN 40. SW-GATEWAY connected VLAN 50 to the Cisco 2911 WTP gateway. SW-CONTROL-CENTRE connected the management PC in VLAN 80 and vendor PC in VLAN 90.

The gateway used 10.10.50.10/24 toward the core and 10.10.99.1/30 toward the ASA. The ASA inside interface used 10.10.99.2/30. VLAN 60 connected the SCADA server at 10.10.60.10/24, and VLAN 70 connected the DMZ jump server at 10.10.70.10/24. The server gateways were the ASA interfaces 10.10.60.1 and 10.10.70.1.

The final baseline confirmed end-to-end connectivity from the gateway and approved endpoint VLANs to the servers. Several configuration issues were identified during validation. First, the ASA 5505 in Packet Tracer uses physical Ethernet ports as Layer 2 switchports and logical VLAN interfaces for Layer 3 addressing. The inside, SCADA and DMZ nameif assignments therefore had to be placed on VLAN interfaces. Second, the WTP gateway required explicit routes to 10.10.60.0/24 and 10.10.70.0/24 through the ASA. Third, the ASA required return routes to the internal VLANs through the gateway. Fourth, the SCADA and DMZ servers required correct default gateways. The ASA could ping a directly connected server without those gateways, but remote endpoints could not receive return traffic.

The final security policy was applied only after the open baseline worked. The ASA inside ACL permitted the RTU at 10.10.30.10 and CCTV endpoint at 10.10.40.10 to reach the SCADA server. It permitted the management PC at 10.10.80.10 to reach SCADA and the DMZ, and the vendor PC at 10.10.90.10 to reach only the DMZ jump server. It permitted gateway troubleshooting to both servers. Explicit deny rules blocked lagoon sensors, process sensors and vendor access to SCADA, followed by a default deny.

The approved tests succeeded: RTU to SCADA, CCTV to SCADA, management to SCADA, management to DMZ, vendor to DMZ, and gateway to both servers. The prohibited tests failed: lagoon sensor to SCADA, process sensor to SCADA, vendor to SCADA, RTU to DMZ and CCTV to DMZ. ACL hit counters confirmed that the expected permit and deny entries processed the traffic.

The implementation provides practical validation of the zone-and-conduit concept, but it remains a teaching simulation. PC-PT devices represented sensors and RTUs, custom icons changed only presentation, and ICMP was used as the main test protocol. The ASA 5505 is a legacy platform with simplified Packet Tracer behaviour. A production design would use current industrial firewalls, application-specific rules, secure management, logging and redundancy.

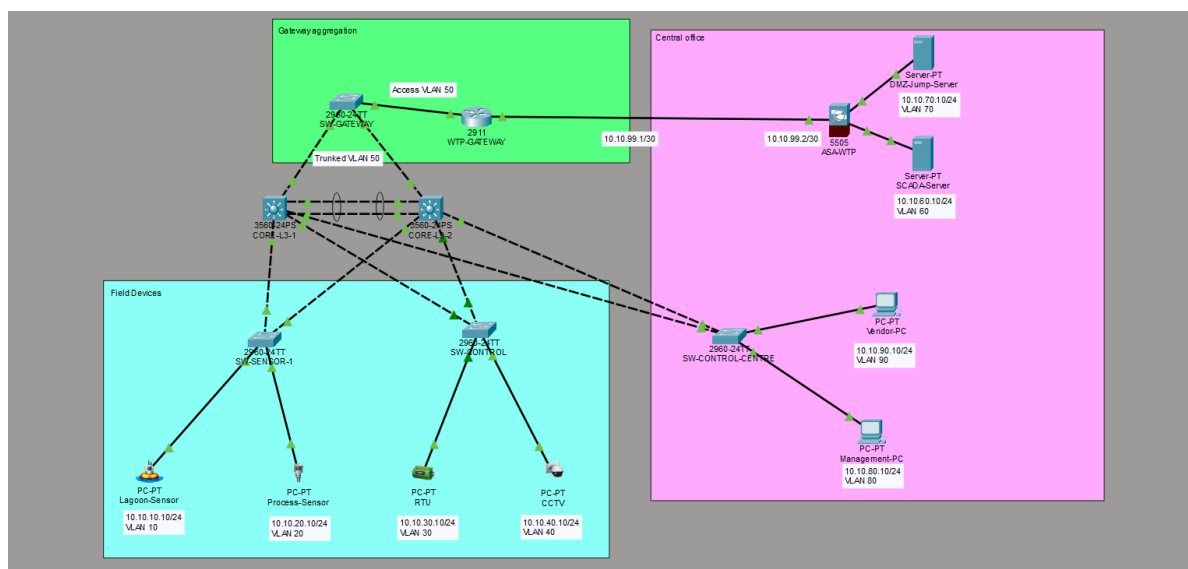


Figure 15: Final Packet Tracer OT/SCADA topology (source: author's work).

#### 4.6.1 Redundancy and remaining single points of failure

The Packet Tracer core used two multilayer switches, dual access uplinks, EtherChannel and HSRP. This demonstrated resilience against a core link or active gateway failure at the LAN layer. The final simplified design still contained single points of failure at the WTP gateway, ASA firewall and individual servers. Earlier design iterations considered dual gateways and a transit switch, but the additional complexity created configuration uncertainty in Packet Tracer and did not improve the core research result.

The final thesis should therefore describe the implementation as partial redundancy. A production design would require redundant gateways, firewall high availability, diverse fibre

paths, clustered or replicated servers and tested failover procedures. Recognising these remaining failure points is stronger than claiming complete resilience from a teaching model.

#### **4.6.2 Verification matrix and evidence**

The final verification matrix contained both positive and negative tests. Positive tests confirmed that required operations remained possible. Negative tests were equally important because they showed that segmentation had an observable effect. The final working file was saved after the restricted ACL was validated, and a separate baseline copy was retained.

Command outputs such as show interfaces trunk, show standby brief, show etherchannel summary, show ip route, show access-list and show interface ip brief provided configuration evidence beyond ping results. A ping can fail for many reasons, and a successful ping does not prove every redundancy feature. Combining traffic tests with device-state commands improved confidence in the implementation.

#### **4.7 Integrated discussion against the research question**

The research question asked how a GIS-assisted hybrid communication architecture and segmented OT network could improve communication suitability, reliability and operational resilience. The results support a qualified positive answer.

Communication suitability improved because devices were not evaluated against one common performance target. LPWAN served low-rate environmental sensing. Wi-Fi HaLow served denser process telemetry. Wi-Fi HaLow backhaul carried aggregated field traffic. High-capacity links served CCTV and similar devices, and fibre supported central backhaul. The baseline results showed performance differences that aligned with these roles. This is more defensible than choosing the technology with the lowest delay for every device, because cost, power and range would be ignored.

Reliability was strong under the baseline and remained high under the tested stress scenarios. Overall PDR stayed above 98.89% in every scenario and reached 98.979% under combined doubled obstruction and tripled traffic. This indicates that the architecture had reasonable simulated headroom. However, reliability was not uniform. LPWAN and wireless backhaul were more sensitive to obstruction, while fibre remained stable. Operational resilience therefore depends on assigning critical traffic to links with suitable margins and providing recovery mechanisms for wireless loss.

Security segmentation improved resilience against the modelled unauthorised traffic. The flat scenario allowed nine attack flows and nearly 60,000 packets reached SCADA. The segmented scenario blocked all candidate flows while normal PDR remained approximately 99.48%. Packet Tracer demonstrated the same principle through explicit allowed and denied paths. The combined evidence is stronger than either experiment alone because it links large-scale policy effects with a concrete implementation.

GIS contributed by making the architecture site aware. The model could distinguish dense sensor areas, long backhaul paths and obstruction classes. This did not produce a final radio design, but it improved the relevance of the simulation and created a reproducible data pipeline. The study therefore supports GIS as an early-stage engineering tool that should be followed by field measurement, not used as a substitute for it.

#### 4.7.1 Relationship between performance and security

Performance and security were not independent outcomes. The gateway and backhaul design concentrated approved traffic into a small number of conduits, which made policy easier to define. At the same time, those conduits became high-value assets. The high-bandwidth attack result showed that devices with greater communication capacity can create greater security impact.

Segmentation can also support performance by separating broadcast and high-bandwidth traffic from low-rate control zones. The Packet Tracer VLAN design isolated CCTV from sensor VLANs and vendor access from SCADA. The study did not measure VLAN-related congestion reduction directly, but the architecture supports future quality-of-service and monitoring policies.

*Table 8: Packet Tracer VLAN and addressing plan.*

| VLAN | Zone                | Subnet        | Gateway          |
|------|---------------------|---------------|------------------|
| 10   | Lagoon sensors      | 10.10.10.0/24 | 10.10.10.1       |
| 20   | Process sensors     | 10.10.20.0/24 | 10.10.20.1       |
| 30   | RTU/field control   | 10.10.30.0/24 | 10.10.30.1       |
| 40   | CCTV/high bandwidth | 10.10.40.0/24 | 10.10.40.1       |
| 50   | Gateway aggregation | 10.10.50.0/24 | 10.10.50.1       |
| 60   | SCADA               | 10.10.60.0/24 | 10.10.60.1 (ASA) |

| VLAN    | Zone           | Subnet        | Gateway          |
|---------|----------------|---------------|------------------|
| 70      | OT DMZ         | 10.10.70.0/24 | 10.10.70.1 (ASA) |
| 80      | Management     | 10.10.80.0/24 | 10.10.80.1       |
| 90      | Vendor         | 10.10.90.0/24 | 10.10.90.1       |
| Transit | Gateway to ASA | 10.10.99.0/30 | 10.10.99.1 / .2  |

*Table 9: Packet Tracer allowed and blocked traffic matrix.*

| Source                     | Destination       | Policy | Reason                             |
|----------------------------|-------------------|--------|------------------------------------|
| RTU 10.10.30.10            | SCADA 10.10.60.10 | Permit | Operational telemetry/control path |
| CCTV 10.10.40.10           | SCADA 10.10.60.10 | Permit | Approved high-bandwidth monitoring |
| Management 10.10.80.10     | SCADA and DMZ     | Permit | Authorised OT administration       |
| Vendor 10.10.90.10         | DMZ 10.10.70.10   | Permit | Controlled jump-server access      |
| Lagoon sensor 10.10.10.10  | SCADA             | Deny   | No direct sensor-to-SCADA path     |
| Process sensor 10.10.20.10 | SCADA             | Deny   | Must use approved aggregation      |
| Vendor 10.10.90.10         | SCADA             | Deny   | No direct vendor access            |
| RTU/CCTV                   | DMZ               | Deny   | No operational requirement         |

*Table 10: Principal limitations and implications*

| Limitation           | Effect                                                                       | Required follow-up                                             |
|----------------------|------------------------------------------------------------------------------|----------------------------------------------------------------|
| Protocol abstraction | Technology classes used configurable links rather than full protocol stacks. | Interpret results comparatively; run protocol-specific pilots. |
| GIS accuracy         | Public and inferred spatial data were not an as-built survey.                | Verify asset coordinates and collect RF measurements.          |

| Limitation     | Effect                                                               | Required follow-up                                               |
|----------------|----------------------------------------------------------------------|------------------------------------------------------------------|
| Delay response | Fixed link delays and unsaturated queues limited congestion effects. | Add queue, contention and burst-traffic models.                  |
| Security model | Segmented attack flows were blocked at policy level.                 | Test real firewalls, identity controls and industrial protocols. |
| Packet Tracer  | Legacy ASA and simplified endpoints were used.                       | Use current production-grade OT equipment for deployment.        |

#### 4.8 Overall results and comparison with previous studies

The overall results suggest that the proposed architecture is both an appropriate design and evaluation model for a lagoon-based wastewater treatment system. The base packet delivery ratio being high indicates that the combination of LPWAN, Wi-Fi HaLow, and 5G/fibre and their respective roles is feasible, and a specific architecture is warranted. The results prevail in the context of previous studies of IoT and industrial communications in which it is established that heterogeneous networks are needed when differing requirements for device coverage, power, bandwidth, and reliability are observed (Al-Fuqaha et al., 2015; Boyes et al., 2018; Gungor et al., 2011).

The results also express the importance of device role in technology selection. LPWAN was found to be suitable for the low-rate sensing role but expressed the highest latency and low throughput. Wi-Fi HaLow took the role of intermediate wireless, while the fibre and 5G/fibre took the high capacity backhaul. This is in line with the literature on LPWAN which emphasizes low power and coverage, and with Wi-Fi HaLow literature in which the IEEE 802.11ah communication is seen as an IoT wireless technology that is more appropriate for longer range and higher capacity than a number of LPWAN sensing links (Austin & Ancillotti, 2018; Institute of Electrical and Electronics Engineers, 2016; Lee et al., 2021; Mekki et al., 2019; Raza et al., 2017).

The sensitivity analysis argues that the architecture can't be evaluated solely on baseline assumptions. Obstruction x2 resulted in a decreased Packet Delivery Ratio, while traffic x3 resulted in an increased throughput and an increased absolute count of lost packets. The stress combined case demonstrated that the architecture was still working, but with even greater losses. This fits within the context of other wireless network research, where physical and environmental factors and the traffic load operate to influence the effect and the reliability of



the network. From this and given obstruction (etc.) in a real deployment, it is also recommended to conduct site surveys, link margin tests, check placement of gateways, and plan redundancy.

The results on security also show that performance of communication is not the only consideration in an OT environment. The flat model permitted undesired communication to SCADA, while the segmented model prevented the same communication. This is in line with the OT security recommendations, where zone-based design, conduits, and enforcement of the least privilege principles and firewalls are proposed, in order to prevent undue communication between the field, vendor, management, and control systems (International Electrotechnical Commission, 2013, 2020; Stouffer et al., 2023).

A high Packet Delivery Ratio is reached, which also is in line with simulation-based studies which postulate that, if less than perfect (controlled) assumptions are made, more stable delivery results can be expected. The fixed-delay behaviour presents the limitation of simplified simulation models. Previous studies undertaken in ns-3, emphasize that the validity of a simulation is highly dependent on the chosen parameters, topological assumptions, and a calibration to the real (observable) conditions (Campanile et al., 2020). For these reasons, the results attained should be considered comparative design constraints and not assurance of a deployable design.

Overall, results support the thesis contribution: a segmented, GIS-based, hybrid architecture assesses communication and OT security. The study is not proposing a one-size-fits-all solution; rather, the study is suggesting communication systems for large-scale wastewater treatment should consider the spatial arrangement, role of the equipment, the nature of the traffic, and security zoning.

## **4.9 Practical implications and deployment sequence**

A practical deployment should be staged. First, the GIS model should be refined using verified asset locations, power availability, existing fibre routes and maintenance access. Second, a radio survey should measure sub-1 GHz and cellular performance across representative lagoon, pipeline, wetland and built zones. Third, a small pilot should test LPWAN and Wi-Fi HaLow devices under seasonal weather and vegetation. Fourth, gateway placement should be optimised for coverage, power, physical security and redundant backhaul.

The central network should be segmented from the beginning. Field sensors, RTUs, CCTV, gateways, management, vendor access, the OT DMZ and SCADA should have separate zones. Vendor connections should terminate at a managed jump server with authentication, logging

and time-limited access. Firewall policy should permit only required protocols and directions. Core and gateway redundancy should be added where outage consequences justify the cost.

The model also supports prioritisation. Fibre investment should focus on gateway and central-backbone paths where it provides the greatest benefit. LPWAN should be used where reporting intervals and loss tolerance are appropriate. Wi-Fi HaLow should be piloted for process telemetry where conventional Wi-Fi range is insufficient, but LPWAN capacity is too low. High-bandwidth wireless should be reserved for devices that need it.

#### **4.10 Threats to validity**

Construct validity was limited by the use of simplified technology classes. PDR, delay and throughput measured the configured links, but they did not capture every protocol mechanism. Internal validity was strengthened by using the same topology across scenarios, retaining raw outputs and checking aggregation, although the experiments did not include multiple random seeds for every reported result. External validity was limited because the case study used public and inferred data rather than an as-built plant model. Conclusion validity was strongest for relative differences and policy outcomes and weaker for absolute service guarantees.

The study addressed these threats through transparent assumptions, sensitivity testing, comparison with literature, preservation of code and raw CSV outputs, and a separate Packet Tracer validation. The remaining uncertainty is why the recommendations require field testing before operational adoption.

## 5. Conclusion and Recommendations

### 5.1 Conclusion

This study investigated how a GIS-assisted hybrid communication architecture and segmented OT network could support a large lagoon-based wastewater treatment plant. The research question was answered positively, with important qualifications. A hybrid design was more suitable than a uniform network because the plant contained device classes with different range, energy, bandwidth, latency and security requirements. GIS improved the relationship between communication design and infrastructure location. ns-3 provided repeatable comparative evidence, and Packet Tracer demonstrated that the architecture could be translated into VLANs, routing, redundancy, firewall zones and controlled communication paths.

The final GIS model contained 423 nodes and 395 links. Most access links used LPWAN or Wi-Fi HaLow, while RTU aggregation, high-bandwidth communication and central backhaul used progressively higher-capacity tiers. The baseline simulation achieved 99.482% overall PDR. Fibre produced the lowest delay and no loss. The 5G-or-fibre tier provided high throughput and low delay. Wi-Fi HaLow offered a useful middle tier for process telemetry. LPWAN had the highest delay and lowest throughput but remained suitable for small, periodic environmental messages.

Sensitivity testing showed that obstruction was more damaging to delivery than the selected traffic increase. Overall PDR fell to 98.890% when obstruction-related error doubled and remained 99.500% when traffic tripled. Under combined stress, PDR was 98.979%. These results indicate simulated headroom but do not establish real-world capacity limits. The stable delay across scenarios reflected fixed-delay and queue assumptions, which is a major limitation for congestion analysis.

The security results were clear within the model. Nine unauthorised flows reached the SCADA zone in the flat scenario, and 59,807 attack packets were observed. The segmented scenario blocked all nine candidates before application traffic was installed, while normal PDR remained approximately 99.48%. Packet Tracer confirmed the intended policy: RTU and CCTV traffic could reach SCADA, management could reach SCADA and the DMZ, vendor access was restricted to the DMZ, and direct sensor or vendor access to SCADA was blocked.

The study's main contribution was the integration of spatial planning, hybrid technology allocation, performance simulation, sensitivity analysis, security modelling and practical network validation. It did not invent a new protocol or prove a production deployment. Its value

is a structured engineering method for deciding which links require low power, moderate capacity, high capacity, fibre stability or stronger security control.

### **5.1.1 Degree of confidence in the answer**

Confidence is highest in the comparative architecture conclusion and the logical segmentation result. It is moderate for the numerical performance values because they depend on configured link assumptions. It is lowest for direct field performance, battery life and congestion delay because those were not measured. This graded confidence should guide how the thesis is used: as a design and research foundation, not as final commissioning evidence.

## **5.2 Recommendations for implementation**

A real deployment should begin with an asset and communications survey. GIS coordinates, existing conduits, fibre routes, antenna heights, power availability and maintenance access should be verified. Radio measurements should be collected at representative lagoon, pipeline, wetland and built locations across different weather and vegetation conditions.

LPWAN should be considered for delay-tolerant lagoon, level and environmental telemetry. Application-layer acknowledgements, retries and data buffering should be used where missing observations would be significant. Wi-Fi HaLow should be piloted for process and pipeline monitoring that needs greater responsiveness and payload capacity. Private 5G or another managed high-bandwidth technology should be evaluated for CCTV, mobile assets and difficult-to-fibre locations. Fibre should remain the preferred option for fixed critical backbone links where installation is feasible.

Gateways should be installed at powered, accessible and physically secured locations. Critical gateways and central links should have redundancy, backup power and monitored failover. The architecture should avoid one gateway or one fibre route becoming an unrecognised common failure point.

Security zoning should be treated as a design requirement. Lagoon sensors, process sensors, RTUs, CCTV, gateways, management, vendor access, the OT DMZ and SCADA should remain separate. Firewall rules should specify required source, destination, protocol and direction rather than broad IP access. Vendor access should use a controlled jump server, multifactor authentication, logging, approval and time limits. Current supported firewalls should replace the legacy ASA platform used for the teaching model.

Operations should monitor communication health, packet loss, gateway state, firewall events and configuration changes. Baseline traffic profiles should be established so that abnormal device behaviour can be detected. Backups, incident-response procedures and tested recovery paths should complement segmentation.

### **5.2.1 Recommended acceptance criteria for a pilot**

A pilot should define acceptance criteria before testing. Suggested criteria include minimum PDR for routine and alarm traffic, maximum alarm delay, coverage margin at seasonal worst-case locations, gateway failover time, backhaul recovery behaviour and successful enforcement of the approved traffic matrix. The pilot should also measure battery consumption and maintenance workload. These criteria would convert the comparative findings of this thesis into operational evidence.

## **5.3 Recommendations for future research**

Future research should implement full LoRaWAN and IEEE 802.11ah protocol models, including contention, retransmission, adaptive data rate and shared-channel effects. Queue and scheduling models should be added so that delay responds realistically to load. Multiple random seeds and confidence intervals should be reported. Future work should also integrate production-grade OT security tools to test routing, return-path behaviour, and firewall policy. The Packet Tracer validation indicated that default server gateways, ASA return paths, and access-control rules affect if and how authorised traffic is successful. These issues should be addressed in any real pilot or hardware validation phase.

The GIS model should be linked to measured terrain, antenna and propagation data. Gateway placement could then be optimised using coverage, cost, resilience and maintenance constraints. Energy models should be introduced for battery-powered sensors, and seasonal operation should be evaluated.

Edge processing should be tested using measured data-reduction and alarm-prioritisation workloads. A future study could compare raw forwarding with local filtering, aggregation and anomaly detection. The security experiment should be extended using a production-grade emulator, industrial protocols, firewall failover, intrusion detection and controlled attack traffic. Integration with a digital-twin platform would then allow the communication architecture to be assessed as part of a complete operational information system.

## 5.4 Final statement

A lagoon-based wastewater treatment plant should not be designed around the assumption that every device needs the same network. The study showed that a spatially grounded hybrid architecture can allocate communication resources according to operational need and that segmentation can remove unnecessary paths without disrupting authorised traffic. The results provide a credible design foundation. Field measurements, protocol-specific trials and organisational security validation are the next steps required to turn that foundation into an operational system.

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## 7. APPENDICES

### 7.1 Appendix A: Research artefact inventory

*Appendix Table A: Research artefact inventory*

| Artefact            | File                                               | Purpose                                                             |
|---------------------|----------------------------------------------------|---------------------------------------------------------------------|
| GIS nodes           | wtp_nodes_ns3_zoned.csv                            | 423 nodes with operational and security-zone attributes             |
| GIS links           | wtp_links_ns3_zoned.csv                            | 395 links with distance, technology, obstruction and quality fields |
| Baseline source     | wtp-hybrid-ns3.cc                                  | ns-3 architecture and traffic implementation                        |
| Baseline results    | wtp_ns3_thesis_summary_by_technology.csv           | Technology-level PDR, delay, throughput and loss                    |
| Sensitivity source  | wtp-hybrid-sensitivity.cc                          | Obstruction and traffic stress scenarios                            |
| Sensitivity summary | wtp_ns3_sensitivity_overall_summary.csv            | Overall scenario comparison                                         |
| Security source     | wtp-security-flat-vs-segmented.cc                  | Flat and segmented policy scenarios                                 |
| Security comparison | wtp_security_flat_vs_segmented_comparison.csv      | Attack reach and normal traffic results                             |
| Packet Tracer guide | WTP_OT_SCADA_Packet_Tracer_Configuration_Guide.pdf | Device configuration and verification guide                         |

## 7.2 Appendix B: Baseline simulation parameters

*Appendix Table B: Baseline simulation parameters*

| Parameter                    | Value                                           |
|------------------------------|-------------------------------------------------|
| Simulation duration          | 120 s                                           |
| Normal traffic start         | 10 s                                            |
| Normal traffic stop          | 115 s                                           |
| Active normal-traffic period | 105 s                                           |
| Nodes                        | 423                                             |
| Links                        | 395                                             |
| Performance outputs          | FlowMonitor XML, by-flow CSV and grouped CSV    |
| Primary metrics              | PDR, average delay, throughput and lost packets |

## 7.3 Appendix C: Security policy summary

The security model grouped lagoon sensors, process sensors, RTUs, CCTV, gateways and SCADA into separate zones. Nine candidate unauthorised flows were created for the flat scenario. In the segmented scenario, the same candidates were rejected by the policy before application installation. The Packet Tracer implementation added management, vendor and DMZ zones and validated the authorised and blocked paths listed in Table 8.

## 7.4 Appendix D: Packet Tracer verification commands

```
Core switches:
show vlan brief
show interfaces trunk
show etherchannel summary
show spanning-tree root
show standby brief
show ip route
```

```
WTP gateway:
show ip interface brief
show ip route
show arp
```

```

ASA:
show interface ip brief
show route
show access-list
show service-policy
show running-config

```

## 7.5 Appendix F to I: Screenshot Evidence Added to the Draft

The screenshots support the QGIS data preparation, ns-3 simulation output, Packet Tracer validation, and Chapter 4 result interpretation. They are included as supporting evidence rather than replacing the main argument in the body of the thesis.

*Appendix Table F1: Screenshot evidence register.*

| Appendix section | Evidence included                                                                                                     | Purpose in thesis                                                                                                                  | Recommended placement                                                        |
|------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Appendix F       | QGIS map, node table, link table, obstruction fields and obstruction-zone layer.                                      | Shows the spatial data model and traceable QGIS-to-ns-3 preparation process.                                                       | Methodology support and appendix evidence.                                   |
| Appendix G       | Labelled ns-3 plots for overall PDR, technology-level PDR, lost packets and throughput across all uploaded scenarios. | Shows the additional performance evidence generated from the uploaded ns-3 result summaries and supports Chapter 4 interpretation. | Appendix G evidence, with key trends referenced in Chapter 4 where relevant. |
| Appendix H       | Packet Tracer topology, ASA/gateway/core verification and endpoint tests.                                             | Shows VLAN/routing/security validation and allowed versus blocked traffic behaviour.                                               | Results validation and appendix evidence.                                    |
| Appendix I       | PDR, delay, throughput, sensitivity and security result graphs.                                                       | Supports the numerical discussion in Chapter 4 with presentation-ready graph evidence.                                             | Chapter 4 and appendix support.                                              |

## 7.5.1 Appendix F: QGIS spatial model and attribute evidence

These screenshots show the QGIS model used to prepare the node and communication-link datasets. They demonstrate that the method was based on structured spatial layers rather than only a conceptual drawing.

### GIS technology map



Appendix Figure F1: Technology-coloured QGIS communication-link map showing the hybrid technology layout used for the WTP-style model.

### GIS node table

Nodes old — Features Total: 423, Filtered: 423, Selected: 0

| fid | type     | name   | height_m | height_ref | height_note                                                                                     |
|-----|----------|--------|----------|------------|-------------------------------------------------------------------------------------------------|
| 1   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 2   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 3   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 4   | gas      | p55    |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 5   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 6   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 7   | gas      | p55-5a |          | 2          | Industrial field ... Gas/process sensor assumed mounted on post, cabinet or process structure.  |
| 8   | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 9   | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 10  | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 11  | water    | p55-5a |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 12  | level    | p55-5a |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 13  | level    | p55-5a |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 14  | level    | p55-5  |          | 1.5        | LoRa field meas... Level sensor assumed low-mounted near lagoon/channel structure.              |
| 15  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 16  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 17  | water    | p55-5  |          | 1.5        | LoRa field meas... Water sensor assumed near lagoon edge or low field post.                     |
| 18  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 19  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 20  | pipeline | p55-5  |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |
| 21  | pipeline | p55-5a |          | 1.5        | Low field instru... Pipeline or metering sensor assumed mounted close to pipe or field cabinet. |

Show All Features

Appendix Figure F2: QGIS node attribute table showing 423 node features with device type, name and height-assumption fields.



## QGIS link table, basic fields

auto\_connections\_tech copy — Features Total: 395, Filtered: 395, Selected: 395

|    | from_id | to_id | from_type | to_type | technology  | distance_m        | final_error | fid | freq_mhz | snr_db            | tx_power | noise_fig | bandwidth_hz | noise_floor       | path_loss         | total_loss        |
|----|---------|-------|-----------|---------|-------------|-------------------|-------------|-----|----------|-------------------|----------|-----------|--------------|-------------------|-------------------|-------------------|
| 1  | 1       | 38    | gas       | rtu     | Wi-Fi HaLow | 67.805624761625   | 0.0035      | 1   | 915      | 56.70626368038... | 20       | 8         | 1000000      | -106              | 68.29373631961... | 69.29373631961... |
| 2  | 2       | 38    | gas       | rtu     | Wi-Fi HaLow | 20.17191599736... | 0.0035      | 2   | 915      | 67.23663509981... | 20       | 8         | 1000000      | -106              | 57.76336490018... | 58.76336490018... |
| 3  | 3       | 330   | gas       | rtu     | Wi-Fi HaLow | 65.22516845003... | 0.0035      | 3   | 915      | 57.04327393108... | 20       | 8         | 1000000      | -106              | 67.956726068917   | 68.956726068917   |
| 4  | 4       | 38    | gas       | rtu     | Wi-Fi HaLow | 80.62509271068... | 0.0035      | 4   | 915      | 55.20217357790... | 20       | 8         | 1000000      | -106              | 69.79782642209... | 70.79782642209... |
| 5  | 5       | 330   | gas       | rtu     | Wi-Fi HaLow | 80.5660253537478  | 0.0035      | 5   | 915      | 55.20859394511... | 20       | 8         | 1000000      | -106              | 69.79146065488... | 70.79146065488... |
| 6  | 6       | 346   | gas       | rtu     | Wi-Fi HaLow | 75.2867120871...  | 0.0035      | 6   | 915      | 55.7972114854388  | 20       | 8         | 1000000      | -106              | 69.2027885045612  | 70.2027885045612  |
| 7  | 7       | 330   | gas       | rtu     | Wi-Fi HaLow | 105.6944845225... | 0.0035      | 7   | 915      | 52.85053161829... | 20       | 8         | 1000000      | -106              | 72.14946838170... | 73.14946838170... |
| 8  | 8       | 330   | water     | rtu     | LPWAN       | 103.2310335533... | 0.0105      | 8   | 915      | 58.08627247786... | 14       | 6         | 125000       | -117.030899869... | 71.9446273920588  | 72.9446273920588  |
| 9  | 10      | 330   | water     | rtu     | LPWAN       | 117.1333312266... | 0.0105      | 9   | 915      | 56.9886809601...  | 14       | 6         | 125000       | -117.030899869... | 73.0420317739025  | 74.0420317739025  |
| 10 | 12      | 330   | water     | rtu     | LPWAN       | 61.76407339028... | 0.0105      | 10  | 915      | 62.54775910048... | 14       | 6         | 125000       | -117.030899869... | 67.4831407694362  | 68.4831407694362  |
| 11 | 13      | 330   | level     | rtu     | LPWAN       | 18.33409990213... | 0.0105      | 11  | 915      | 73.09728611894... | 14       | 6         | 125000       | -117.030899869... | 56.93361375097... | 57.93361375097... |
| 12 | 14      | 400   | level     | rtu     | LPWAN       | 222.401353732283  | 0.02        | 12  | 915      | 42.41972946018... | 14       | 6         | 125000       | -117.030899869... | 78.6111704097336  | 88.6111704097336  |
| 13 | 15      | 38    | water     | rtu     | LPWAN       | 169.2723348425... | 0.0105      | 13  | 915      | 53.79075829563... | 14       | 6         | 125000       | -117.030899869... | 76.24014157428... | 77.24014157428... |
| 14 | 16      | 38    | water     | rtu     | LPWAN       | 502.8698159120... | 0.0105      | 14  | 915      | 44.3336661943...  | 14       | 6         | 125000       | -117.030899869... | 85.69753325048... | 86.69753325048... |
| 15 | 17      | 400   | water     | rtu     | LPWAN       | 414.904760665563  | 0.02        | 15  | 915      | 37.00350962852... | 14       | 6         | 125000       | -117.030899869... | 84.02739024138... | 94.02739024138... |
| 16 | 18      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 172.1562386921... | 0.0035      | 16  | 915      | 48.6131228081222  | 20       | 8         | 1000000      | -106              | 76.3868771918778  | 77.3868771918778  |
| 17 | 19      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 139.1797980066... | 0.0035      | 17  | 915      | 50.46050408141... | 20       | 8         | 1000000      | -106              | 74.53994591858... | 75.53994591858... |
| 18 | 20      | 38    | pipeline  | rtu     | Wi-Fi HaLow | 115.5264242410... | 0.0035      | 18  | 915      | 52.07795149219... | 20       | 8         | 1000000      | -106              | 72.92204850780... | 73.92204850780... |
| 19 | 21      | 330   | pipeline  | rtu     | Wi-Fi HaLow | 46.10629708675... | 0.0035      | 19  | 915      | 60.05637323202... | 20       | 8         | 1000000      | -106              | 64.94362676797... | 65.94362676797... |
| 20 | 22      | 330   | pipeline  | rtu     | Wi-Fi HaLow | 43.56721347635... | 0.0035      | 20  | 915      | 60.54838244464... | 20       | 8         | 1000000      | -106              | 64.451617553512   | 65.451617553512   |

Show All Features

Appendix Figure F3: QGIS link attribute table showing 395 generated communication links with endpoint IDs, device types, technology classes and distances.

## QGIS link table, obstruction fields

auto\_connections\_tech copy — Features Total: 395, Filtered: 395, Selected: 395

|    | power | noise_fig | bandwidth_hz | noise_floor       | path_loss         | total_loss        | rx_power          | obs_cross | obs_density | veg_loss_db | veg_error | obs_ref | obs_note           | from_name           | to_name | endpoint_check |
|----|-------|-----------|--------------|-------------------|-------------------|-------------------|-------------------|-----------|-------------|-------------|-----------|---------|--------------------|---------------------|---------|----------------|
| 1  | 20    | 8         | 1000000      | -106              | 68.29373631961... | 69.29373631961... | -49.2937363196... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55     | OK             |
| 2  | 20    | 8         | 1000000      | -106              | 57.76336490018... | 58.76336490018... | -38.7633649001... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55     | OK             |
| 3  | 20    | 8         | 1000000      | -106              | 67.956726068917   | 68.956726068917   | -48.9567260689... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 4  | 20    | 8         | 1000000      | -106              | 69.79782642209... | 70.79782642209... | -50.7978264220... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55     | OK             |
| 5  | 20    | 8         | 1000000      | -106              | 69.79146065488... | 70.79146065488... | -50.7914606548... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 6  | 20    | 8         | 1000000      | -106              | 69.2027885045612  | 70.2027885045612  | -50.2027885045... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 7  | 20    | 8         | 1000000      | -106              | 72.14946838170... | 73.14946838170... | -53.1494683817... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 8  | 14    | 6         | 125000       | -117.030899869... | 71.9446273920588  | 72.9446273920588  | -58.9446273920... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 9  | 14    | 6         | 125000       | -117.030899869... | 73.0420317739025  | 74.0420317739025  | -60.0420317739... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 10 | 14    | 6         | 125000       | -117.030899869... | 67.4831407694362  | 68.4831407694362  | -54.4831407694... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 11 | 14    | 6         | 125000       | -117.030899869... | 56.93361375097... | 57.93361375097... | -43.9336137509... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 12 | 14    | 6         | 125000       | -117.030899869... | 78.6111704097336  | 88.6111704097336  | -74.6111704097... | yes       | high        |             | 10        | 0.01    | ITU-R P.833; SA... | Dense trees or s... | p55-5   | OK             |
| 13 | 14    | 6         | 125000       | -117.030899869... | 76.24014157428... | 77.24014157428... | -63.2401415742... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5   | OK             |
| 14 | 14    | 6         | 125000       | -117.030899869... | 85.69753325048... | 86.69753325048... | -72.6975332504... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5   | OK             |
| 15 | 14    | 6         | 125000       | -117.030899869... | 84.02739024138... | 94.02739024138... | -80.0273902413... | yes       | high        |             | 10        | 0.01    | ITU-R P.833; SA... | Dense trees or s... | p55-5   | OK             |
| 16 | 20    | 8         | 1000000      | -106              | 76.3868771918778  | 77.3868771918778  | -57.3868771918... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5   | OK             |
| 17 | 20    | 8         | 1000000      | -106              | 74.53994591858... | 75.53994591858... | -55.5399459185... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5   | OK             |
| 18 | 20    | 8         | 1000000      | -106              | 72.92204850780... | 73.92204850780... | -53.9220485078... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5   | OK             |
| 19 | 20    | 8         | 1000000      | -106              | 64.94362676797... | 65.94362676797... | -45.9436267679... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |
| 20 | 20    | 8         | 1000000      | -106              | 64.451617553512   | 65.451617553512   | -45.4516175535... | yes       | low         |             | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... | p55-5a  | OK             |

Show All Features

Appendix Figure F4: QGIS link attribute table showing obstruction crossing, vegetation loss, vegetation error and endpoint-check fields.

## QGIS obstruction layer

wtproject\_layers — land\_vegetation\_obstruction\_zones\_editable — Features Total: 250, Filtered: 250, Selected: 250

| ire | globalid | veg_type         | density           | obstruction | loss_db    | error_add | ref_code | method_note        | obs_cross           | obs_type | obs_density         | obs_level  | veg_loss_db | veg_error | obs_ref | obs_note           |                     |
|-----|----------|------------------|-------------------|-------------|------------|-----------|----------|--------------------|---------------------|----------|---------------------|------------|-------------|-----------|---------|--------------------|---------------------|
| 1   | 71369    | 3C4BC0F9-8BC...  | open_water_or...  | low         | low        | 1         | 0.0005   | Open water/bar...  | Borrow pit or ex... | yes      | Borrow pit          | low        | low         | 1         | 0.0005  | Open water/bar...  | Borrow pit or ex... |
| 2   | 69764    | 3D4875F7-BB5...  | open_water_or...  | low         | low        | 1         | 0.0005   | Open water/bar...  | Borrow pit or ex... | yes      | Borrow pit          | low        | low         | 1         | 0.0005  | Open water/bar...  | Borrow pit or ex... |
| 3   | 40778    | C057E1D3-ECF...  | open_water_or...  | low         | low        | 1         | 0.0005   | Open water/bar...  | Borrow pit or ex... | yes      | NULL                | high       | high        | 10        | 0.01    | ITU-R P.833; SA... | Dense trees or s... |
| 4   | 56672    | 93EF83F8-971E... | open_water_or...  | low         | low        | 1         | 0.0005   | Open water/bar...  | Borrow pit or ex... | yes      | Borrow pit          | low        | low         | 1         | 0.0005  | Open water/bar...  | Borrow pit or ex... |
| 5   | 56949    | DD55A681-BF7...  | open_water_or...  | low         | low        | 1         | 0.0005   | Open water/bar...  | Borrow pit or ex... | yes      | Borrow pit          | low        | low         | 1         | 0.0005  | Open water/bar...  | Borrow pit or ex... |
| 6   | 39718    | 420F8AF9-7E7...  | wetland_reeds     | medium      | medium     | 6         | 0.005    | ITU-R P.833; we... | Temporary fres...   | yes      | Temporary fres...   | medium     | medium      | 6         | 0.005   | ITU-R P.833; we... | Temporary fres...   |
| 7   | 73314    | 8C37B515-4E7...  | wetland_reeds     | medium      | medium     | 6         | 0.005    | ITU-R P.833; we... | Temporary fres...   | yes      | Temporary fres...   | medium     | medium      | 6         | 0.005   | ITU-R P.833; we... | Temporary fres...   |
| 8   | 69375    | 5A1431A7-6B3...  | wetland_reeds     | medium      | medium     | 6         | 0.005    | ITU-R P.833; we... | Temporary fres...   | yes      | Temporary fres...   | medium     | medium      | 6         | 0.005   | ITU-R P.833; we... | Temporary fres...   |
| 9   | 37139    | 36C5F4F-F4A...   | wet_grassland     | low_medium  | low_medium | 4         | 0.003    | ITU-R P.833; we... | Paddock wetla...    | yes      | Paddock wetland     | low_medium | low_medium  | 4         | 0.003   | ITU-R P.833; we... | Paddock wetla...    |
| 10  | 27051    | 11FA1384-C5B...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Experimental p...   | yes      | Experimental p...   | low        | low         | 1         | 0.0005  | Open water/lag...  | Experimental p...   |
| 11  | 31076    | 08EB8642-B41...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Experimental p...   | yes      | Experimental p...   | low        | low         | 1         | 0.0005  | Open water/lag...  | Experimental p...   |
| 12  | 88187    | F6192688-6E4F... | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 13  | 99632    | 22926011-83A...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 14  | 00951    | 27B3EE72-5BD...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 15  | 10212    | EA9FF83D-1839... | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 16  | 41505    | 08F5C5E0-55E...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 17  | 88784    | 5A7B2E7F-E32...  | open_water_or...  | low         | low        | 1         | 0.0005   | Open water GIS...  | Farm pond or d...   | yes      | Farm pond           | low        | low         | 1         | 0.0005  | Open water GIS...  | Farm pond or d...   |
| 18  | 34923    | 258687BE-450...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 19  | 11934    | 6E674521-C5E...  | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |
| 20  | 64933    | 5D93A9D6-AB...   | open_water_lag... | low         | low        | 1         | 0.0005   | Open water/lag...  | Waste stabilisat... | yes      | Waste stabilisat... | low        | low         | 1         | 0.0005  | Open water/lag...  | Waste stabilisat... |

Show All Features

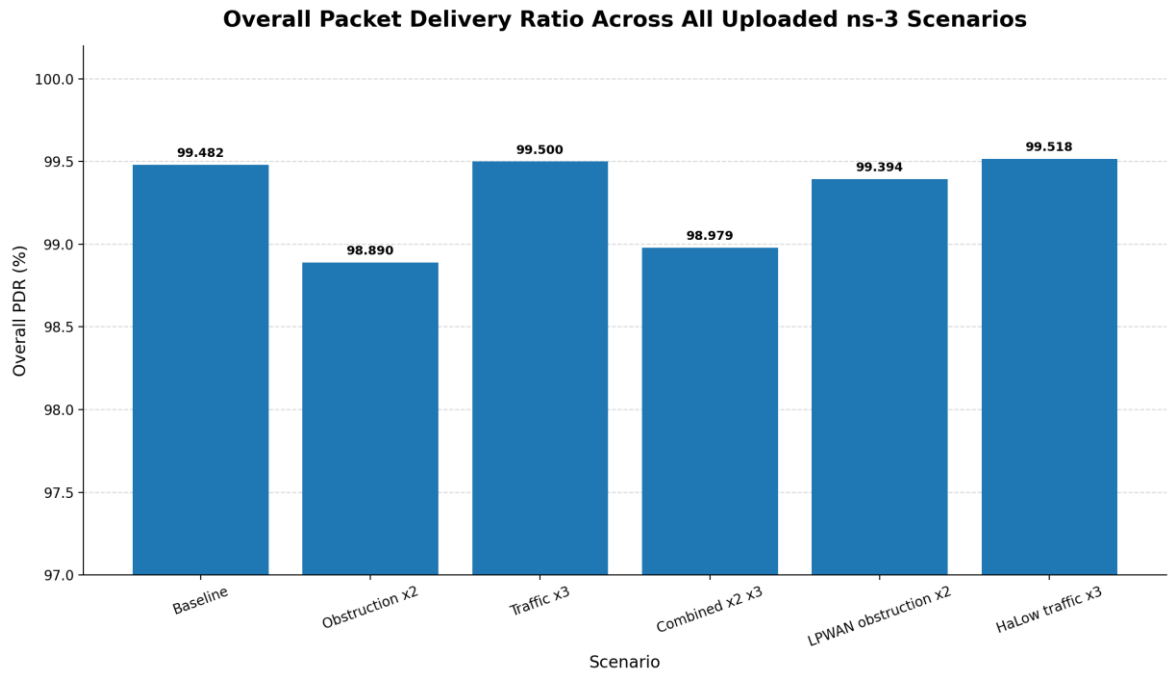
Appendix Figure F5: QGIS obstruction and vegetation zone layer used to support additional loss and error assumptions.

### 7.5.2 Appendix G: Labelled ns-3 performance plot evidence

The following figures are arranged from overall scenario performance to technology-level detail. They use the labelled ns-3 plots supplied in the missing ns-3 plots package and provide clearer appendix evidence for Chapter 4. The scenario set includes the baseline, obstruction x2, traffic x3, combined x2 x3, LPWAN obstruction x2 and HaLow traffic x3 tests.

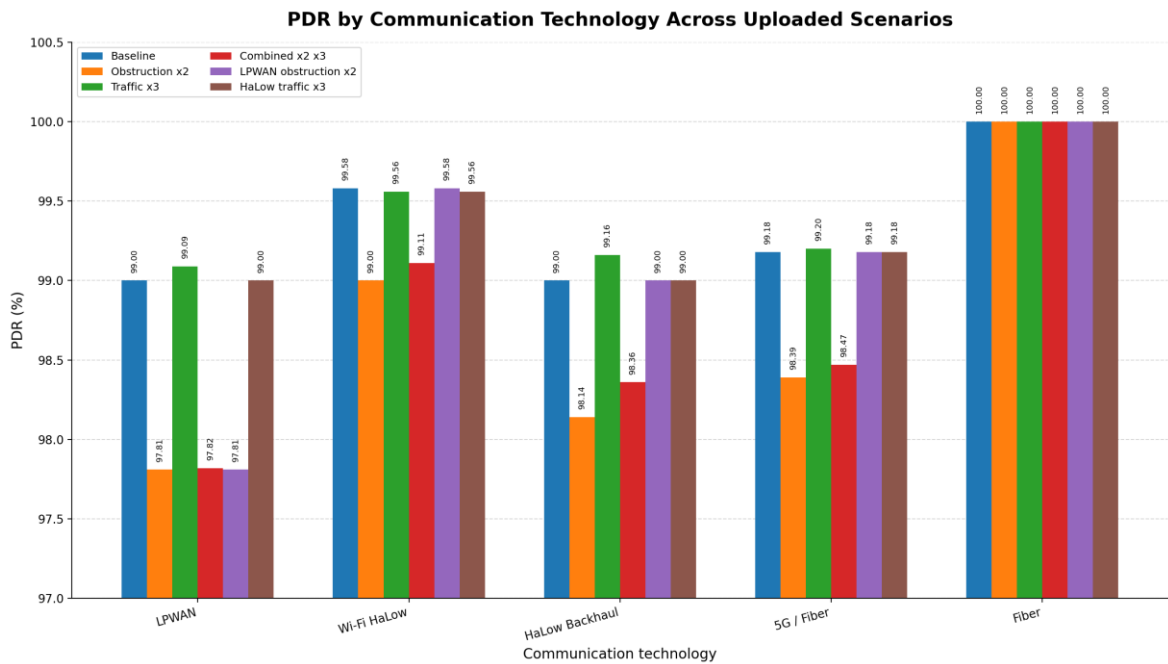
Overall packet delivery ratio across all uploaded ns-3 scenarios.





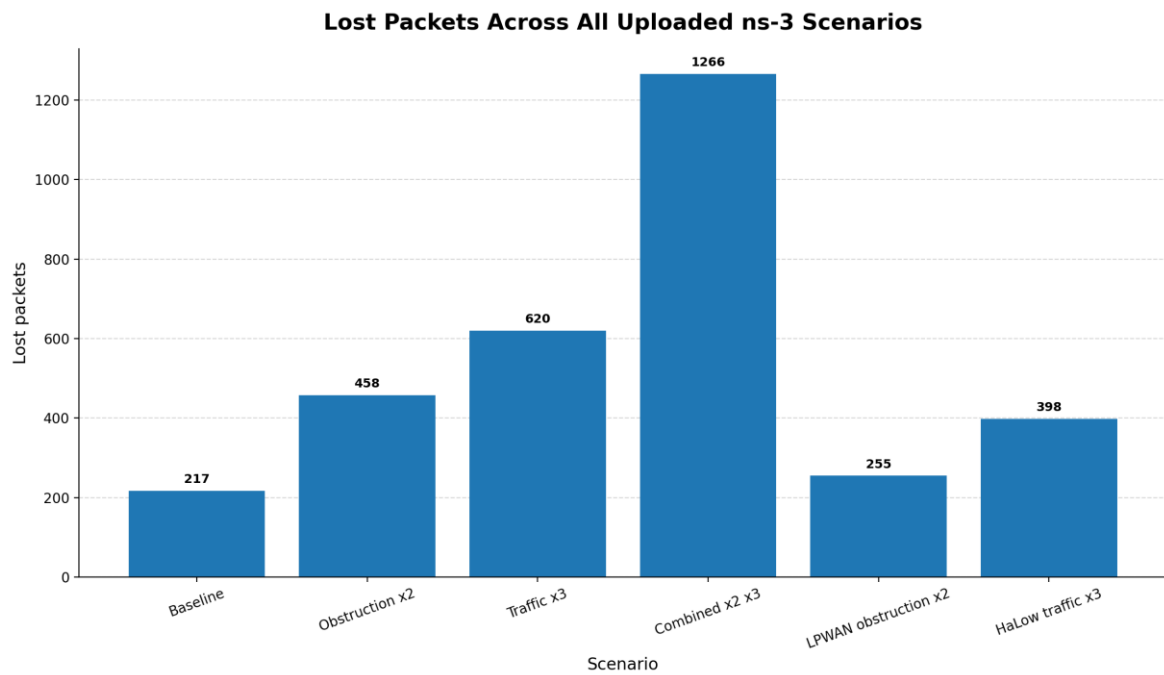
*Appendix Figure G1: Overall packet delivery ratio across all uploaded ns-3 scenarios.*

## Packet delivery ratio by communication technology



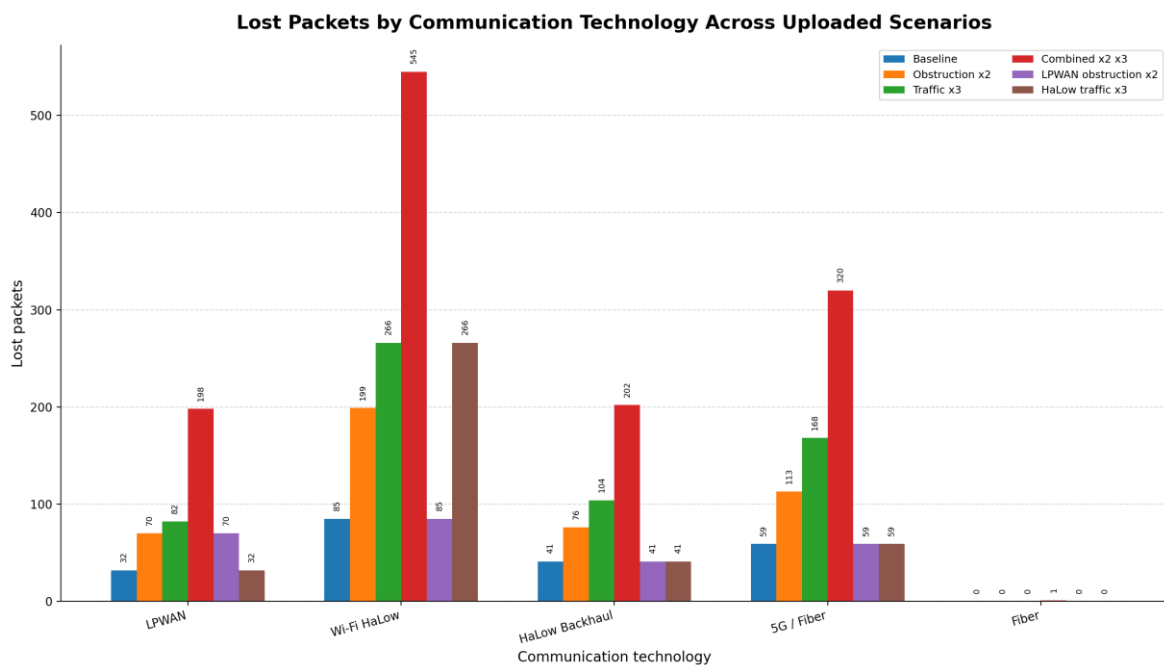
*Appendix Figure G2: Packet delivery ratio by communication technology across all uploaded scenarios.*

## Lost packets across all uploaded ns-3 scenarios



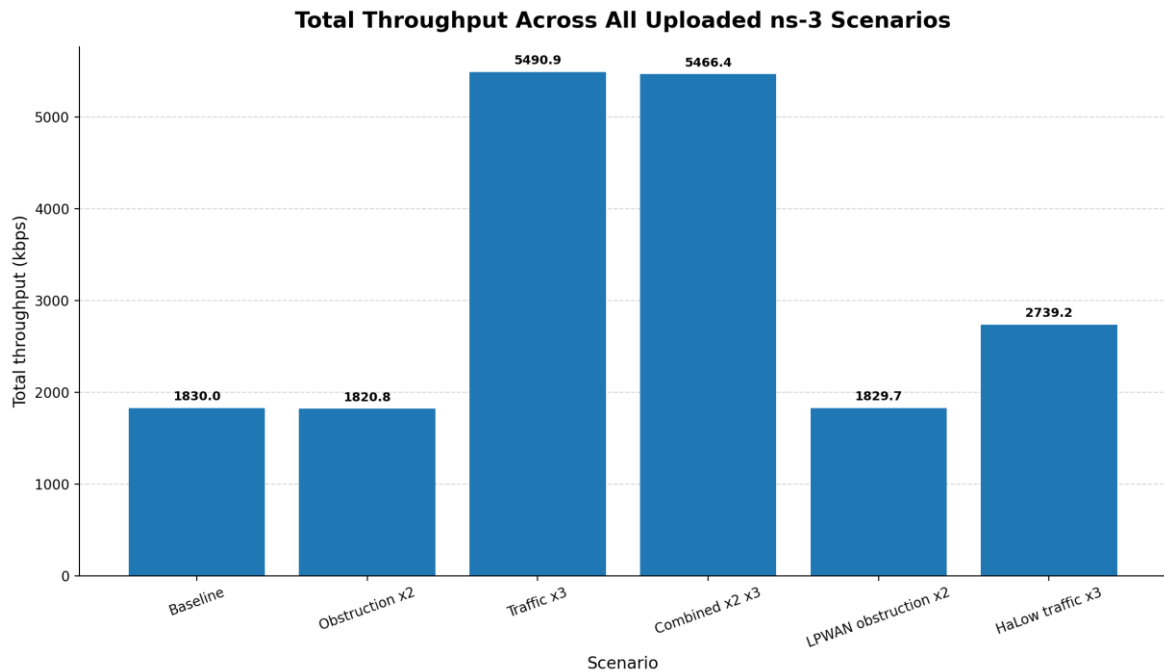
Appendix Figure G3: Lost packets across all uploaded ns-3 scenarios.

## Lost packets by communication technology



Appendix Figure G4: Lost packets by communication technology across all uploaded scenarios.

## Total throughput across all uploaded ns-3 scenarios



*Appendix Figure G5: Total throughput across all uploaded ns-3 scenarios.*

## Summary values behind the labelled plots

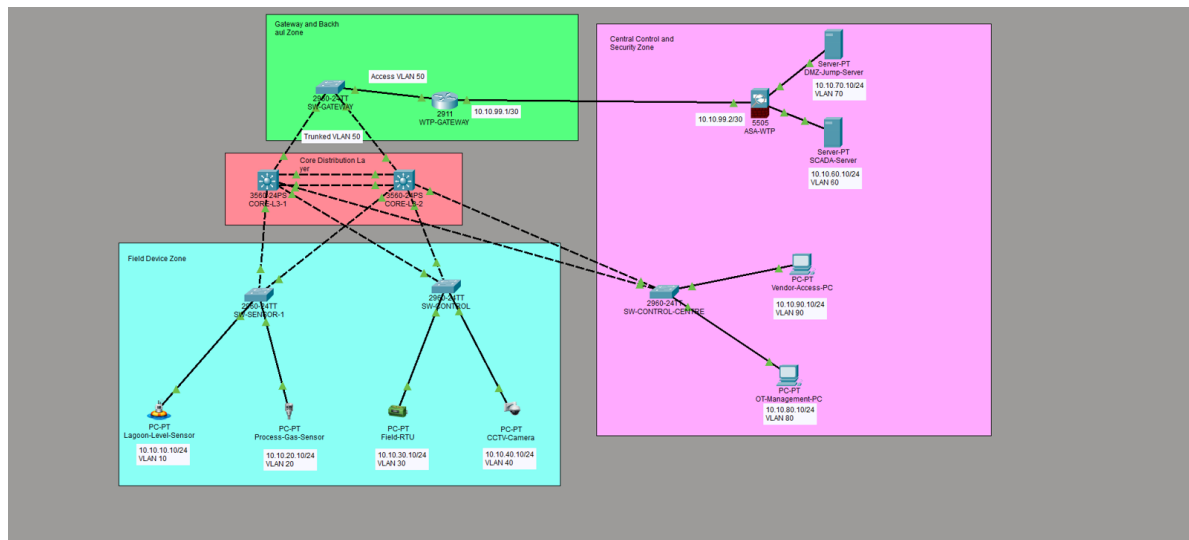
*Appendix Table G1: Overall scenario values used to generate the labelled ns-3 appendix plots.*

| Scenario                | TX packets | RX packets | Lost packets | PDR     | Throughput     |
|-------------------------|------------|------------|--------------|---------|----------------|
| Baseline                | 43,407     | 43,182     | 217          | 99.482% | 1,829.957 kbps |
| Obstruction x2          | 43,407     | 42,925     | 458          | 98.890% | 1,820.817 kbps |
| Traffic x3              | 130,221    | 129,570    | 620          | 99.500% | 5,490.901 kbps |
| Combined x2<br>x3       | 130,221    | 128,891    | 1,266        | 98.979% | 5,466.448 kbps |
| LPWAN<br>obstruction x2 | 43,407     | 43,144     | 255          | 99.394% | 1,829.691 kbps |
| HaLow traffic<br>x3     | 85,617     | 85,204     | 398          | 99.518% | 2,739.233 kbps |

### **7.5.3 Appendix H: Packet Tracer and ASA verification evidence**

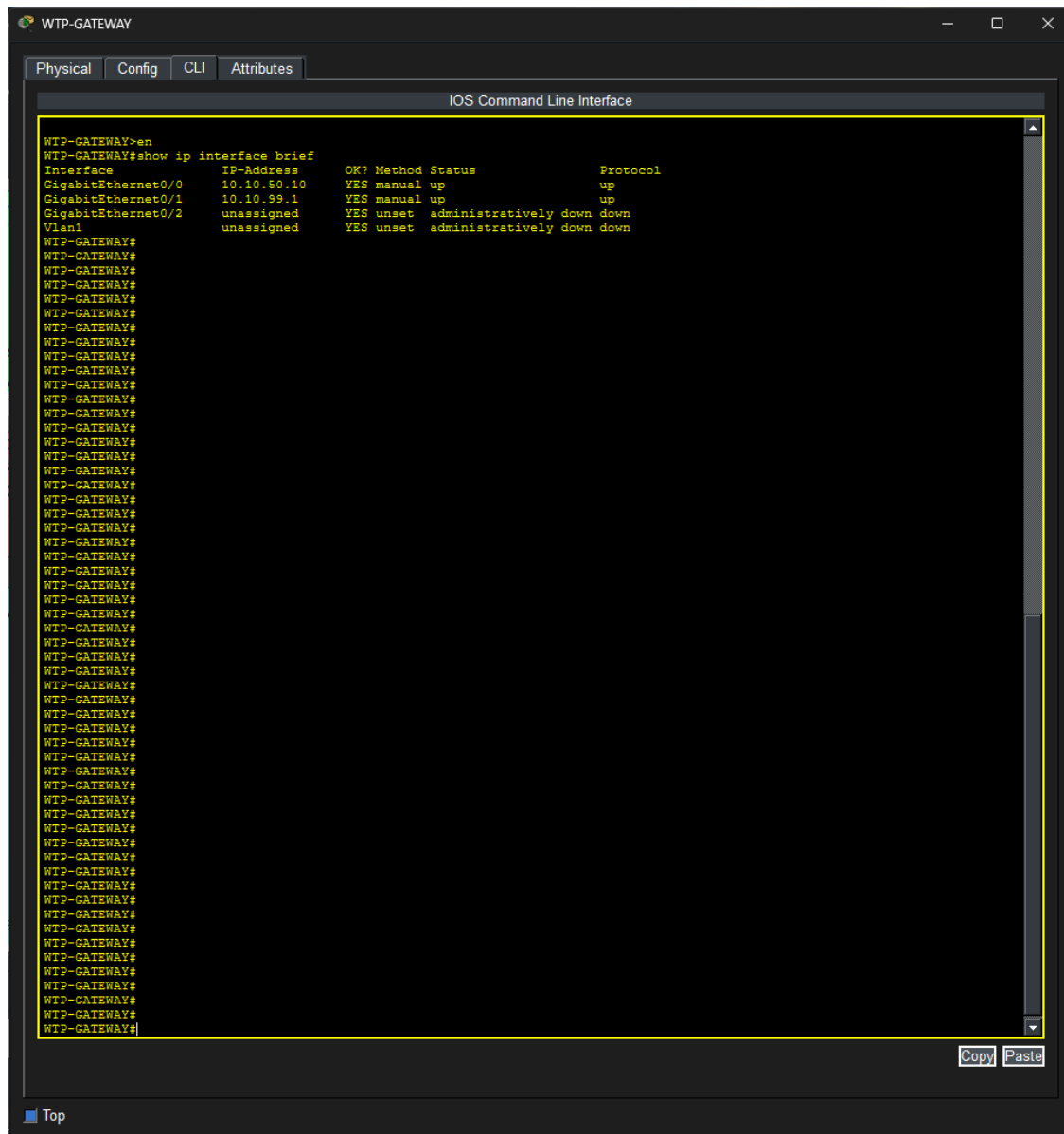
These screenshots support the Packet Tracer validation stage. They show the topology, gateway and ASA addressing, core redundancy, VLAN evidence, ASA policy hit counts, and endpoint traffic tests.

## Packet Tracer logical topology



Appendix Figure H1: Final Packet Tracer logical topology for the segmented WTP OT/SCADA network.

## Gateway interface verification



The screenshot shows a terminal window titled "WTP-GATEWAY" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The user has entered the command "show ip interface brief", and the output is as follows:

```
WTP-GATEWAY>en
WTP-GATEWAY#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.10.60.10     YES manual up          up
GigabitEthernet0/1       10.10.99.1      YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
```

Below the output, there are 30 lines of "WTP-GATEWAY#" prompts, followed by a "WTP-GATEWAY#" prompt at the bottom. At the bottom right of the terminal window, there are "Copy" and "Paste" buttons. At the bottom left, there is a "Top" button.

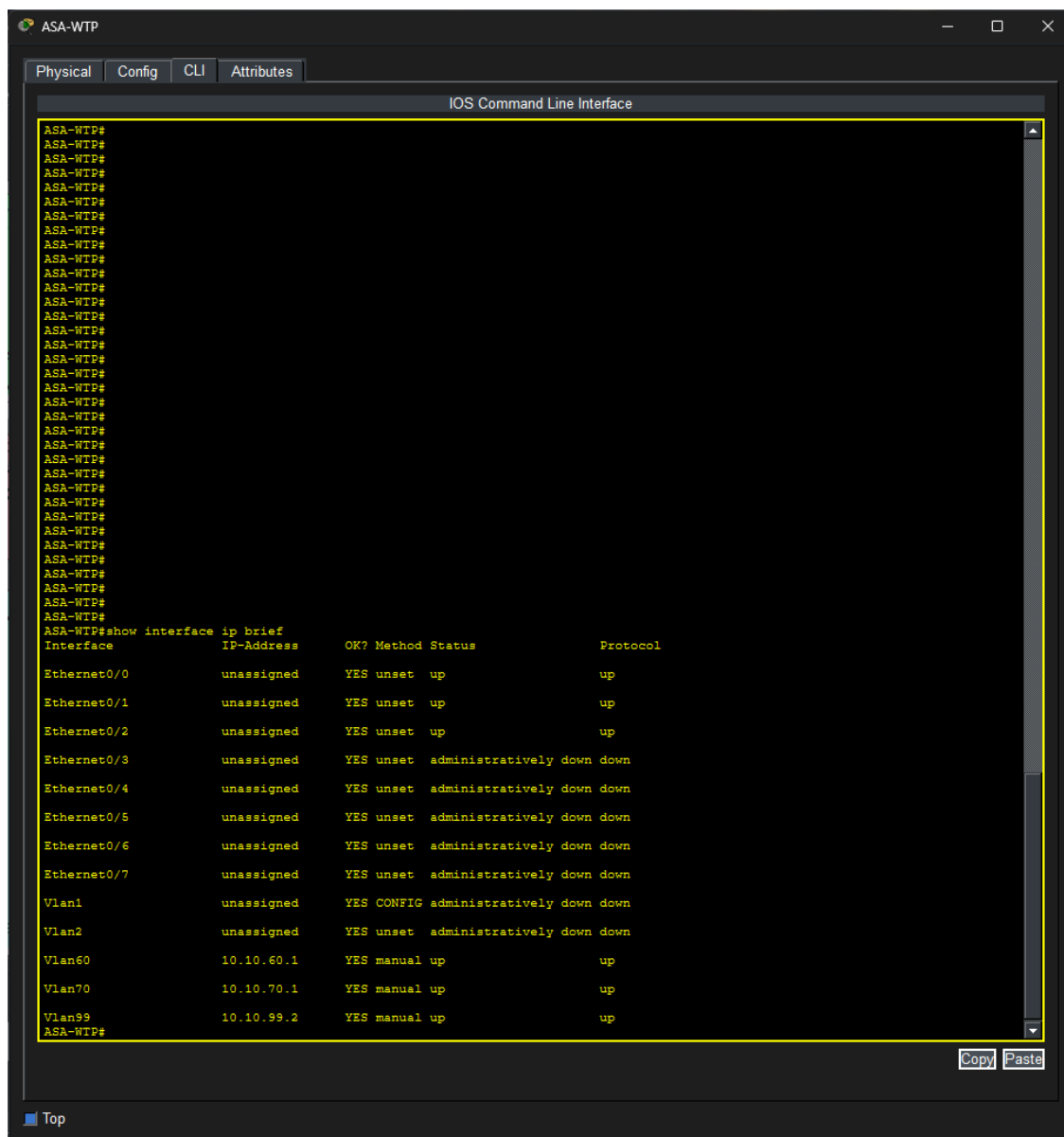
Appendix Figure H2: WTP-GATEWAY interface status and IP addressing verification.

## Gateway route verification

```
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#  
WTP-GATEWAY#show ip route  
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
        * - candidate default, U - per-user static route, o - ODR  
        P - periodic downloaded static route  
  
Gateway of last resort is 10.10.99.2 to network 0.0.0.0  
  
    10.0.0.0/8 is variably subnetted, 12 subnets, 3 masks  
S       10.10.10.0/24 [1/0] via 10.10.50.1  
S       10.10.20.0/24 [1/0] via 10.10.50.1  
S       10.10.30.0/24 [1/0] via 10.10.50.1  
S       10.10.40.0/24 [1/0] via 10.10.50.1  
C       10.10.50.0/24 is directly connected, GigabitEthernet0/0  
L       10.10.50.10/32 is directly connected, GigabitEthernet0/0  
S       10.10.60.0/24 [1/0] via 10.10.99.2  
S       10.10.70.0/24 [1/0] via 10.10.99.2  
S       10.10.80.0/24 [1/0] via 10.10.50.1  
S       10.10.90.0/24 [1/0] via 10.10.50.1  
C       10.10.99.0/30 is directly connected, GigabitEthernet0/1  
L       10.10.99.1/32 is directly connected, GigabitEthernet0/1  
S*     0.0.0.0/0 [1/0] via 10.10.99.2  
  
WTP-GATEWAY#
```

Appendix Figure H3: WTP-GATEWAY routing table verification.

## ASA interface verification



*Appendix Figure H4: ASA interface and VLAN IP verification.*



```
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#  
ASA-WTP#show route  
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
I - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route  
  
Gateway of last resort is not set  
  
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks  
C    10.0.0.0 255.255.255.0 is directly connected, scada, Vlan60  
      is directly connected, dmz, Vlan70  
C    10.0.0.0 255.255.255.252 is directly connected, inside, Vlan99  
S    10.10.10.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
S    10.10.20.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
S    10.10.30.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
S    10.10.40.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
S    10.10.50.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
C    10.10.60.0 255.255.255.0 is directly connected, scada, Vlan60  
S    10.10.70.0 255.255.255.0 is directly connected, dmz, Vlan70  
S    10.10.80.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
S    10.10.90.0 255.255.255.0 [1/0] via 10.10.99.1, inside  
C    10.10.99.0 255.255.255.252 is directly connected, inside, Vlan99  
ASA-WTP#
```

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## Core HSRP verification

```

CORE-L3-1
Physical Config CLI Attributes
IOS Command Line Interface

%HSRP-6-STATECHANGE: Vlan50 Grp 50 state Standby -> Active
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan80, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan30, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan40, changed state to up
%HSRP-6-STATECHANGE: Vlan10 Grp 10 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan90 Grp 90 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan80 Grp 80 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan40 Grp 40 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan40 Grp 40 state Standby -> Active
%HSRP-6-STATECHANGE: Vlan20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: Vlan40 Grp 40 state Speak -> Standby

CORE-L3-1>
CORE-L3-1>
CORE-L3-1>en
CORE-L3-1#show standby brief
P indicates configured to preempt.
|
Interface Grp Pri P State Active Standby Virtual IP
Vl10 10 110 P Active local 10.10.10.3 10.10.10.1
Vl20 20 100 P Standby 10.10.20.3 local 10.10.20.1
Vl30 30 110 P Active local 10.10.30.3 10.10.30.1
Vl40 40 100 P Standby 10.10.40.3 local 10.10.40.1
Vl50 50 110 P Active local 10.10.50.3 10.10.50.1
Vl80 80 110 P Active local 10.10.80.3 10.10.80.1
Vl90 90 100 P Standby 10.10.90.3 local 10.10.90.1
CORE-L3-1#
Copy Paste
Top

```

Appendix Figure H6: CORE-L3-1 HSRP standby summary showing gateway redundancy evidence.

CORE-L3-1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#  
CORE-L3-1#show etherchannel summary  
Flags: D - down          P - in port-channel  
       I - stand-alone s - suspended  
       H - Hot-standby (LACP only)  
       R - Layer3        S - Layer2  
       U - in use         f - failed to allocate aggregator  
       u - unsuitable for bundling  
       w - waiting to be aggregated  
       d - default port  
  
Number of channel-groups in use: 1  
Number of aggregators:           1  
  
Group Port-channel Protocol Ports  
-----+-----+-----+-----  
1      Po1(SU)            LACP Fa0/23(P) Fa0/24(P)  
CORE-L3-1#
```

CopyPaste

## Core VLAN verification



```

CORE-L3-1
Physical Config CLI Attributes
IOS Command Line Interface

w - waiting to be aggregated
d - default port

Number of channel-groups in use: 1
Number of aggregators: 1

Group Port-channel Protocol Ports
-----
1 Po1(SU) LACP Fa0/23(P) Fa0/24(P)
CORE-L3-1#

CORE-L3-1 con0 is now available

Press RETURN to get started.

CORE-L3-1>en
CORE-L3-1#show vlan brief

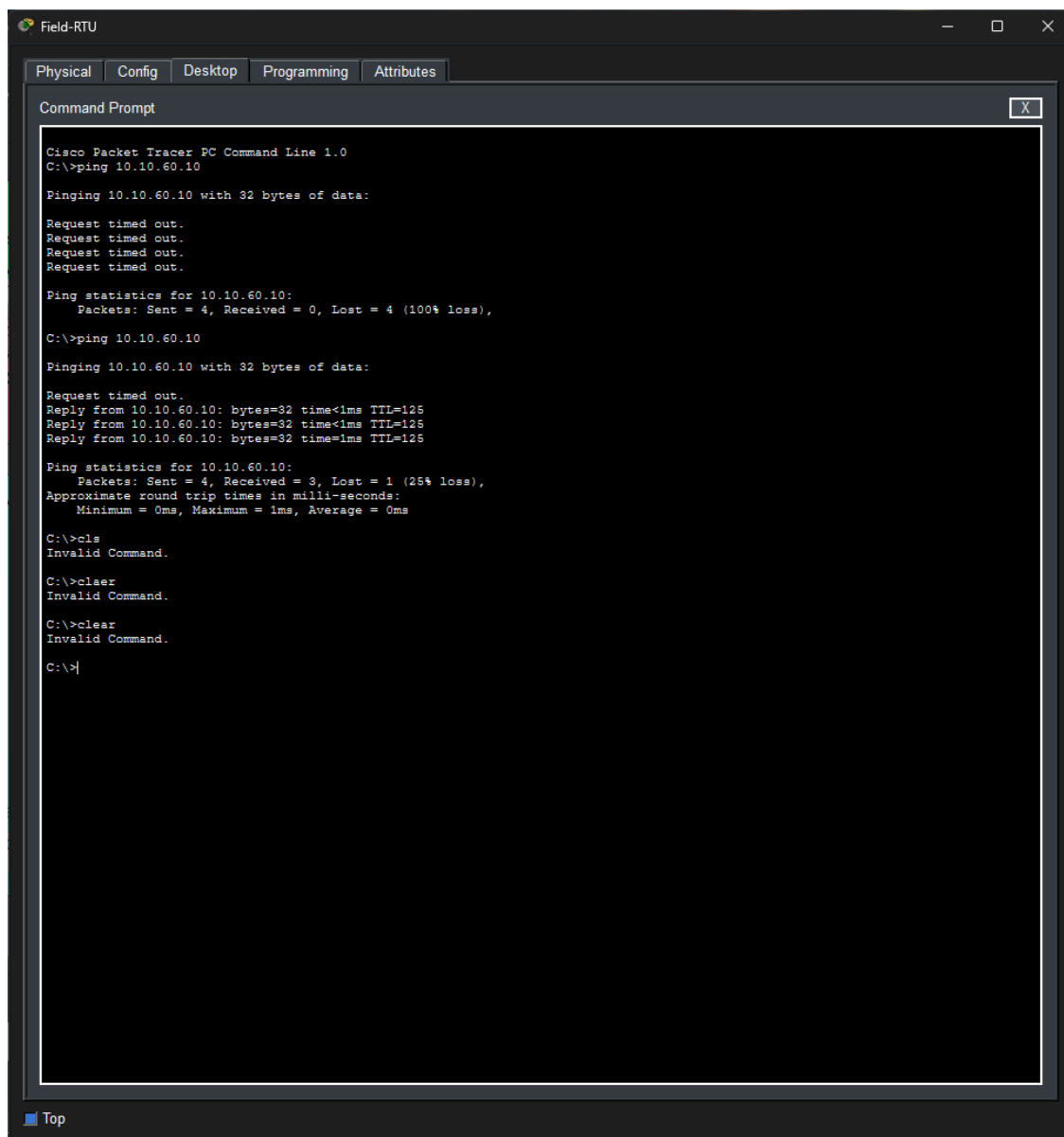
VLAN Name                Status    Ports
-----
1  default                active    Po1, Fa0/5, Fa0/6, Fa0/7,
Fa0/8, Fa0/9, Fa0/10, Fa0/11,
Fa0/12, Fa0/13, Fa0/14, Fa0/15,
Fa0/16, Fa0/17, Fa0/18, Fa0/19,
Fa0/20, Fa0/21, Fa0/22, Fa0/23,
Fa0/24, Gig0/1, Gig0/2
10  LAGOON_SENSOR          active
20  PROCESS_SENSOR         active
30  FIELD_CONTROL_RTU      active
40  CCTV_HIGH_BW         active
50  GATEWAY_BACKHAUL       active    Fa0/4
80  OT_MANAGEMENT          active
90  VENDOR_REMOTE          active
1002 fddi-default          active
1003 token-ring-default    active
1004 fddinet-default       active
1005 trnet-default         active
CORE-L3-1#
Copy Paste
Top
```

Appendix Figure H8: CORE-L3-1 VLAN brief showing VLAN creation and segmentation evidence.

The screenshot shows a network device's configuration interface for 'ASA-WTP'. It features four tabs: 'Physical', 'Config', 'CLI', and 'Attributes'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The terminal window shows a series of 'ASA-WTP#' prompts followed by the command 'show access-list'. The output lists various access control entries (ACEs) for different protocols and ports, including BASELINE\_IN, SCADA\_RETURN, DMZ\_RETURN, and OT\_IN. Each entry specifies the action (permit or deny), protocol, source/destination IP addresses, and hit counts. At the bottom right of the terminal window are 'Copy' and 'Paste' buttons.

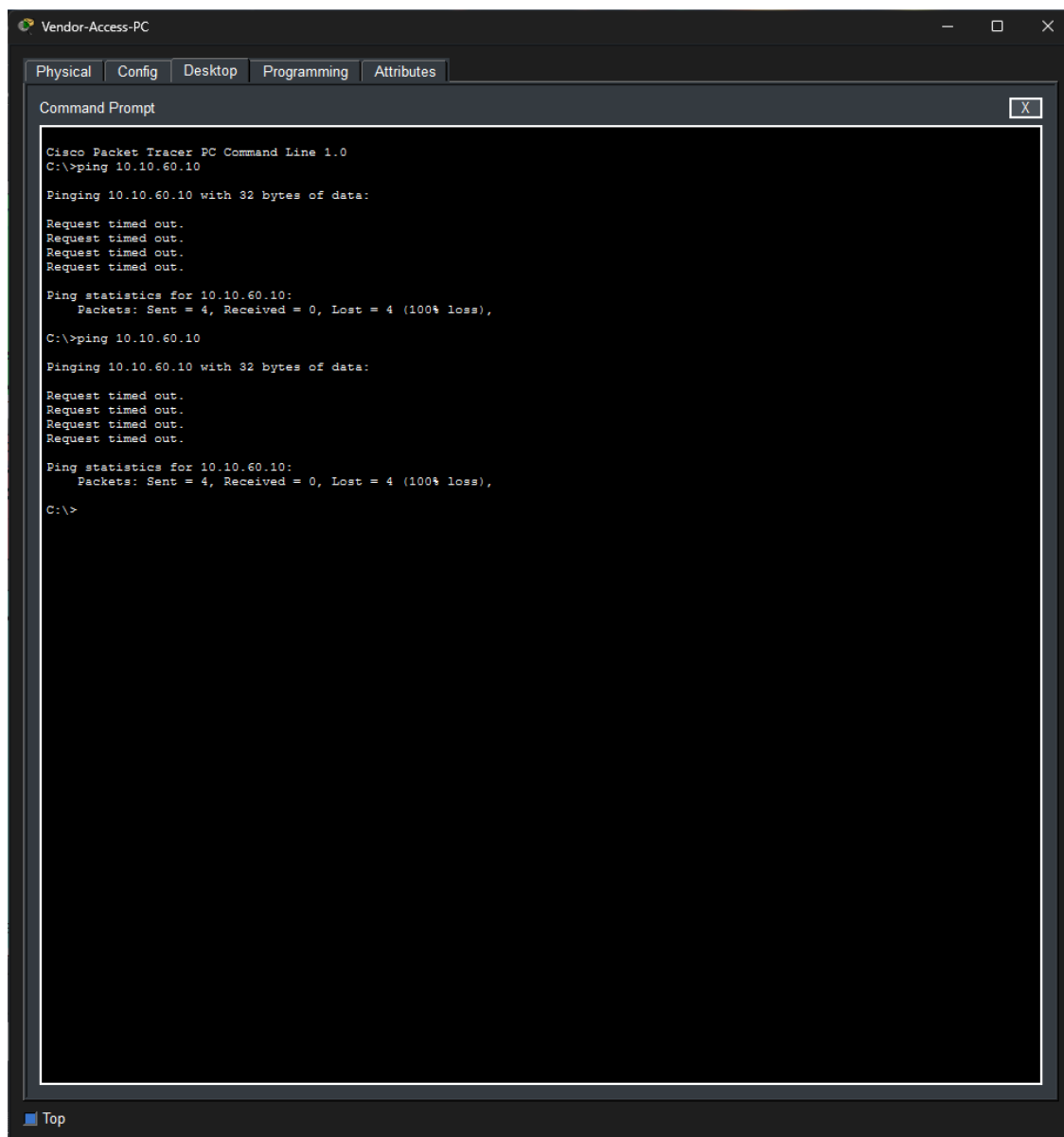
93

## Allowed ping test



*Appendix Figure H10: Allowed RTU-to-SCADA ping test demonstrating permitted operational traffic.*

## Blocked ping test

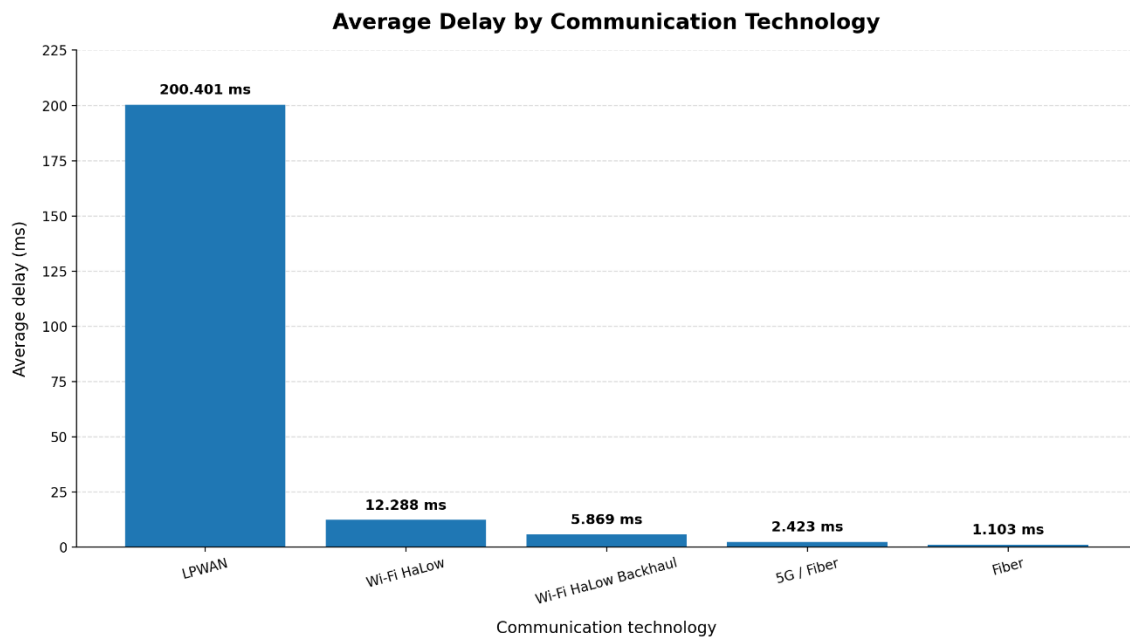


*Appendix Figure H11: Blocked vendor-to-SCADA ping test demonstrating enforcement of segmentation policy.*

### 7.5.4 Appendix I: Result graph screenshots

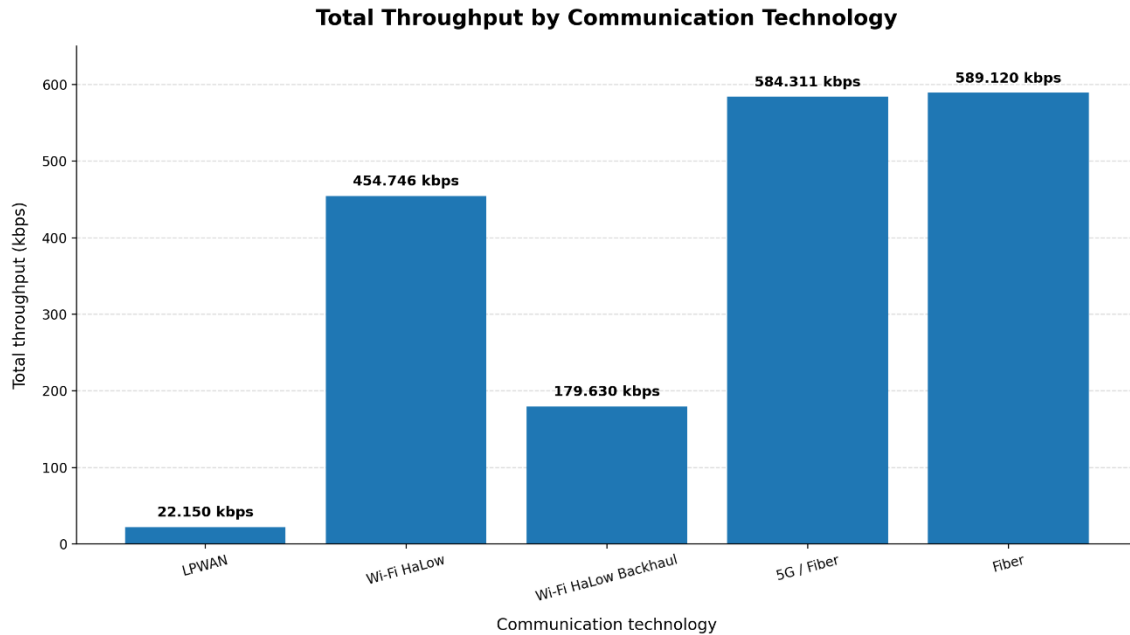
These result screenshots support the numerical results presented in Chapter 4. They are included here as an additional evidence set for the baseline, sensitivity and security findings.

## Baseline average delay



Appendix Figure 11: Baseline average delay by communication technology (source: author's ns-3 results).

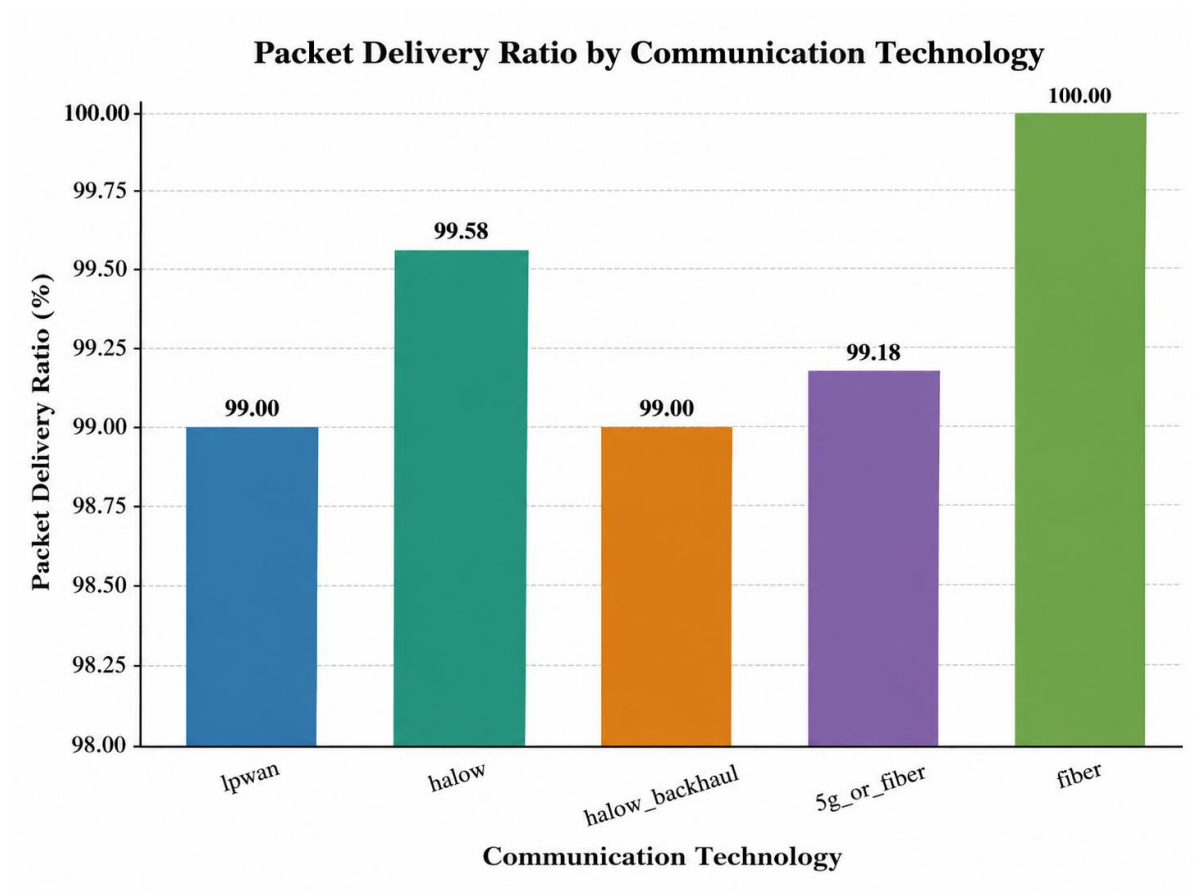
## Baseline total throughput



Appendix Figure 12: Baseline total throughput by communication technology (source: author's ns-3 results).

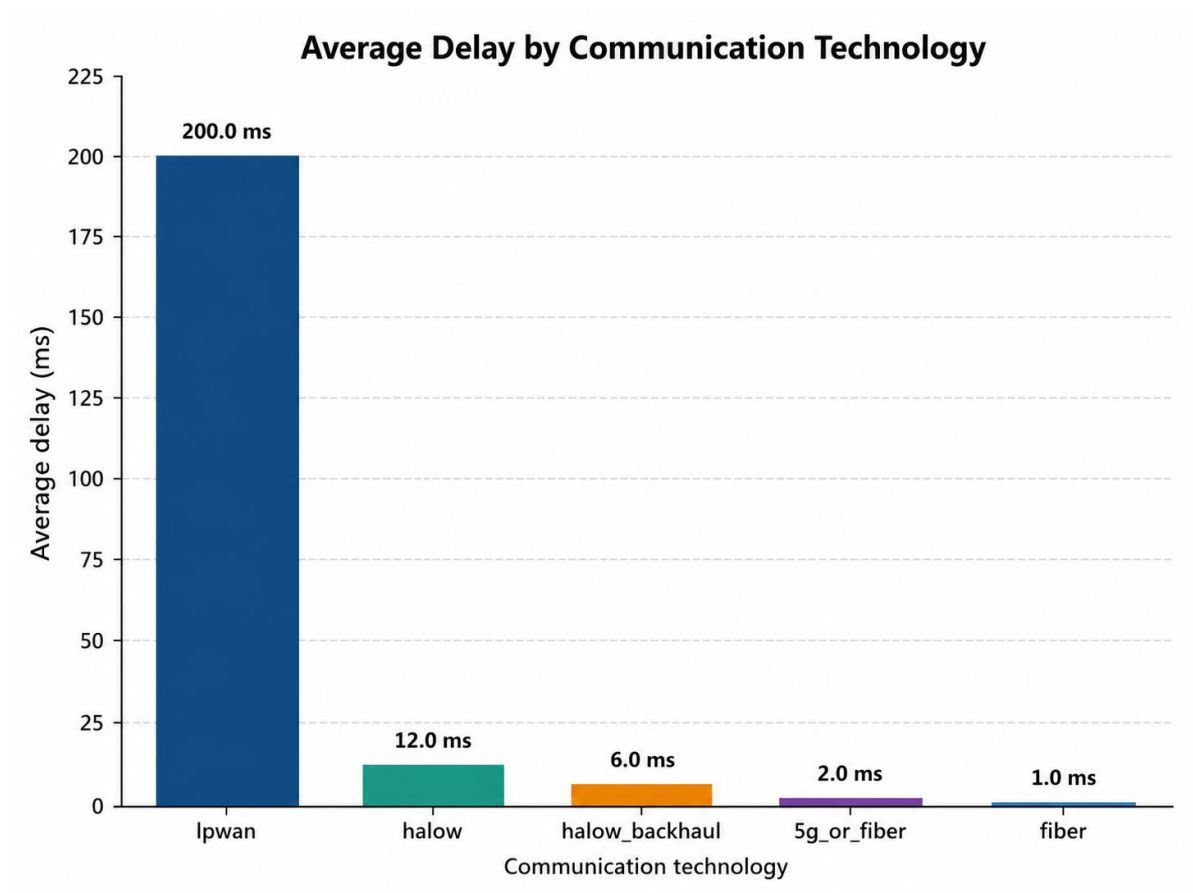


## Baseline PDR graph



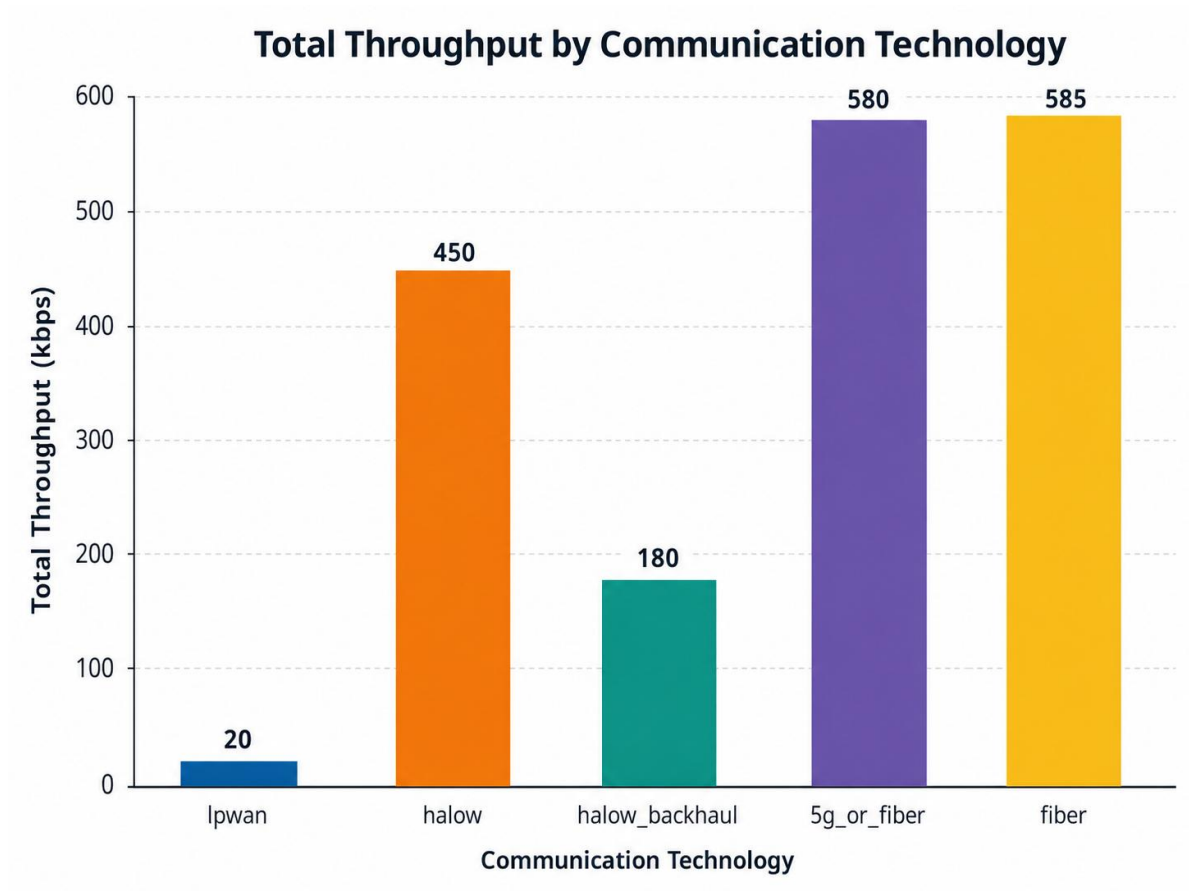
*Appendix Figure I3: Baseline packet delivery ratio by communication technology.*

## Baseline delay graph



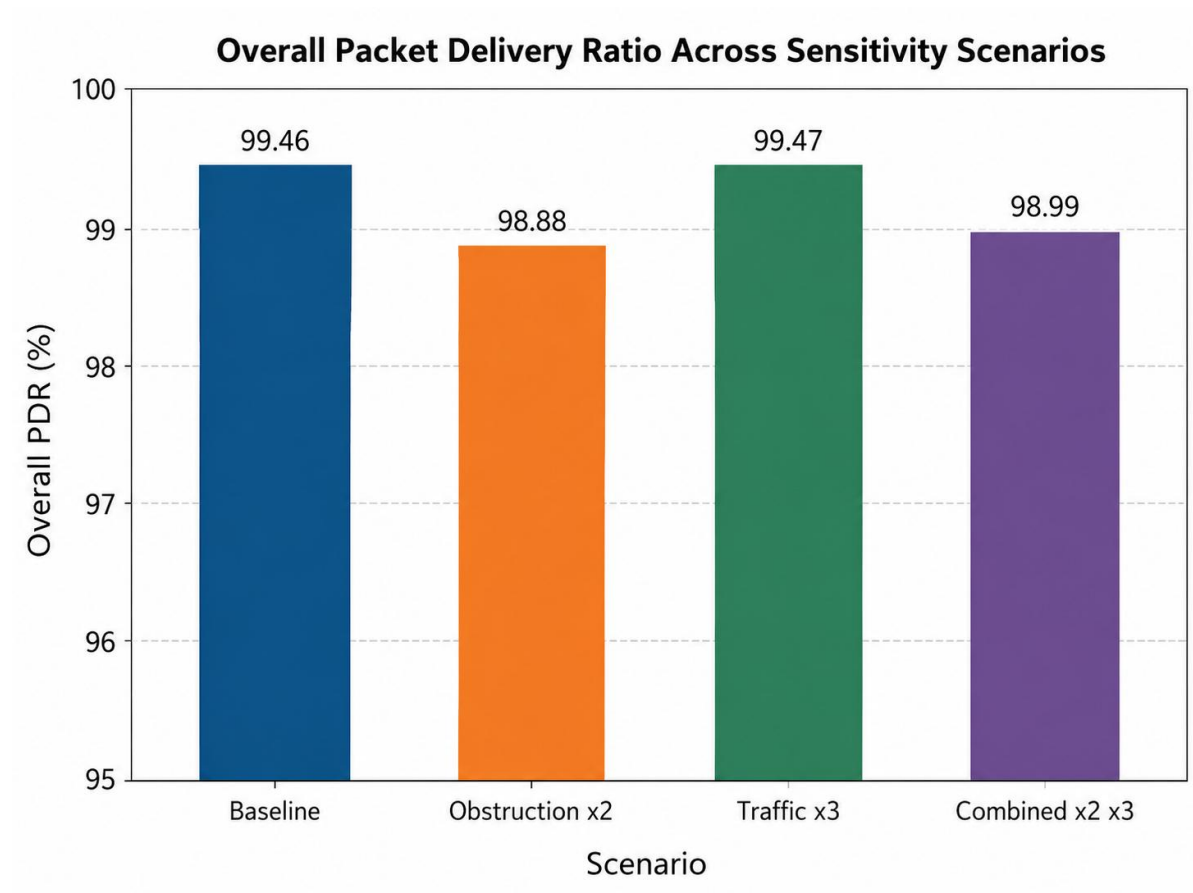
*Appendix Figure I4: Baseline average delay by communication technology.*

## Baseline throughput graph



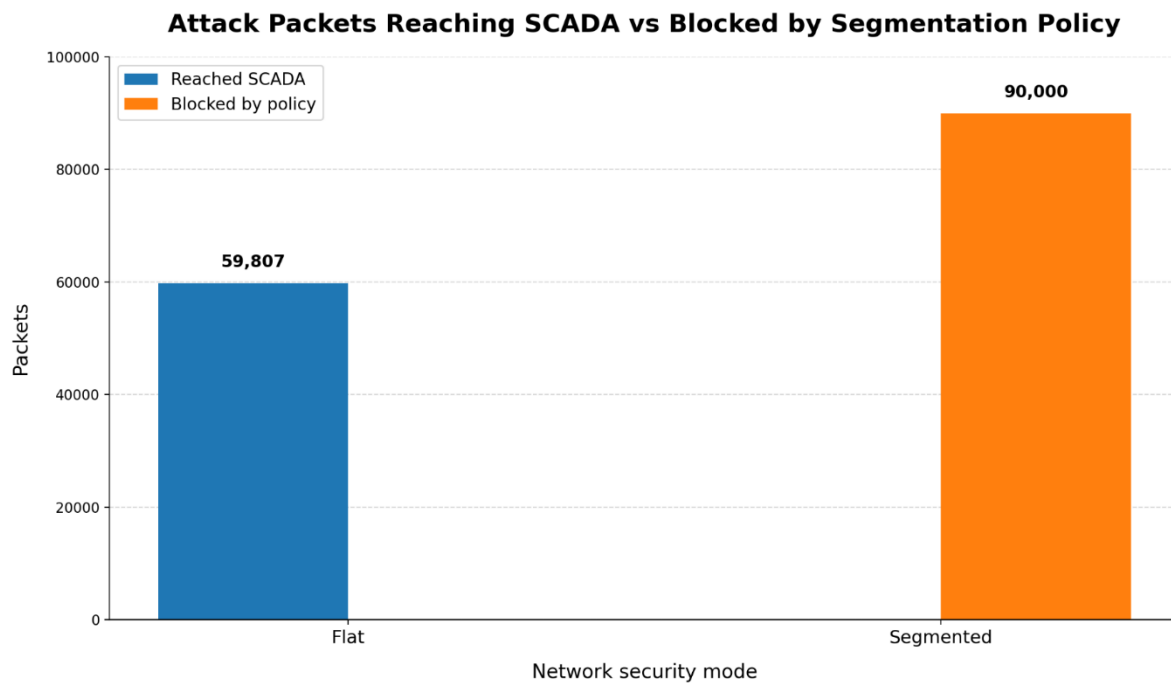
*Appendix Figure I5: Baseline total throughput by communication technology.*

### Sensitivity PDR graph



*Appendix Figure I6: Overall packet delivery ratio across sensitivity scenarios.*

## Security result graph



*Appendix Figure I7: Attack packets reaching SCADA compared with packets blocked by segmentation policy.*

### 7.5.5 Appendix evidence notes

The screenshots provide verification evidence but should not be treated as the main research result by themselves. The main claims in the thesis should continue to be based on the structured QGIS datasets, ns-3 CSV/XML outputs, Packet Tracer configuration evidence, and the interpreted results in Chapter 4.